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STEAM DRUM (WHB)
(A150NRJ10EQ001)

DESIGN CALCULATION

In Accordance with ASME Section VIII Division 2

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\Steam drum (WHB) - division 2.pvdb

Date of Analysis : Sep 16,2026 6:47pm

PV Elite 28 U2, April 2026



Vessel Design Summary:

ASME Code, Section VIII Division 2, 2025

Diameter Spec : 2450.000 x 2860.000 mm. OD	
Vessel Design Length, Tangent to Tangent	15200.00 mm.
Distance of Bottom Tangent above Grade	14500.00 mm.
Specified Datum Line Distance	-50.00 mm.
Internal Design Temperature	286 °C
Internal Design Pressure	51.000 bars
External Design Temperature	286 °C
External Design Pressure	1.034 bars
Maximum Allowable Working Pressure	47 bars
External Max. Allowable Working Pressure	18 bars
Shop Test Pressure	59.165 bars
Required Minimum Design Metal Temperature	-10.0 °C
Warmest Computed Minimum Design Metal Temperature	39.0 °C

Warning: Computed overall MDMT was higher than the required value !

Wind Design Code	ASCE/SEI 7-22
Earthquake Design Code	ASCE/SEI 7-22

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-387 11	2	...	K11789	
Head	SA-387 11	2	...	K11789	
Cone	SA-387 11	2	...	K11789	
Nozzle	SA-182 F11	1	...	K11597	
Nozzle	SA-336 F12				
Nozzle Flg	SA-105	...	...	K03504	
Nozzle Flg	SA-182 F11	1	...	K11597	
Sup. Lugs	SA-387 11	2	...	K11789	
Lug Bolting	SA-193 B7				

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
shell 6	51.510	1.03	52.000	1.5000	N/A	No
shell 5	51.510	1.03	52.000	1.5000	N/A	No
shell 4	51.510	1.03	52.000	1.5000	N/A	No
cone transition	51.216	1.03	53.000	1.5000	N/A	No
Shell-3	51.098	1.03	52.000	1.5000	N/A	No
Shell-2	51.000	1.03	52.000	1.5000	N/A	No
Shell-1	51.000	1.03	52.000	1.5000	N/A	No
Top Dish	51.000	1.03	52.000	1.5000	N/A	No

Liquid Level: 5200.00 mm. Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000

Element Types and Properties:



Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
Cylinder	2050.0	2000.0	40.0	40.0	39.1	13.2	1.00	1.00
Cylinder	4050.0	2000.0	40.0	40.0	39.1	13.2	1.00	1.00
Cylinder	7050.0	3000.0	40.0	40.0	39.1	13.2	1.00	1.00
Conical	8250.0	1200.0	58.0	48.0	45.9	7.8	1.00	1.00
Cylinder	9250.0	1000.0	46.0	46.0	45.0	14.7	1.00	1.00
Cylinder	12250.0	3000.0	46.0	46.0	45.0	14.7	1.00	1.00
Cylinder	15250.0	3000.0	46.0	46.0	45.0	14.7	1.00	1.00
Sphere	15250.0	0.0	28.0	24.0	23.4	5.4	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
A2-6"	WRC-297	0.902	Shell	Passed
B2-14"	WRC-107/537	0.904	n/a	Passed

Loads for Foundation/Support Design:

Total Wind Shear on Support	10858. Kg
Total Earthquake Shear on Support	22902. Kg
Wind Moment on Support	20805. Kg-m.
Earthquake Moment on Support	163991. Kg-m.
Wind Load Case Test Moment on Support	6866. Kg-m.
Test Case Wind Shear on Support	3583. Kg
Maximum Initial Bolt Load	18141. Kg

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	126750.0 kg.
Shop Test - Fabricated + water (full)	263629.8 kg.
Shipping - Fab. + rem. intls.+ shipping app.	130017.3 kg.
Erected - Fab. + rem. intls.+ insul. (etc)	130687.3 kg.
Empty - Fab. + intls. + details + wghts.	130687.3 kg.
Operating - Empty + operating liquid (no ca)	156103.1 kg.
Field Test - Empty weight + test medium (full)	266897.1 kg.

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Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	Tbl 4.5.2	Weld	Pass/Fail
A2-6"	52	OK	...	OK	OK	Passed
L2-2"	47	OK	...	OK	OK	Passed
L4-2"	47	OK	...	OK	OK	Passed
L6-2"	47	OK	...	OK	OK	Passed
L8-2"	47	OK	...	OK	OK	Passed
L10-2"	47	OK	...	OK	OK	Passed
L1-2"	47	OK	...	OK	OK	Passed
L3-2"	47	OK	...	OK	OK	Passed
L5-2"	47	OK	...	OK	OK	Passed
L7-2"	47	OK	...	OK	OK	Passed
L9-2"	47	OK	...	OK	OK	Passed
M2-24"	52	OK	...	...	OK	Passed
B2-14"	52	OK	...	OK	OK	Passed
P1-2"	52	OK	...	OK	OK	Passed
P2-2"	52	OK	...	OK	OK	Passed
V1-2"	52	OK	...	OK	OK	Passed
S1-2"	52	OK	...	OK	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 47 Nozzle : L9-2"
Minimum MAWP Shells/Flanges : 47 Element : Shell-3
Minimum MAPnc Shells/Flanges : 68 Element : Shell-1

Computed Vessel M.A.W.P. : 47 bars

VIII-2 Nozzle to Gross Structural Discontinuity Check:

No Nozzles were too close to any gross structural discontinuities per 4.5.13.1

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm.	Layout Angle deg	Inside Radius mm.
70	A2-6"	10699.999	270.000	69.680
70	L2-2"	10700.000	90.000	26.900
70	L4-2"	10700.000	120.000	26.900
70	L6-2"	10700.000	150.000	26.900
70	L8-2"	10700.000	180.000	26.900
70	L10-2"	10700.000	210.000	26.900
80	L1-2"	13699.999	90.000	26.900
80	L3-2"	13699.999	120.000	26.900
80	L5-2"	13699.999	150.000	26.900
80	L7-2"	13699.999	180.000	26.900
80	L9-2"	13699.999	210.000	26.900
80	M2-24"	13699.999	270.000	286.300
90	B2-14"	0.000	0.000	154.986
90	P1-2"	0.000	90.000	26.900
90	P2-2"	0.000	270.000	26.900
90	V1-2"	0.000	180.000	26.900
90	S1-2"	0.000	0.000	39.600

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
lc - Arc length along the inside vessel surface in the circ. direction

STEAM DRUM (WHB)
(A150NRJ10EQ001)



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FileName : Steam drum (WHB) - division 2

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Nozzle Summary: Step: 48 6:47pm Sep 16,2026



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If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
L2-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	243.49	600
L4-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	243.49	600
L6-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	243.49	600
L8-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	243.49	600
L10-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	243.49	600
L1-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	243.49	600
L3-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	243.49	600
L5-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	243.49	600
L7-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	243.49	600
L9-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	243.49	600
P1-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	447.25	...
P2-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	447.25	...
V1-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	447.25	...
S1-2"	3.000 in	Actual	LWN	4.620	20.574	...	...	246.37	...
A2-6"	6.000 in	160	WNF	6.625	18.237	...	...	251.05	600
B2-14"	14.000 in	120	None	14.000	27.788	...	...	343.52	...
M2-24"	24.000 in	Actual	LWN	24.000	20.000	...	...	312.92	600

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
L2-2"	SA-182 F11	52.000	10.000	...	...	...
L4-2"	SA-182 F11	52.000	10.000	...	...	...
L6-2"	SA-182 F11	52.000	10.000	...	...	...
L8-2"	SA-182 F11	52.000	10.000	...	...	...
L10-2"	SA-182 F11	52.000	10.000	...	...	...
L1-2"	SA-182 F11	52.000	10.000	...	...	...
L3-2"	SA-182 F11	52.000	10.000	...	...	...
L5-2"	SA-182 F11	52.000	10.000	...	...	...
L7-2"	SA-182 F11	52.000	10.000	...	...	...
L9-2"	SA-182 F11	52.000	10.000	...	...	...
P1-2"	SA-182 F11	27.000	10.000	...	...	...
P2-2"	SA-182 F11	27.000	10.000	...	...	...
V1-2"	SA-182 F11	27.000	10.000	...	...	...
S1-2"	SA-182 F11	27.000	10.000	...	...	...
A2-6"	SA-182 F11	52.000	10.000	...	...	...
B2-14"	SA-182 F11	27.000	10.000	...	...	...
M2-24"	SA-336 F12	52.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
L2-2"	10750.000	90.0	196.85	0.00	Shell-2

STEAM DRUM (WHB)
(A150NRJ10EQ001)



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FileName : Steam drum (WHB) - division 2

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Nozzle Schedule: Step: 47 6:47pm Sep 16,2026

L4-2"	10750.000	120.0	196.85	0.00	Shell-2
L6-2"	10750.000	150.0	196.85	0.00	Shell-2
L8-2"	10750.000	180.0	196.85	0.00	Shell-2
L10-2"	10750.000	210.0	196.85	0.00	Shell-2
L1-2"	13750.000	90.0	196.85	0.00	Shell-1
L3-2"	13750.000	120.0	196.85	0.00	Shell-1
L5-2"	13750.000	150.0	196.85	0.00	Shell-1
L7-2"	13750.000	180.0	196.85	0.00	Shell-1
L9-2"	13750.000	210.0	196.85	0.00	Shell-1
P1-2"	...	90.0	400.00	0.00	Top Dish
P2-2"	...	270.0	400.00	0.00	Top Dish
V1-2"	...	180.0	400.00	0.00	Top Dish
S1-2"	...	0.0	190.50	0.00	Top Dish
A2-6"	10749.999	270.0	200.00	0.00	Shell-2
B2-14"	...	0.0	300.00	0.00	Top Dish
M2-24"	13750.000	270.0	196.85	0.00	Shell-1

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	SPHERICAL HEAD: 28.0mm. THK X 2812.0mm. ID X 0.0mm.	SA-387 11
2	INSULATION: 2000mm X 109mm THK	
8	GUSSET PLATES: 500.0mm X 300.0mm X 500.0mm X 40.0mm	SA-387 11
1	TOP RING: 2410.0mm X 3010.0mm X 66.0mm	SA-387 11
1	BOTTOM RING: 2410mm X 3410mm X 62mm	SA-387 11
4	LUG BOLTS: 48.0mm TEMA Metric	SA-193 B7
3	INSULATION: 3000mm X 109mm THK	
1	INSULATION: 1200mm X 109mm THK	
1	INSULATION: 1000mm X 109mm THK	
1	INSULATION: 1430mm X 109mm THK	
1	CLASS 600 GR 1.1, 24.0" BLIND FLANGE(S)	SA-182 F11
2	CYLINDRICAL SEGMENT 40.0mm. THK X 2957.5mm. X 2000.0mm.	SA-387 11
1	CYLINDRICAL SEGMENT 40.0mm. THK X 4508.6mm. X 2000.0mm.	SA-387 11
1	CYLINDRICAL SEGMENT 40.0mm. THK X 20.7mm. X 2000.0mm.	SA-387 11
1	CYLINDRICAL SEGMENT 40.0mm. THK X 2936.8mm. X 2000.0mm.	SA-387 11
1	CYLINDRICAL SEGMENT 40.0mm. THK X 1551.1mm. X 2000.0mm.	SA-387 11
2	CYLINDRICAL SEGMENT 40.0mm. THK X 2957.5mm. X 3000.0mm.	SA-387 11
1	CYLINDRICAL SEGMENT 40.0mm. THK X 1530.5mm. X 3000.0mm.	SA-387 11
2	CON. CONICAL SEGMENT 58.0mm. THK X 2503.9mm. X 1200.0mm. X 2940.	SA-387 11
1	CON. CONICAL SEGMENT 58.0mm. THK X 2380.7mm. X 1200.0mm. X 2795.	SA-387 11
1	CYLINDRICAL SEGMENT 46.0mm. THK X 2949.9mm. X 1000.0mm.	SA-387 11
1	CYLINDRICAL SEGMENT 46.0mm. THK X 5745.9mm. X 1000.0mm.	SA-387 11
1	CYLINDRICAL SEGMENT 46.0mm. THK X 0.0mm. X 1000.0mm.	SA-387 11
2	CYLINDRICAL SEGMENT 46.0mm. THK X 2949.9mm. X 3000.0mm.	SA-387 11
1	CYLINDRICAL SEGMENT 46.0mm. THK X 2796.0mm. X 3000.0mm.	SA-387 11
2	CYLINDRICAL SEGMENT 46.0mm. THK X 3454.2mm. X 3000.0mm.	SA-387 11
1	CYLINDRICAL SEGMENT 46.0mm. THK X 1787.5mm. X 3000.0mm.	SA-387 11
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	PWHT	Thickness ratio	Gov Thk mm.	E*
shell 6	[8]	A	26	14	Yes	0.777	40.000	1.00
shell 5	[8]	A	26	14	Yes	0.777	40.000	1.00
shell 4	[8]	A	26	14	Yes	0.777	40.000	1.00
cone transitio	[8]	A	49	39	No	0.760	48.000	1.00
Shell-3	[8]	A	48	39	No	0.778	46.000	1.00
Shell-2	[8]	A	48	39	No	0.777	46.000	1.00
Shell-1	[8]	A	48	39	No	0.777	46.000	1.00
Top Dish	[9]	A	34	25	No	0.775	24.000	1.00
A2-6"	[1]	A	48	36	No	0.720	46.000	1.00
Nozzle Flg	[5]	A	-18	-48	No			
L2-2"	[1]	A	-104	-104	No	0.061	16.764	1.00
Nozzle Flg	[5]	A	-18	-48	No			
L4-2"	[1]	A	-104	-104	No	0.061	16.764	1.00
Nozzle Flg	[5]	A	-18	-48	No			
L6-2"	[1]	A	-104	-104	No	0.061	16.764	1.00
Nozzle Flg	[5]	A	-18	-48	No			
L8-2"	[1]	A	-104	-104	No	0.061	16.764	1.00
Nozzle Flg	[5]	A	-18	-48	No			
L10-2"	[1]	A	-104	-104	No	0.061	16.764	1.00
Nozzle Flg	[5]	A	-18	-48	No			
L1-2"	[1]	A	-104	-104	No	0.061	16.764	1.00
Nozzle Flg	[5]	A	-18	-48	No			
L3-2"	[1]	A	-104	-104	No	0.061	16.764	1.00
Nozzle Flg	[5]	A	-18	-48	No			
L5-2"	[1]	A	-104	-104	No	0.061	16.764	1.00
Nozzle Flg	[5]	A	-18	-48	No			
L7-2"	[1]	A	-104	-104	No	0.061	16.764	1.00
Nozzle Flg	[5]	A	-18	-48	No			
L9-2"	[1]	A	-104	-104	No	0.061	16.764	1.00
Nozzle Flg	[5]	A	-18	-48	No			
M2-24"	[1]	A	48	36	No	0.720	46.000	1.00
Nozzle Flg	[5]	A	-18	-48	No			
B2-14"	[1]	A	34	22	No	0.719	24.000	1.00
P1-2"	[1]	A	-104	-104	No	0.061	16.764	1.00
P2-2"	[1]	A	-104	-104	No	0.061	16.764	1.00
V1-2"	[1]	A	-104	-104	No	0.061	16.764	1.00
S1-2"	[1]	A	-104	-104	No	0.072	20.574	1.00
Nozl. Bolting[22]			-48		No	1.000	64.000	1.00

Warmest MDMT: 49 39

Required Minimum Design Metal Temperature -10.0 °C
Warmest Computed Minimum Design Metal Temperature 39.0 °C

Warning: Computed overall MDMT was higher than the required value !

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.1.
- [5] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.3.
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per 3.11.2.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material

STEAM DRUM (WHB)
(A150NRJ10EQ001)

PV Elite 28 U2

 **v28**

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FileName : Steam drum (WHB) - division 2

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MDMT Summary:

Step: 49 6:47pm Sep 16,2026



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- [16] - Shell or Head Material
- [17] - Impact Testing required
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

Note: Lug supports should be modeled on an element of their own whose length is just slightly greater than the height of the lug assembly. The vessel does not have to be constructed with this extra element, just modeled this way. This is to insure there is proper mass distribution for these types of models.

Warning:

This is a vertical vessel on support lugs and the earthquake type selected is a building code. Building codes assume all of the mass is located above the point of support. This makes their use on lug supported vessels questionable. It is strongly suggested that based on the seismic zone and applicable parameters, an appropriate lateral (Gx) and (+-) vertical (Gy) acceleration be applied using the G loading option.

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.

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Units used in this Analysis (bars):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

STEAM DRUM (WHB)
(A150NRJ10EQ001)

Design Internal Pressure (for Hydrotest)	51 bars
Design Internal Temperature	286.0 °C
Projection of Nozzle from Vessel Top	0 mm.
Projection of Nozzle from Vessel Bottom	0 mm.
Minimum Design Metal Temperature	-10.0 °C
Type of Construction	Welded
Special Service	None
Use Higher Longitudinal Stresses (Flag)	N
Select t for Internal Pressure (Flag)	N
Select t for External Pressure (Flag)	N
Select t for Axial Stress (Flag)	N
Select Location for Stiff. Rings (Flag)	N
Consider Vortex Shedding	N

Shop Pressure Test:

Type of Pressure Test	Hydrostatic
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	Hydrostatic
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI

Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-22	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	[V] 180	Km/hr
Surface Roughness Category	D: Flat, unobstructed	
Type of Surface	Moderately Smooth	
Base Elevation	7500	mm.
Percent Wind for Hydrotest	33.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment	H or Hh	0 mm.
Distance Upwind of Crest	Lh	0 mm.
Distance from Crest to the Vessel	x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	mm.

Seismic Design Code	ASCE/SEI 7-22	
Importance Factor	1.500	
Force Modification Factor R	3.000	
Site Class	D	
Component Elevation Ratio	z/h	0.500
Amplification Factor	Ap	1.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Component resonance duct. fact.	[CAR]	1.000
Component strength factor	[Rpo]	1.500
Overstrength factor	[Omega0]	2.000
Response Acceleration Factor	[SMS]	0.326
Response Acceleration Factor	[SM1]	0.139
Building Importance Factor	[Ie]	1.000
Site Latitude		0.000 deg
Site Longitude		0.000 deg

M.A.W.P + Static Head to Nozzle	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Using Metric Material Databases, ASME II D	Yes
Using ASME VIII-1 Appendix 46	No
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/8
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Element From Node	20	
Element To Node	30	
Element Type	Cylinder	
Description	shell 6	
Distance "FROM" to "TO"	2000	mm.
Element Outside Diameter	2450	mm.
Element Thickness	40	mm.
Internal Corrosion Allowance	1.5	mm.
Nominal Thickness	40	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	51	bars
Design Temperature Internal Pressure	286	°C
Design External Pressure	1.0342	bars
Design Temperature External Pressure	286	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11	
Allowable Stress, Ambient	207	N./mm ²
Allowable Stress, Operating	165.12	N./mm ²
Allowable Stress, Hydrotest	294.49	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	0	mm.
Yield Stress, Operating	247.68	N./mm ²
MDMT Curve	A	
External Pressure Chart Name	CS-3	
UNS Number	K11789	
Class / Thickness / Grade	2 ::	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	20	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	2000	mm.
Thickness of Insulation	110	mm.
Density	0.0002	kg./cm ³

Element From Node	20	
Detail Type	Weight	
Detail ID	Boiler-fab weig	
Dist. from "FROM" Node / Offset dist	0	mm.
Miscellaneous Weight	55349	Kgf
Offset from Element Centerline	0	mm.

Element From Node	20	
Detail Type	Weight	
Detail ID	Boiler -Hydro	
Dist. from "FROM" Node / Offset dist	0	mm.
Miscellaneous Weight	51247	Kgf
Offset from Element Centerline	0	mm.

Element From Node	20	
Detail Type	Weight	
Detail ID	Boiler SF	
Dist. from "FROM" Node / Offset dist	0	mm.
Miscellaneous Weight	1900	Kgf
Offset from Element Centerline	0	mm.

Element#	2/8	
Element From Node	30	
Element To Node	40	
Element Type	Cylinder	
Description	shell 5	
Distance "FROM" to "TO"	2000	mm.



Element Outside Diameter	2450	mm.
Element Thickness	40	mm.
Internal Corrosion Allowance	1.5	mm.
Nominal Thickness	40	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	51	bars
Design Temperature Internal Pressure	286	°C
Design External Pressure	1.0342	bars
Design Temperature External Pressure	286	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	30	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	2000	mm.
Thickness of Insulation	110	mm.
Density	0.0002	kg./cm³

Element From Node	30	
Detail Type	Lug	
Detail ID	Support Lugs	
Dist. from "FROM" Node / Offset dist	1500	mm.
Number of Lugs	4	
Dist. from OD to Lug Cntrline(dlug)	300	mm.
Height of Gusset Plates (hgp)	500	mm.
Force Bearing Width (wfb)	300	mm.
Weight of Lug	973.38	Kgf
Lug Start Angle (degrees)	90.0	

Element#	3/8	
Element From Node	40	
Element To Node	50	
Element Type	Cylinder	
Description	shell 4	
Distance "FROM" to "TO"	3000	mm.
Element Outside Diameter	2450	mm.
Element Thickness	40	mm.
Internal Corrosion Allowance	1.5	mm.
Nominal Thickness	40	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	51	bars
Design Temperature Internal Pressure	286	°C
Design External Pressure	1.0342	bars
Design Temperature External Pressure	286	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	3000	mm.
Liquid Density	0.0009984	kg./cm³

Element From Node	40	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm.



Height/Length of Insulation	3000	mm.
Thickness of Insulation	110	mm.
Density	0.0002	kg./cm ³

Element#	4/8
Element From Node	50
Element To Node	60
Element Type	Conical
Description	cone transition
Distance "FROM" to "TO"	1200 mm.
Element Outside Diameter	2450 mm.
Element Thickness	48 mm.
Internal Corrosion Allowance	1.5 mm.
Nominal Thickness	58 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	51 bars
Design Temperature Internal Pressure	286 °C
Design External Pressure	1.0342 bars
Design Temperature External Pressure	286 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Cone Diameter at "To" End	2860 mm.
Design Length of Cone	1200 mm.
Half Apex Angle of Cone	12.0 degrees
Toriconical (Y/N)	N
Weld is pre-Heated	No

Element From Node	50
Detail Type	Liquid
Detail ID	Liquid: 50
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	1200 mm.
Liquid Density	0.0009984 kg./cm ³

Element From Node	50
Detail Type	Insulation
Detail ID	Ins: 20
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	1200 mm.
Thickness of Insulation	110 mm.
Density	0.0002 kg./cm ³

Element#	5/8
Element From Node	60
Element To Node	70
Element Type	Cylinder
Description	Shell-3
Distance "FROM" to "TO"	1000 mm.
Element Outside Diameter	2860 mm.
Element Thickness	46 mm.
Internal Corrosion Allowance	1.5 mm.
Nominal Thickness	46 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	51 bars
Design Temperature Internal Pressure	286 °C
Design External Pressure	1.0342 bars
Design Temperature External Pressure	286 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No



Element From Node	60
Detail Type	Liquid
Detail ID	Liquid: 60
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	1000 mm.
Liquid Density	0.0009984 kg./cm ³

Element From Node	60
Detail Type	Insulation
Detail ID	Ins: 20
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	1000 mm.
Thickness of Insulation	110 mm.
Density	0.0002 kg./cm ³

Element#	6/8
Element From Node	70
Element To Node	80
Element Type	Cylinder
Description	Shell-2
Distance "FROM" to "TO"	3000 mm.
Element Outside Diameter	2860 mm.
Element Thickness	46 mm.
Internal Corrosion Allowance	1.5 mm.
Nominal Thickness	46 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	51 bars
Design Temperature Internal Pressure	286 °C
Design External Pressure	1.0342 bars
Design Temperature External Pressure	286 °C
Effective Diameter Multiplier	1.2
Material Name	SA-387 11
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	70
Detail Type	Insulation
Detail ID	Ins: 20
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	3000 mm.
Thickness of Insulation	110 mm.
Density	0.0002 kg./cm ³

Element From Node	70
Detail Type	Nozzle
Detail ID	A2-6"
Dist. from "FROM" Node / Offset dist	1500 mm.
Nozzle Diameter	6 in.
Nozzle Schedule	160
Nozzle Class	600
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-182 F11

Element From Node	70
Detail Type	Nozzle
Detail ID	L2-2"
Dist. from "FROM" Node / Offset dist	1500 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	90.0



Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 0 Kgf
Grade of Attached Flange GR 1.1
Nozzle Matl SA-182 F11

Element From Node 70
Detail Type Nozzle
Detail ID L4-2"
Dist. from "FROM" Node / Offset dist 1500 mm.
Nozzle Diameter 2 in.
Nozzle Schedule None
Nozzle Class 600
Layout Angle 120.0
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 0 Kgf
Grade of Attached Flange GR 1.1
Nozzle Matl SA-182 F11

Element From Node 70
Detail Type Nozzle
Detail ID L6-2"
Dist. from "FROM" Node / Offset dist 1500 mm.
Nozzle Diameter 2 in.
Nozzle Schedule None
Nozzle Class 600
Layout Angle 150.0
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 0 Kgf
Grade of Attached Flange GR 1.1
Nozzle Matl SA-182 F11

Element From Node 70
Detail Type Nozzle
Detail ID L8-2"
Dist. from "FROM" Node / Offset dist 1500 mm.
Nozzle Diameter 2 in.
Nozzle Schedule None
Nozzle Class 600
Layout Angle 180.0
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 0 Kgf
Grade of Attached Flange GR 1.1
Nozzle Matl SA-182 F11

Element From Node 70
Detail Type Nozzle
Detail ID L10-2"
Dist. from "FROM" Node / Offset dist 1500 mm.
Nozzle Diameter 2 in.
Nozzle Schedule None
Nozzle Class 600
Layout Angle 210.0
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 0 Kgf
Grade of Attached Flange GR 1.1
Nozzle Matl SA-182 F11

Element# 7/8
Element From Node 80
Element To Node 90
Element Type Cylinder
Description Shell-1
Distance "FROM" to "TO" 3000 mm.
Element Outside Diameter 2860 mm.
Element Thickness 46 mm.
Internal Corrosion Allowance 1.5 mm.



Nominal Thickness	46	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	51	bars
Design Temperature Internal Pressure	286	°C
Design External Pressure	1.0342	bars
Design Temperature External Pressure	286	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	80	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	3000	mm.
Thickness of Insulation	110	mm.
Density	0.0002	kg./cm ³

Element From Node	80	
Detail Type	Nozzle	
Detail ID	L1-2"	
Dist. from "FROM" Node / Offset dist	1500	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-182 F11	

Element From Node	80	
Detail Type	Nozzle	
Detail ID	L3-2"	
Dist. from "FROM" Node / Offset dist	1500	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	120.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-182 F11	

Element From Node	80	
Detail Type	Nozzle	
Detail ID	L5-2"	
Dist. from "FROM" Node / Offset dist	1500	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	150.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-182 F11	

Element From Node	80	
Detail Type	Nozzle	
Detail ID	L7-2"	
Dist. from "FROM" Node / Offset dist	1500	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	180.0	
Blind Flange (Y/N)	N	



Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-182 F11	

Element From Node	80	
Detail Type	Nozzle	
Detail ID	L9-2"	
Dist. from "FROM" Node / Offset dist	1500	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	210.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-182 F11	

Element From Node	80	
Detail Type	Nozzle	
Detail ID	M2-24"	
Dist. from "FROM" Node / Offset dist	1500	mm.
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	270.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-336 F12	

Element From Node	80	
Detail Type	Weight	
Detail ID	Demister	
Dist. from "FROM" Node / Offset dist	0	mm.
Miscellaneous Weight	670	Kgf
Offset from Element Centerline	0	mm.

Element From Node	80	
Detail Type	Weight	
Detail ID	demister suppor	
Dist. from "FROM" Node / Offset dist	0	mm.
Miscellaneous Weight	70	Kgf
Offset from Element Centerline	0	mm.

Element From Node	80	
Detail Type	Weight	
Detail ID	10% misc	
Dist. from "FROM" Node / Offset dist	0	mm.
Miscellaneous Weight	10000	Kgf
Offset from Element Centerline	0	mm.

Element From Node	80	
Detail Type	Weight	
Detail ID	Clip weight	
Dist. from "FROM" Node / Offset dist	0	mm.
Miscellaneous Weight	1550	Kgf
Offset from Element Centerline	0	mm.

Element#	8/8	
Element From Node	90	
Element To Node	100	
Element Type	Spherical	
Description	Top Dish	
Distance "FROM" to "TO"	0	mm.
Element Outside Diameter	2860	mm.
Element Thickness	24	mm.



Internal Corrosion Allowance	1.5	mm.
Nominal Thickness	28	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	51	bars
Design Temperature Internal Pressure	286	°C
Design External Pressure	1.0342	bars
Design Temperature External Pressure	286	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-387 11	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	90	
Detail Type	Insulation	
Detail ID	Ins: 20	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	1430	mm.
Thickness of Insulation	110	mm.
Density	0.0002	kg./cm ³
Element From Node	90	
Detail Type	Nozzle	
Detail ID	B2-14"	
Dist. from "FROM" Node / Offset dist	0	mm.
Nozzle Diameter	14	in.
Nozzle Schedule	120	
Nozzle Class	None	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-182 F11	
Element From Node	90	
Detail Type	Nozzle	
Detail ID	P1-2"	
Dist. from "FROM" Node / Offset dist	600	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-182 F11	
Element From Node	90	
Detail Type	Nozzle	
Detail ID	P2-2"	
Dist. from "FROM" Node / Offset dist	600	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-182 F11	
Element From Node	90	
Detail Type	Nozzle	
Detail ID	V1-2"	
Dist. from "FROM" Node / Offset dist	600	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	180.0	

STEAM DRUM (WHB)
(A150NRJ10EQ001)

Elite® v28

PV Elite 28 U2 Licensee: SL User
FileName : Steam drum (WHB) - division 2
Input Echo: Step: 1 6:47pm Sep 16,2026

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HEXAGON

Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kgf
Grade of Attached Flange	None
Nozzle Matl	SA-182 F11
Element From Node	90
Detail Type	Nozzle
Detail ID	S1-2"
Dist. from "FROM" Node / Offset dist	600 mm.
Nozzle Diameter	3 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kgf
Grade of Attached Flange	None
Nozzle Matl	SA-182 F11
Element From Node	90
Detail Type	Weight
Detail ID	Trunioin
Dist. from "FROM" Node / Offset dist	0 mm.
Miscellaneous Weight	3000 Kgf
Offset from Element Centerline	0 mm.
Element From Node	90
Detail Type	Weight
Detail ID	SF weight
Dist. from "FROM" Node / Offset dist	0 mm.
Miscellaneous Weight	3000 Kgf
Offset from Element Centerline	0 mm.

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XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
shell 6		...	2000	...	2000
shell 5		...	4000	...	2000
shell 4		...	7000	...	3000
cone transition		...	8200	...	1200
Shell-3		...	9200	...	1000
Shell-2		...	12200	...	3000
Shell-1		...	15200	...	3000
Top Dish		...	15200	...	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
shell 6		51.51	40	1.5	2370	165.12	1
shell 5		51.51	40	1.5	2370	165.12	2
shell 4		51.51	40	1.5	2370	165.12	3
cone transition		51.216	58	1.5	2761.9	165.12	4
Shell-3		51.098	46	1.5	2768	165.12	5
Shell-2		51	46	1.5	2768	165.12	6
Shell-1		51	46	1.5	2768	165.12	7
Top Dish		51	28	1.5	2812	165.12	8

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
shell 6		51	52.2148	68.7148	40	39.0991	1
shell 5		51	52.2148	68.7148	40	39.0991	2
shell 4		51	52.2148	68.7148	40	39.0991	3
cone transition		51	53.3508	69.2041	48	45.9229	4
Shell-3		51	52.0985	67.6768	46	45.0488	5
Shell-2		51	52.1965	67.6768	46	44.964	6
Shell-1		51	52.1965	67.6768	46	44.964	7
Top Dish		51	52.3704	70.0668	24	23.4066	8

Minimum

47

68

MAWP: 47 bars, limited by: Nozzle Reinforcement.

Warning: The computed MAWP is less than the design pressure.

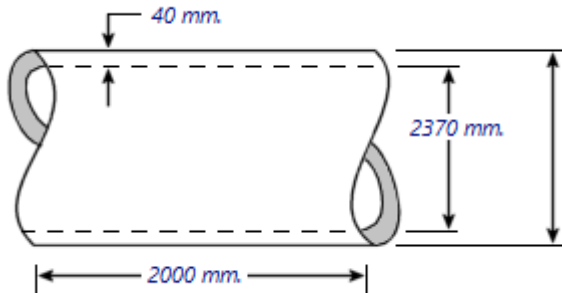
Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: shell 6

Cylindrical Shell Design Information:					
Design Pressure	P	51.510 bars	Int. Design Temperature	286 °C	
Material		SA-387 11	Ext. Design Temperature	286 °C	
Length		2000.000 mm.	External Pressure Chart	CS-3	
			UNS Number	K11789	
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	165.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	207.0 N./mm ²
Inside Diameter	Di	2370.000 mm.	Actual Stress	158.8 N./mm ²	
Spec. Min. Thk t,e, or sa		40.0000 mm.	Spec. Nominal Thk.	40.000 mm.	
Surface Area		153938.0 cm ²	SG of Contents	0.000	
Element Volume		8824587.0 cm ³	Weight of Contents	0.0 Kg	
Empty Weight		4694.5 Kg	Operating Weight	4694.5 Kg	
			Comp. All. Ope. Stress	85.2 N./mm ²	

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 2373.0 \cdot (\exp(51.51/(165.119 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 37.5991 + 1.5 + 0.0 = 39.0991 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 165.119 \cdot 1.0 \cdot \ln((2 \cdot 38.5 + 2373.0) / 2373.0) - 0.51$$

$$= 52.725 - 0.51 = 52.2148 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 206.996 \cdot 1.0 \cdot \ln((2 \cdot 40.0 + 2370.0) / 2370.0)$$

$$= 68.7148 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 51.51 \cdot 2373.0 / (1.0 (2450.0 - 2373.0))$$

$$= 158.7529 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 51.51 / (1.0 \cdot \ln((2 \cdot 38.5 + 2373.0) / 2373.0))$$

$$= 161.3145 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o) \quad 1.660 \%$$

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: 14 °C
User entered Minimum Design Temperature	: -10 °C
The Curve for this Material is	: A
From Figure 3.8M- with PWHT, the MDMT is	: 26 °C
Temperature reduction per Graph 3.13	: 12 °C
The computed stress ratio is	: 0.777

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (29.898 \cdot 1.0) / (40.0 - 1.5 - 0.0)$$

$$= 0.777$$

Warning: Impact Testing Is Mandatory!

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	51.510 bars
Governing Thk.	40.000 mm.	Min. Des. Mtl. Temp.	-10 °C
Stress/Thk. Ratio	0.777	Basic MDMT	26 °C
		Temperature Difference	12 °C
		Computed Min. Temp.	14 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: shell 6

Cylindrical Shell From 20 to 30 Ext. Chart: CS-2 at 286 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{R_o \cdot t}$$

$$= 7000.0 / \sqrt{1225.0 \cdot 38.5}$$

$$= 32.233$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 13 \text{ and } M_x < 2 \cdot (D_o/t)^{0.94}, \text{ Ch} = 1.12 \cdot M_x^{-1.058}$$

$$= 1.12 \cdot 32.233^{-1.058}$$

$$= 0.0284$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y(t / D_o)$$

$$= 1.6 \cdot 0.028 \cdot 0.18730E+09(38.5/2450.0)$$

$$= 133.787 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 133.8/247.7$$

$$= 0.540$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 247.7/517.1$$

$$= 0.479$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 133.8 / 1.8730E+09$$

$$= 0.000714$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.479) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.160$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 247.7 (1 + 0.002) / \ln(1 + 0.002)^{0.160318}$$

$$= 672.232 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.479)$$

$$= 0.313$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (517.1 \cdot \exp(0.313)) / (0.313^{0.313})$$

$$= 1016.803 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4791.5 - 0.5 \cdot 0.4792.5 - 0.4793.5$$

$$= 0.342$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	258	-1.764	0.311E-05	0.262E-05	0.775E-05	-0.580E-05	0.122E+08	0.205E-01
2	129	-4.571	0.825E-07	0.825E-07	0.169E-05	-0.169E-05	0.143E+09	0.190E-03
3	64	-5.975	0.219E-08	0.219E-08	0.368E-06	-0.368E-06	0.186E+09	0.366E-03
4	96	-5.273	0.183E-07	0.183E-07	0.897E-06	-0.897E-06	0.175E+09	0.161E-03
5	113	-4.922	0.410E-07	0.410E-07	0.126E-05	-0.126E-05	0.162E+09	0.172E-04
6	121	-4.747	0.588E-07	0.588E-07	0.146E-05	-0.146E-05	0.153E+09	0.761E-04
7	117	-4.834	0.493E-07	0.493E-07	0.136E-05	-0.136E-05	0.158E+09	0.272E-04
8	115	-4.878	0.450E-07	0.450E-07	0.131E-05	-0.131E-05	0.160E+09	0.445E-05
9	114	-4.900	0.429E-07	0.429E-07	0.128E-05	-0.128E-05	0.161E+09	0.651E-05
10	114	-4.889	0.440E-07	0.439E-07	0.130E-05	-0.130E-05	0.161E+09	0.106E-05
11	114	-4.884	0.445E-07	0.445E-07	0.130E-05	-0.130E-05	0.160E+09	0.169E-05
12	114	-4.887	0.442E-07	0.442E-07	0.130E-05	-0.130E-05	0.161E+09	0.313E-06
13	114	-4.888	0.441E-07	0.441E-07	0.130E-05	-0.130E-05	0.161E+09	0.373E-06
14	114	-4.887	0.441E-07	0.441E-07	0.130E-05	-0.130E-05	0.161E+09	0.300E-07
15	114	-4.887	0.442E-07	0.442E-07	0.130E-05	-0.130E-05	0.161E+09	0.142E-06
16	114	-4.887	0.442E-07	0.441E-07	0.130E-05	-0.130E-05	0.161E+09	0.558E-07
17	114	-4.887	0.442E-07	0.441E-07	0.130E-05	-0.130E-05	0.161E+09	0.129E-07

Factor of Safety [FS]:

$$\text{Since } F_{ic} [114.73] < 0.55 \cdot 247.68, FS = 2$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic} / FS$$

$$= 114.735 / 2.0$$

$$= 57.367 \text{ N./mm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha} (t / D_o)$$

$$= 2 \cdot 573.64 (38.5 / 2450.0)$$

$$= 18.029 \text{ bars}$$

STEAM DRUM (WHB)
(A150NRJ10EQ001)

PV Elite® v28

PV Elite 28 U2 Licensee: SL User

FileName : Steam drum (WHB) - division 2

Element: shell 6

Step: 4 6:47pm Sep 16, 2026

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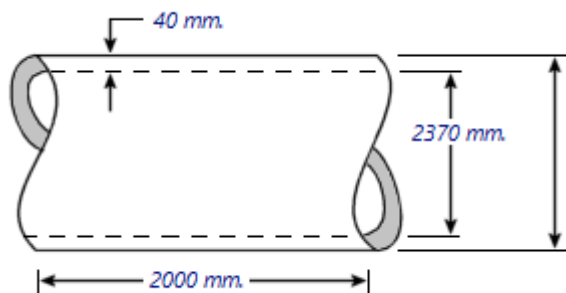
HEXAGON

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: shell 5

Cylindrical Shell Design Information:					
Design Pressure	P	51.510 bars	Int. Design Temperature		286 °C
Material		SA-387 11	Ext. Design Temperature		286 °C
Length		2000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K11789
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	165.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	207.0 N./mm ²
Inside Diameter	Di	2370.000 mm.	Actual Stress		158.8 N./mm ²
Spec. Min. Thk t,e, or sa		40.0000 mm.	Spec. Nominal Thk.		40.000 mm.
Surface Area		153938.0 cm ²	SG of Contents		0.000
Element Volume		8824587.0 cm ³	Weight of Contents		0.0 Kg
Empty Weight		4694.5 Kg	Operating Weight		4694.5 Kg
			Comp. All. Ope. Stress		85.2 N./mm ²
Radiography Information:					
...			...		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 2373.0 \cdot (\exp(51.51/(165.119 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 37.5991 + 1.5 + 0.0 = 39.0991 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 165.119 \cdot 1.0 \cdot \ln((2 \cdot 38.5 + 2373.0) / 2373.0) - 0.51$$

$$= 52.725 - 0.51 = 52.2148 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 206.996 \cdot 1.0 \cdot \ln((2 \cdot 40.0 + 2370.0) / 2370.0)$$

$$= 68.7148 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 51.51 \cdot 2373.0 / (1.0 (2450.0 - 2373.0))$$

$$= 158.7529 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 51.51 / (1.0 \cdot \ln((2 \cdot 38.5 + 2373.0) / 2373.0))$$

$$= 161.3145 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o) \quad 1.660 \%$$

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: 14 °C
User entered Minimum Design Temperature	: -10 °C
The Curve for this Material is	: A
From Figure 3.8M- with PWHT, the MDMT is	: 26 °C
Temperature reduction per Graph 3.13	: 12 °C
The computed stress ratio is	: 0.777

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (29.898 \cdot 1.0) / (40.0 - 1.5 - 0.0)$$

$$= 0.777$$

Warning: Impact Testing Is Mandatory!

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	51.510 bars
Governing Thk.	40.000 mm.	Min. Des. Mtl. Temp.	-10 °C
Stress/Thk. Ratio	0.777	Basic MDMT	26 °C
		Temperature Difference	12 °C
		Computed Min. Temp.	14 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: shell 5

Cylindrical Shell From 30 to 40 Ext. Chart: CS-2 at 286 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{R_o \cdot t}$$

$$= 7000.0 / \sqrt{1225.0 \cdot 38.5}$$

$$= 32.233$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 13 \text{ and } M_x < 2 \cdot (D_o/t)^{0.94}, \text{ Ch} = 1.12 \cdot M_x^{-1.058}$$

$$= 1.12 \cdot 32.233^{-1.058}$$

$$= 0.0284$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y(t / D_o)$$

$$= 1.6 \cdot 0.028 \cdot 0.18730E+09(38.5/2450.0)$$

$$= 133.787 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / F_y$$

$$= 133.8/247.7$$

$$= 0.540$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 247.7/517.1$$

$$= 0.479$$



Ratio [AE]:

$$= F_{he} / E$$

$$= 133.8 / 1.8730E+09$$

$$= 0.000714$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.479) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.160$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 247.7 (1 + 0.002) / \ln(1 + 0.002)^{0.160318}$$

$$= 672.232 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.479)$$

$$= 0.313$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (517.1 \cdot \exp(0.313)) / (0.313^{0.313})$$

$$= 1016.803 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4791.5 - 0.5 \cdot 0.4792.5 - 0.4793.5$$

$$= 0.342$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	258	-1.764	0.311E-05	0.262E-05	0.775E-05	-0.580E-05	0.122E+08	0.205E-01
2	129	-4.571	0.825E-07	0.825E-07	0.169E-05	-0.169E-05	0.143E+09	0.190E-03
3	64	-5.975	0.219E-08	0.219E-08	0.368E-06	-0.368E-06	0.186E+09	0.366E-03
4	96	-5.273	0.183E-07	0.183E-07	0.897E-06	-0.897E-06	0.175E+09	0.161E-03
5	113	-4.922	0.410E-07	0.410E-07	0.126E-05	-0.126E-05	0.162E+09	0.172E-04
6	121	-4.747	0.588E-07	0.588E-07	0.146E-05	-0.146E-05	0.153E+09	0.761E-04
7	117	-4.834	0.493E-07	0.493E-07	0.136E-05	-0.136E-05	0.158E+09	0.272E-04
8	115	-4.878	0.450E-07	0.450E-07	0.131E-05	-0.131E-05	0.160E+09	0.445E-05
9	114	-4.900	0.429E-07	0.429E-07	0.128E-05	-0.128E-05	0.161E+09	0.651E-05
10	114	-4.889	0.440E-07	0.439E-07	0.130E-05	-0.130E-05	0.161E+09	0.106E-05
11	114	-4.884	0.445E-07	0.445E-07	0.130E-05	-0.130E-05	0.160E+09	0.169E-05
12	114	-4.887	0.442E-07	0.442E-07	0.130E-05	-0.130E-05	0.161E+09	0.313E-06
13	114	-4.888	0.441E-07	0.441E-07	0.130E-05	-0.130E-05	0.161E+09	0.373E-06
14	114	-4.887	0.441E-07	0.441E-07	0.130E-05	-0.130E-05	0.161E+09	0.300E-07
15	114	-4.887	0.442E-07	0.442E-07	0.130E-05	-0.130E-05	0.161E+09	0.142E-06
16	114	-4.887	0.442E-07	0.441E-07	0.130E-05	-0.130E-05	0.161E+09	0.558E-07
17	114	-4.887	0.442E-07	0.441E-07	0.130E-05	-0.130E-05	0.161E+09	0.129E-07

Factor of Safety [FS]:

$$\text{Since } F_{ic} [114.73] < 0.55 \cdot 247.68, FS = 2$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic} / FS$$

$$= 114.735 / 2.0$$

$$= 57.367 \text{ N./mm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha} (t / D_o)$$

$$= 2 \cdot 573.64 (38.5 / 2450.0)$$

$$= 18.029 \text{ bars}$$

STEAM DRUM (WHB)
(A150NRJ10EQ001)

PV Elite® v28

PV Elite 28 U2 Licensee: SL User

FileName : Steam drum (WHB) - division 2

Element: shell 5

Step: 5 6:47pm Sep 16, 2026

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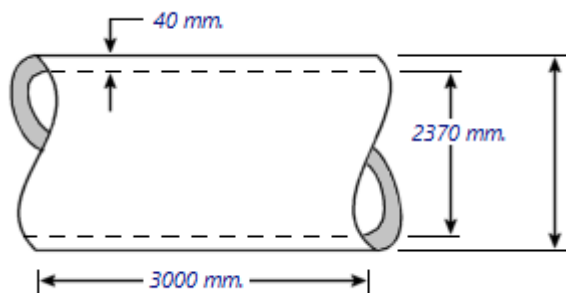
HEXAGON

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: shell 4

Cylindrical Shell Design Information:					
Design Pressure	P	51.510 bars	Int. Design Temperature		286 °C
Material		SA-387 11	Ext. Design Temperature		286 °C
Length		3000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K11789
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	165.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	207.0 N./mm ²
Inside Diameter	Di	2370.000 mm.	Actual Stress		158.8 N./mm ²
Spec. Min. Thk t,e, or sa		40.0000 mm.	Spec. Nominal Thk.		40.000 mm.
Surface Area		230907.1 cm ²	SG of Contents		1.000
Element Volume		13236881.0 cm ³	Weight of Contents		13228.8 Kg
Empty Weight		7041.7 Kg	Operating Weight		20270.5 Kg
			Comp. All. Ope. Stress		85.2 N./mm ²
Radiography Information:					
...			...		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 2373.0 \cdot (\exp(51.51/(165.119 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 37.5991 + 1.5 + 0.0 = 39.0991 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 165.119 \cdot 1.0 \cdot \ln((2 \cdot 38.5 + 2373.0) / 2373.0) - 0.51$$

$$= 52.725 - 0.51 = 52.2148 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 206.996 \cdot 1.0 \cdot \ln((2 \cdot 40.0 + 2370.0) / 2370.0)$$

$$= 68.7148 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 51.51 \cdot 2373.0 / (1.0 (2450.0 - 2373.0))$$

$$= 158.7529 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 51.51 / (1.0 \cdot \ln((2 \cdot 38.5 + 2373.0) / 2373.0))$$

$$= 161.3145 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o) \quad 1.660 \%$$

Minimum Design Metal Temperature Results:

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	: 14 °C
User entered Minimum Design Temperature	: -10 °C
The Curve for this Material is	: A
From Figure 3.8M- with PWHT, the MDMT is	: 26 °C
Temperature reduction per Graph 3.13	: 12 °C
The computed stress ratio is	: 0.777

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (29.898 \cdot 1.0) / (40.0 - 1.5 - 0.0)$$

$$= 0.777$$

Warning: Impact Testing Is Mandatory!

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	51.510 bars
Governing Thk.	40.000 mm.	Min. Des. Mtl. Temp.	-10 °C
Stress/Thk. Ratio	0.777	Basic MDMT	26 °C
		Temperature Difference	12 °C
		Computed Min. Temp.	14 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: shell 4

Cylindrical Shell From 40 to 50 Ext. Chart: CS-2 at 286 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{R_o \cdot t}$$

$$= 7000.0 / \sqrt{1225.0 \cdot 38.5}$$

$$= 32.233$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 13 \text{ and } M_x < 2 \cdot (D_o/t)^{0.94}, \text{ Ch} = 1.12 \cdot M_x^{-1.058}$$

$$= 1.12 \cdot 32.233^{-1.058}$$

$$= 0.0284$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y(t / D_o)$$

$$= 1.6 \cdot 0.028 \cdot 0.18730E+09(38.5/2450.0)$$

$$= 133.787 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / F_y$$

$$= 133.8/247.7$$

$$= 0.540$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 247.7/517.1$$

$$= 0.479$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 133.8 / 1.8730E+09$$

$$= 0.000714$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.479) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.160$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 247.7 (1 + 0.002) / \ln(1 + 0.002)^{0.160318}$$

$$= 672.232 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.479)$$

$$= 0.313$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (517.1 \cdot \exp(0.313)) / (0.313^{0.313})$$

$$= 1016.803 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4791.5 - 0.5 \cdot 0.4792.5 - 0.4793.5$$

$$= 0.342$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	258	-1.764	0.311E-05	0.262E-05	0.775E-05	-0.580E-05	0.122E+08	0.205E-01
2	129	-4.571	0.825E-07	0.825E-07	0.169E-05	-0.169E-05	0.143E+09	0.190E-03
3	64	-5.975	0.219E-08	0.219E-08	0.368E-06	-0.368E-06	0.186E+09	0.366E-03
4	96	-5.273	0.183E-07	0.183E-07	0.897E-06	-0.897E-06	0.175E+09	0.161E-03
5	113	-4.922	0.410E-07	0.410E-07	0.126E-05	-0.126E-05	0.162E+09	0.172E-04
6	121	-4.747	0.588E-07	0.588E-07	0.146E-05	-0.146E-05	0.153E+09	0.761E-04
7	117	-4.834	0.493E-07	0.493E-07	0.136E-05	-0.136E-05	0.158E+09	0.272E-04
8	115	-4.878	0.450E-07	0.450E-07	0.131E-05	-0.131E-05	0.160E+09	0.445E-05
9	114	-4.900	0.429E-07	0.429E-07	0.128E-05	-0.128E-05	0.161E+09	0.651E-05
10	114	-4.889	0.440E-07	0.439E-07	0.130E-05	-0.130E-05	0.161E+09	0.106E-05
11	114	-4.884	0.445E-07	0.445E-07	0.130E-05	-0.130E-05	0.160E+09	0.169E-05
12	114	-4.887	0.442E-07	0.442E-07	0.130E-05	-0.130E-05	0.161E+09	0.313E-06
13	114	-4.888	0.441E-07	0.441E-07	0.130E-05	-0.130E-05	0.161E+09	0.373E-06
14	114	-4.887	0.441E-07	0.441E-07	0.130E-05	-0.130E-05	0.161E+09	0.300E-07
15	114	-4.887	0.442E-07	0.442E-07	0.130E-05	-0.130E-05	0.161E+09	0.142E-06
16	114	-4.887	0.442E-07	0.441E-07	0.130E-05	-0.130E-05	0.161E+09	0.558E-07
17	114	-4.887	0.442E-07	0.441E-07	0.130E-05	-0.130E-05	0.161E+09	0.129E-07

Factor of Safety [FS]:

$$\text{Since } F_{ic} [114.73] < 0.55 \cdot 247.68, FS = 2$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic} / FS$$

$$= 114.735 / 2.0$$

$$= 57.367 \text{ N./mm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha} (t / D_o)$$

$$= 2 \cdot 573.64 (38.5 / 2450.0)$$

$$= 18.029 \text{ bars}$$

STEAM DRUM (WHB)
(A150NRJ10EQ001)

PV Elite® v28

PV Elite 28 U2 Licensee: SL User

FileName : Steam drum (WHB) - division 2

Element: shell 4

Step: 6 6:47pm Sep 16, 2026

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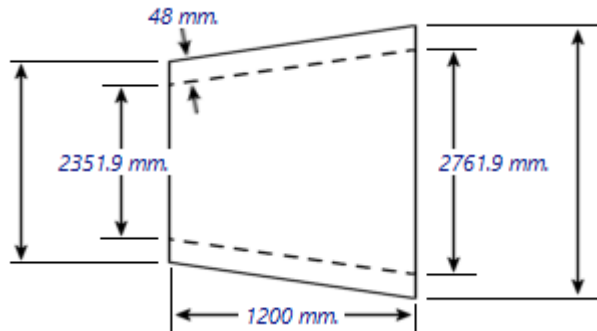


HEXAGON

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: cone transition

Conical Head/Section Design Information:					
Design Pressure	P	51.098 bars	Int. Design Temperature		286 °C
Material		SA-387 11	Ext. Design Temperature		286 °C
Length		1200.000 mm.	External Pressure Chart		CS-3
			UNS Number		K11789
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	165.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	207.0 N./mm ²
Inside Diameter	Di	2761.855 mm.	Actual Stress		155.3 N./mm ²
Spec. Min. Thk t,e, or sa		48.0000 mm.	Spec. Nominal Thk.		58.000 mm.
Surface Area		102323.2 cm ²	SG of Contents		1.000
Element Volume		6175766.0 cm ³	Weight of Contents		6172.0 Kg
Empty Weight		4532.6 Kg	Operating Weight		10704.6 Kg
			Comp. All. Ope. Stress		88.1 N./mm ²
			To End Diameter		2351.855 mm.

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Thickness due to Internal Pressure: Equation 4.3.4.1 [tr]:

$$= 0.5 \cdot D_c / \cos(\alpha) \cdot (\exp(P / (S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 2764.855 / \cos(12.0) \cdot (\exp(51.098 / (165.119 \cdot 1.0)) - 1) + c_i + c_o$$

$$= 44.4229 + 1.5 + 0.0 = 45.9229 \text{ mm.}$$

Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln(2 \cdot t_{cor} \cdot \cos(\alpha) + D_{cor}) / D_{cor}$$

$$= 165.119 \cdot 1.0 \cdot \ln(2 \cdot 46.5 \cdot \cos(12.0) + 2764.855) / 2764.855$$

$$= 53.4488 \text{ bars}$$

Maximum Allowable Pressure New and Cold [MAPnc]:

$$= S_a \cdot E \cdot \ln(2 \cdot t \cdot \cos(\alpha) + D) / D$$

$$= 206.996 \cdot 1.0 \cdot \ln(2 \cdot 48.0 \cdot \cos(12.0) + 2761.855) / 2761.855$$

$$= 69.2041 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D) \cos(\alpha))$$

$$= 51.098 \cdot 2764.855 / (1.0 (2857.855 - 2764.855) \cos(12.0))$$

$$= 155.315 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t_{cor} \cdot \cos(\alpha) + D_{cor}) / D_{cor}))$$

$$= 51.098 / (1.0 \cdot \ln((2 \cdot 46.5 \cdot \cos(12.0) + 2764.855) / 2764.855))$$

$$= 157.857 \text{ N./mm}^2$$



Percent Fiber Elongation per Table 6.1

Cones formed from Plate $(50 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 2.405 %

Note: The Pressure at the Large Diameter is used in the TR calculation.

Minimum Design Metal Temperature Results:

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature : 39 °C
User entered Minimum Design Temperature : -10 °C
The Curve for this Material is : A
From Figure 3.7M- Non PWHT, the MDMT is : 49 °C
Temperature reduction per Graph 3.12 : 10 °C
The computed stress ratio is : 0.760

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (35.325 \cdot 1.0) / (48.0 - 1.5 - 0.0)$$

$$= 0.760$$

Warning: Impact Testing Is Mandatory!

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	51.098 bars
Governing Thk.	48.000 mm.	Min. Des. Mtl. Temp.	-10 °C
Stress/Thk. Ratio	0.760	Basic MDMT	49 °C
		Temperature Difference	10 °C
		Computed Min. Temp.	39 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: cone transition

Cone From 50 to 60 Ext. Chart: CS-2 at 286 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{Ro \cdot t}$$

$$= 1226.809 / \sqrt{1461.947 \cdot 46.5}$$

$$= 4.705$$

Compute the Factor [Ch]:

$$\text{Since } Mx > 1.5 \text{ and } Mx < 13, Ch = 0.92 / (Mx - 0.579)$$

$$= 0.92 / (4.705 - 0.579)$$

$$= 0.2230$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y(t / D_o)$$

$$= 1.6 \cdot 0.223 \cdot 0.18730E+09(46.5/2923.894)$$

$$= 1062.686 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$

$$= 1062.7/247.7$$

$$= 4.291$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 247.7/517.1$$

$$= 0.479$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 1062.7 / 1.8730E+09$$

$$= 0.005673$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.479) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.160$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m1$$

$$= 247.7 (1 + 0.002) / \ln(1 + 0.002) 0.160318$$

$$= 672.232 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.479)$$

$$= 0.313$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2 m2)$$

$$= (517.1 \cdot \exp(0.313)) / (0.313 0.313)$$

$$= 1016.803 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4791.5 - 0.5 \cdot 0.4792.5 - 0.4793.5$$

$$= 0.342$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	258	-1.764	0.311E-05	0.262E-05	0.775E-05	-0.580E-05	0.122E+08	0.156E-01
2	129	-4.571	0.825E-07	0.825E-07	0.169E-05	-0.169E-05	0.143E+09	0.477E-02
3	193	-3.167	0.690E-06	0.684E-06	0.412E-05	-0.406E-05	0.510E+08	0.187E-02
4	226	-2.466	0.155E-05	0.149E-05	0.578E-05	-0.544E-05	0.256E+08	0.316E-02
5	210	-2.816	0.105E-05	0.103E-05	0.491E-05	-0.477E-05	0.364E+08	0.998E-04
6	202	-2.992	0.854E-06	0.844E-06	0.450E-05	-0.442E-05	0.432E+08	0.994E-03
7	206	-2.904	0.948E-06	0.934E-06	0.470E-05	-0.460E-05	0.396E+08	0.477E-03
8	208	-2.860	0.997E-06	0.981E-06	0.481E-05	-0.469E-05	0.380E+08	0.197E-03
9	209	-2.838	0.102E-05	0.101E-05	0.486E-05	-0.473E-05	0.372E+08	0.504E-04
10	209	-2.827	0.104E-05	0.102E-05	0.488E-05	-0.475E-05	0.368E+08	0.242E-04
11	209	-2.833	0.103E-05	0.101E-05	0.487E-05	-0.474E-05	0.370E+08	0.132E-04
12	209	-2.830	0.103E-05	0.102E-05	0.488E-05	-0.475E-05	0.369E+08	0.547E-05
13	209	-2.832	0.103E-05	0.101E-05	0.487E-05	-0.474E-05	0.369E+08	0.387E-05
14	209	-2.831	0.103E-05	0.101E-05	0.487E-05	-0.475E-05	0.369E+08	0.799E-06
15	209	-2.831	0.103E-05	0.101E-05	0.487E-05	-0.474E-05	0.369E+08	0.154E-05
16	209	-2.831	0.103E-05	0.101E-05	0.487E-05	-0.474E-05	0.369E+08	0.370E-06
17	209	-2.831	0.103E-05	0.101E-05	0.487E-05	-0.475E-05	0.369E+08	0.214E-06

Factor of Safety [FS]:

$$\text{Since } F_{ic} > 0.55 F_y \text{ and } F_{ic} < F_y$$

$$= 2.407 - 0.741 \cdot F_{ic} / F_y$$

$$= 2.407 - 0.741 \cdot 209.42 / 247.68$$

$$= 1.780$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic} / FS$$

$$= 209.415 / 1.78$$

$$= 117.618 \text{ N./mm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha} (t / D_o)$$

$$= 2 \cdot 1176.108 (46.5 / 2923.894)$$

STEAM DRUM (WHB)
(A150NRJ10EQ001)

PV Elite[®] v28

PV Elite 28 U2 Licensee: SL User

FileName : Steam drum (WHB) - division 2

Element: cone transition

Step: 7 6:47pm Sep 16, 2026

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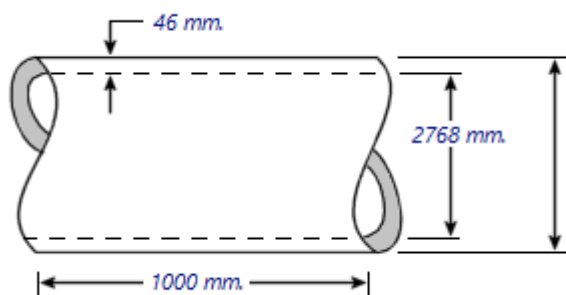
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= 37.408 bars

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Shell-3

Cylindrical Shell Design Information:					
Design Pressure	P	51.098 bars	Int. Design Temperature		286 °C
Material		SA-387 11	Ext. Design Temperature		286 °C
Length		1000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K11789
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	165.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	207.0 N./mm ²
Inside Diameter	Di	2768.000 mm.	Actual Stress		159.1 N./mm ²
Spec. Min. Thk t,e, or sa		46.0000 mm.	Spec. Nominal Thk.		46.000 mm.
Surface Area		89849.6 cm ²	SG of Contents		1.000
Element Volume		6018661.5 cm ³	Weight of Contents		6015.0 Kgf
Empty Weight		3151.8 Kgf	Operating Weight		9166.8 Kgf
			Comp. All. Ope. Stress		85.0 N./mm ²
Radiography Information:					
...			...		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 2771.0 \cdot (\exp(51.098/(165.119 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 43.5488 + 1.5 + 0.0 = 45.0488 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 165.119 \cdot 1.0 \cdot \ln((2 \cdot 44.5 + 2771.0) / 2771.0) - 0.098$$

$$= 52.197 - 0.098 = 52.0985 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 206.996 \cdot 1.0 \cdot \ln((2 \cdot 46.0 + 2768.0) / 2768.0)$$

$$= 67.6768 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 51.098 \cdot 2771.0 / (1.0 (2860.0 - 2771.0))$$

$$= 159.1021 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 51.098 / (1.0 \cdot \ln((2 \cdot 44.5 + 2771.0) / 2771.0))$$

$$= 161.6439 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o) \quad 1.635 \%$$

Minimum Design Metal Temperature Results:

This element is Not Post Weld Heat Treated
 Computed Minimum Design Metal Temperature : 39 °C
 User entered Minimum Design Temperature : -10 °C
 The Curve for this Material is : A
 From Figure 3.7M- Non PWHT, the MDMT is : 48 °C
 Temperature reduction per Graph 3.12 : 9 °C
 The computed stress ratio is : 0.778

The computed thickness ratio [sr]:
 $= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$
 $= (34.629 \cdot 1.0) / (46.0 - 1.5 - 0.0)$
 $= 0.778$

Warning: Impact Testing Is Mandatory!

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	51.098 bars
Governing Thk.	46.000 mm.	Min. Des. Mtl. Temp.	-10 °C
Stress/Thk. Ratio	0.778	Basic MDMT	48 °C
		Temperature Difference	9 °C
		Computed Min. Temp.	39 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-3

Cylindrical Shell From 60 to 70 Ext. Chart: CS-2 at 286 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:
 $= L / \sqrt{R_o \cdot t}$
 $= 7468.667 / \sqrt{1430.0 \cdot 44.5}$
 $= 29.607$

Compute the Factor [Ch]:
 Since $M_x > 13$ and $M_x < 2 \cdot (D_o/t)^{0.94}$, $Ch = 1.12 \cdot M_x^{-1.058}$
 $= 1.12 \cdot 29.607^{-1.058}$
 $= 0.0311$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:
 $= 1.6 \cdot Ch \cdot E_y(t / D_o)$
 $= 1.6 \cdot 0.031 \cdot 0.18730E+09(44.5/2860.0)$
 $= 144.930 \text{ N./mm}^2$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:
 $= Fhe / F_y$
 $= 144.9/247.7$
 $= 0.585$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:
 $= \sigma_{ys} / \sigma_{uts}$
 $= 247.7/517.1$
 $= 0.479$

Ratio [AE]:

$$= F_{he} / E$$

$$= 144.9 / 18730E+09$$

$$= 0.000774$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.479) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.160$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 247.7 (1 + 0.002) / \ln(1 + 0.002)^{0.160318}$$

$$= 672.232 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.479)$$

$$= 0.313$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (517.1 \cdot \exp(0.313)) / (0.313^{0.313})$$

$$= 1016.803 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4791.5 - 0.5 \cdot 0.4792.5 - 0.4793.5$$

$$= 0.342$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	258	-1.764	0.311E-05	0.262E-05	0.775E-05	-0.580E-05	0.122E+08	0.205E-01
2	129	-4.571	0.825E-07	0.825E-07	0.169E-05	-0.169E-05	0.143E+09	0.131E-03
3	64	-5.975	0.219E-08	0.219E-08	0.368E-06	-0.368E-06	0.186E+09	0.426E-03
4	96	-5.273	0.183E-07	0.183E-07	0.897E-06	-0.897E-06	0.175E+09	0.221E-03
5	113	-4.922	0.410E-07	0.410E-07	0.126E-05	-0.126E-05	0.162E+09	0.767E-04
6	121	-4.747	0.588E-07	0.588E-07	0.146E-05	-0.146E-05	0.153E+09	0.166E-04
7	117	-4.834	0.493E-07	0.493E-07	0.136E-05	-0.136E-05	0.158E+09	0.323E-04
8	119	-4.791	0.539E-07	0.539E-07	0.141E-05	-0.141E-05	0.156E+09	0.846E-05
9	120	-4.769	0.563E-07	0.563E-07	0.144E-05	-0.144E-05	0.155E+09	0.392E-05
10	119	-4.780	0.551E-07	0.551E-07	0.142E-05	-0.142E-05	0.155E+09	0.230E-05
11	119	-4.774	0.557E-07	0.557E-07	0.143E-05	-0.143E-05	0.155E+09	0.800E-06
12	119	-4.777	0.554E-07	0.554E-07	0.143E-05	-0.143E-05	0.155E+09	0.755E-06
13	119	-4.776	0.555E-07	0.555E-07	0.143E-05	-0.143E-05	0.155E+09	0.222E-07
14	119	-4.776	0.555E-07	0.555E-07	0.143E-05	-0.143E-05	0.155E+09	0.366E-06
15	119	-4.776	0.555E-07	0.555E-07	0.143E-05	-0.143E-05	0.155E+09	0.172E-06
16	119	-4.776	0.555E-07	0.555E-07	0.143E-05	-0.143E-05	0.155E+09	0.750E-07
17	119	-4.776	0.555E-07	0.555E-07	0.143E-05	-0.143E-05	0.155E+09	0.264E-07

Factor of Safety [FS]:

$$\text{Since } F_{ic} [119.87] < 0.55 \cdot 247.68, FS = 2$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic} / FS$$

$$= 119.872 / 2.0$$

$$= 59.936 \text{ N./mm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha} (t / D_o)$$

$$= 2 \cdot 599.323 (44.5 / 2860.0)$$

$$= 18.650 \text{ bars}$$

STEAM DRUM (WHB)
(A150NRJ10EQ001)

PV Elite® v28

PV Elite 28 U2 Licensee: SL User

FileName : Steam drum (WHB) - division 2

Element: Shell-3

Step: 8 6:47pm Sep 16, 2026

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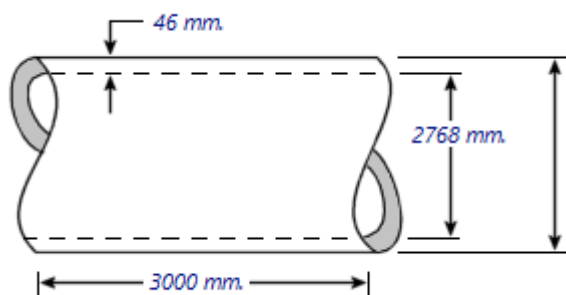


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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Shell-2

Cylindrical Shell Design Information:					
Design Pressure	P	51.000 bars	Int. Design Temperature		286 °C
Material		SA-387 11	Ext. Design Temperature		286 °C
Length		3000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K11789
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	165.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	207.0 N./mm ²
Inside Diameter	Di	2768.000 mm.	Actual Stress		158.8 N./mm ²
Spec. Min. Thk t,e, or sa		46.0000 mm.	Spec. Nominal Thk.		46.000 mm.
Surface Area		269548.7 cm ²	SG of Contents		0.000
Element Volume		18055984.0 cm ³	Weight of Contents		0.0 Kgf
Empty Weight		9455.5 Kgf	Operating Weight		9455.5 Kgf
			Comp. All. Ope. Stress		85.0 N./mm ²
Radiography Information:					
...			...		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 2771.0 \cdot (\exp(51.0/(165.119 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 43.464 + 1.5 + 0.0 = 44.964 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 165.119 \cdot 1.0 \cdot \ln((2 \cdot 44.5 + 2771.0) / 2771.0) - 0.0$$

$$= 52.197 - 0.0 = 52.1965 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 206.996 \cdot 1.0 \cdot \ln((2 \cdot 46.0 + 2768.0) / 2768.0)$$

$$= 67.6768 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 51.0 \cdot 2771.0 / (1.0 (2860.0 - 2771.0))$$

$$= 158.7969 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 51.0 / (1.0 \cdot \ln((2 \cdot 44.5 + 2771.0) / 2771.0))$$

$$= 161.3338 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o) \quad 1.635 \%$$

Minimum Design Metal Temperature Results:

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	:	39 °C
User entered Minimum Design Temperature	:	-10 °C
The Curve for this Material is	:	A
From Figure 3.7M- Non PWHT, the MDMT is	:	48 °C
Temperature reduction per Graph 3.12	:	9 °C
The computed stress ratio is	:	0.777

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{\text{nom}} - c_i - c_{\text{ext}})$$

$$= (34.562 \cdot 1.0) / (46.0 - 1.5 - 0.0)$$

$$= 0.777$$

Warning: Impact Testing Is Mandatory!

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	51.000 bars
Governing Thk.	46.000 mm.	Min. Des. Mtl. Temp.	-10 °C
Stress/Thk. Ratio	0.777	Basic MDMT	48 °C
		Temperature Difference	9 °C
		Computed Min. Temp.	39 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-2

Cylindrical Shell From 70 to 80 Ext. Chart: CS-2 at 286 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{R_o \cdot t}$$

$$= 7468.667 / \sqrt{1430.0 \cdot 44.5}$$

$$= 29.607$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 13 \text{ and } M_x < 2 \cdot (D_o/t)^{0.94}, \text{ Ch} = 1.12 \cdot M_x^{-1.058}$$

$$= 1.12 \cdot 29.607^{-1.058}$$

$$= 0.0311$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot \text{Ch} \cdot E_y(t / D_o)$$

$$= 1.6 \cdot 0.031 \cdot 0.18730 \text{E}+09(44.5/2860.0)$$

$$= 144.930 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / F_y$$

$$= 144.9/247.7$$

$$= 0.585$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 247.7/517.1$$

$$= 0.479$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 144.9 / 18730E+09$$

$$= 0.000774$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.479) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.160$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 247.7 (1 + 0.002) / \ln(1 + 0.002)^{0.160318}$$

$$= 672.232 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.479)$$

$$= 0.313$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (517.1 \cdot \exp(0.313)) / (0.313^{0.313})$$

$$= 1016.803 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4791.5 - 0.5 \cdot 0.4792.5 - 0.4793.5$$

$$= 0.342$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	258	-1.764	0.311E-05	0.262E-05	0.775E-05	-0.580E-05	0.122E+08	0.205E-01
2	129	-4.571	0.825E-07	0.825E-07	0.169E-05	-0.169E-05	0.143E+09	0.131E-03
3	64	-5.975	0.219E-08	0.219E-08	0.368E-06	-0.368E-06	0.186E+09	0.426E-03
4	96	-5.273	0.183E-07	0.183E-07	0.897E-06	-0.897E-06	0.175E+09	0.221E-03
5	113	-4.922	0.410E-07	0.410E-07	0.126E-05	-0.126E-05	0.162E+09	0.767E-04
6	121	-4.747	0.588E-07	0.588E-07	0.146E-05	-0.146E-05	0.153E+09	0.166E-04
7	117	-4.834	0.493E-07	0.493E-07	0.136E-05	-0.136E-05	0.158E+09	0.323E-04
8	119	-4.791	0.539E-07	0.539E-07	0.141E-05	-0.141E-05	0.156E+09	0.846E-05
9	120	-4.769	0.563E-07	0.563E-07	0.144E-05	-0.144E-05	0.155E+09	0.392E-05
10	119	-4.780	0.551E-07	0.551E-07	0.142E-05	-0.142E-05	0.155E+09	0.230E-05
11	119	-4.774	0.557E-07	0.557E-07	0.143E-05	-0.143E-05	0.155E+09	0.800E-06
12	119	-4.777	0.554E-07	0.554E-07	0.143E-05	-0.143E-05	0.155E+09	0.755E-06
13	119	-4.776	0.555E-07	0.555E-07	0.143E-05	-0.143E-05	0.155E+09	0.222E-07
14	119	-4.776	0.555E-07	0.555E-07	0.143E-05	-0.143E-05	0.155E+09	0.366E-06
15	119	-4.776	0.555E-07	0.555E-07	0.143E-05	-0.143E-05	0.155E+09	0.172E-06
16	119	-4.776	0.555E-07	0.555E-07	0.143E-05	-0.143E-05	0.155E+09	0.750E-07
17	119	-4.776	0.555E-07	0.555E-07	0.143E-05	-0.143E-05	0.155E+09	0.264E-07

Factor of Safety [FS]:

$$\text{Since } F_{ic} [119.87] < 0.55 \cdot 247.68, FS = 2$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic} / FS$$

$$= 119.872 / 2.0$$

$$= 59.936 \text{ N./mm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha} (t / D_o)$$

$$= 2 \cdot 599.323 (44.5 / 2860.0)$$

$$= 18.650 \text{ bars}$$

STEAM DRUM (WHB)
(A150NRJ10EQ001)

PV Elite® v28

PV Elite 28 U2 Licensee: SL User

FileName : Steam drum (WHB) - division 2

Element: Shell-2

Step: 9 6:47pm Sep 16, 2026

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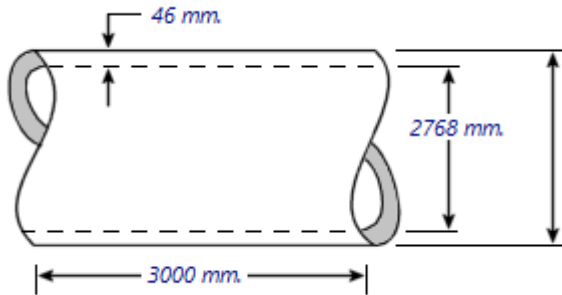


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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	51.000 bars	Int. Design Temperature		286 °C
Material		SA-387 11	Ext. Design Temperature		286 °C
Length		3000.000 mm.	External Pressure Chart		CS-3
			UNS Number		K11789
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	165.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	207.0 N./mm ²
Inside Diameter	Di	2768.000 mm.	Actual Stress		158.8 N./mm ²
Spec. Min. Thk t,e, or sa		46.0000 mm.	Spec. Nominal Thk.		46.000 mm.
Surface Area		269548.7 cm ²	SG of Contents		0.000
Element Volume		18055984.0 cm ³	Weight of Contents		0.0 Kgf
Empty Weight		9455.5 Kgf	Operating Weight		9455.5 Kgf
			Comp. All. Ope. Stress		85.0 N./mm ²
Radiography Information:					
...			...		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 2771.0 \cdot (\exp(51.0/(165.119 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 43.464 + 1.5 + 0.0 = 44.964 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 165.119 \cdot 1.0 \cdot \ln((2 \cdot 44.5 + 2771.0) / 2771.0) - 0.0$$

$$= 52.197 - 0.0 = 52.1965 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 206.996 \cdot 1.0 \cdot \ln((2 \cdot 46.0 + 2768.0) / 2768.0)$$

$$= 67.6768 \text{ bars}$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 51.0 \cdot 2771.0 / (1.0 (2860.0 - 2771.0))$$

$$= 158.7969 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 51.0 / (1.0 \cdot \ln((2 \cdot 44.5 + 2771.0) / 2771.0))$$

$$= 161.3338 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o) \quad 1.635 \%$$

Minimum Design Metal Temperature Results:

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	: 39 °C
User entered Minimum Design Temperature	: -10 °C
The Curve for this Material is	: A
From Figure 3.7M- Non PWHT, the MDMT is	: 48 °C
Temperature reduction per Graph 3.12	: 9 °C
The computed stress ratio is	: 0.777

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{\text{nom}} - c_i - c_{\text{ext}})$$

$$= (34.562 \cdot 1.0) / (46.0 - 1.5 - 0.0)$$

$$= 0.777$$

Warning: Impact Testing Is Mandatory!

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	51.000 bars
Governing Thk.	46.000 mm.	Min. Des. Mtl. Temp.	-10 °C
Stress/Thk. Ratio	0.777	Basic MDMT	48 °C
		Temperature Difference	9 °C
		Computed Min. Temp.	39 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-1

Cylindrical Shell From 80 to 90 Ext. Chart: CS-2 at 286 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{R_o \cdot t}$$

$$= 7468.667 / \sqrt{1430.0 \cdot 44.5}$$

$$= 29.607$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 13 \text{ and } M_x < 2 \cdot (D_o/t)^{0.94}, \text{ Ch} = 1.12 \cdot M_x^{-1.058}$$

$$= 1.12 \cdot 29.607^{-1.058}$$

$$= 0.0311$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot \text{Ch} \cdot E_y(t / D_o)$$

$$= 1.6 \cdot 0.031 \cdot 0.18730 \text{E}+09(44.5/2860.0)$$

$$= 144.930 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / F_y$$

$$= 144.9/247.7$$

$$= 0.585$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 247.7/517.1$$

$$= 0.479$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 144.9 / 18730E+09$$

$$= 0.000774$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.479) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.160$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$

$$= 247.7 (1 + 0.002) / \ln(1 + 0.002)^{0.160318}$$

$$= 672.232 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$

$$= 0.6(1.0 - 0.479)$$

$$= 0.313$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$

$$= (517.1 \cdot \exp(0.313)) / (0.313^{0.313})$$

$$= 1016.803 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.4791.5 - 0.5 \cdot 0.4792.5 - 0.4793.5$$

$$= 0.342$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	258	-1.764	0.311E-05	0.262E-05	0.775E-05	-0.580E-05	0.122E+08	0.205E-01
2	129	-4.571	0.825E-07	0.825E-07	0.169E-05	-0.169E-05	0.143E+09	0.131E-03
3	64	-5.975	0.219E-08	0.219E-08	0.368E-06	-0.368E-06	0.186E+09	0.426E-03
4	96	-5.273	0.183E-07	0.183E-07	0.897E-06	-0.897E-06	0.175E+09	0.221E-03
5	113	-4.922	0.410E-07	0.410E-07	0.126E-05	-0.126E-05	0.162E+09	0.767E-04
6	121	-4.747	0.588E-07	0.588E-07	0.146E-05	-0.146E-05	0.153E+09	0.166E-04
7	117	-4.834	0.493E-07	0.493E-07	0.136E-05	-0.136E-05	0.158E+09	0.323E-04
8	119	-4.791	0.539E-07	0.539E-07	0.141E-05	-0.141E-05	0.156E+09	0.846E-05
9	120	-4.769	0.563E-07	0.563E-07	0.144E-05	-0.144E-05	0.155E+09	0.392E-05
10	119	-4.780	0.551E-07	0.551E-07	0.142E-05	-0.142E-05	0.155E+09	0.230E-05
11	119	-4.774	0.557E-07	0.557E-07	0.143E-05	-0.143E-05	0.155E+09	0.800E-06
12	119	-4.777	0.554E-07	0.554E-07	0.143E-05	-0.143E-05	0.155E+09	0.755E-06
13	119	-4.776	0.555E-07	0.555E-07	0.143E-05	-0.143E-05	0.155E+09	0.222E-07
14	119	-4.776	0.555E-07	0.555E-07	0.143E-05	-0.143E-05	0.155E+09	0.366E-06
15	119	-4.776	0.555E-07	0.555E-07	0.143E-05	-0.143E-05	0.155E+09	0.172E-06
16	119	-4.776	0.555E-07	0.555E-07	0.143E-05	-0.143E-05	0.155E+09	0.750E-07
17	119	-4.776	0.555E-07	0.555E-07	0.143E-05	-0.143E-05	0.155E+09	0.264E-07

Factor of Safety [FS]:

$$\text{Since } F_{ic} [119.87] < 0.55 \cdot 247.68, FS = 2$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic} / FS$$

$$= 119.872 / 2.0$$

$$= 59.936 \text{ N./mm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha} (t / D_o)$$

$$= 2 \cdot 599.323 (44.5 / 2860.0)$$

$$= 18.650 \text{ bars}$$

STEAM DRUM (WHB)
(A150NRJ10EQ001)

PV Elite® v28

PV Elite 28 U2 Licensee: SL User

FileName : Steam drum (WHB) - division 2

Element: Shell-1

Step: 10 6:47pm Sep 16, 2026

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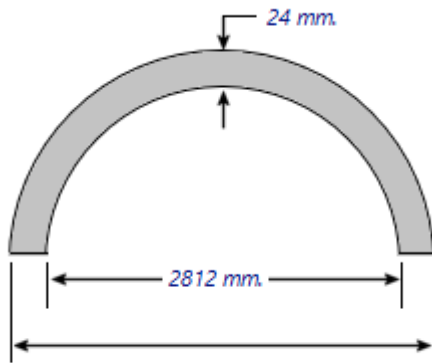
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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: Top Dish

Spherical Head Design Information:					
Design Pressure	P	51.000 bars	Int. Design Temperature		286 °C
Material		SA-387 11	Ext. Design Temperature		286 °C
Length		0.000 mm.	External Pressure Chart		CS-3
			UNS Number		K11789
Int. Corr. All.	c	1.5000 mm.	Allowable Stress (ope)	S	165.1 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	207.0 N./mm ²
Inside Diameter	Di	2812.000 mm.	Actual Stress		158.3 N./mm ²
Spec. Min. Thk t,e, or sa		24.0000 mm.	Spec. Nominal Thk.		28.000 mm.
Surface Area		129204.7 cm ²	SG of Contents		0.000
Element Volume		5822269.0 cm ³	Weight of Contents		0.0 Kgf
Empty Weight		2749.5 Kgf	Operating Weight		2749.5 Kgf
			Comp. All. Ope. Stress		75.4 N./mm ²

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 35.150 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 17.575 mm.

Spherical Shell Calculation - Section 4.3.5.1

Computed Minimum Required Thickness [t]:

$$= D/2 \left(\exp(0.5P/(S \cdot E)) - 1 \right) + ci + co$$

$$= 2815.0/2 \left(\exp(0.5 \cdot 51.0/(165.119 \cdot 1.0)) - 1 \right) + 1.5 + 0.0$$

$$= 21.9066 + 1.5 + 0.0 = 23.4066 \text{ mm.}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D)/D) - P_{liq}$$

$$= 2 \cdot 165.1 \cdot 1.0 \cdot \ln((2 \cdot 22.5 + 2815.0)/2815.0) - 0.0$$

$$= 52.37 - 0.0 = 52.3704 \text{ bars}$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= 2 \cdot S \cdot E \cdot \ln((2 \cdot t + D)/D)$$

$$= 2 \cdot 206.996 \cdot 1.0 \cdot \ln((2 \cdot 24.0 + 2812.0)/2812.0)$$

$$= 70.0668 \text{ bars}$$



Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D^2 / (E (D_o^2 - D^2))$$

$$= 51.0 \cdot 2815.0^2 / (1.0 (2860.0^2 - 2815.0^2))$$

$$= 158.261 \text{ N./mm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= 0.5 \cdot P / (E \cdot \ln((2 \cdot t + D) / D))$$

$$= 0.5 \cdot 51.0 / (1.0 \cdot \ln((2 \cdot 22.5 + 2815.0) / 2815.0))$$

$$= 160.798 \text{ N./mm}^2$$

Percent Fiber Elongation per Table 6.1

For Heads formed from Segments $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.479 %
 For one Piece doubly Curved Heads $100 \cdot \ln(D_b / (D_f - 2t))$ 36.629 %

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:

$$= \text{square root}(4 \cdot \text{Head Surface Area} / 3.14159)$$

$$= \text{square root}(4 \cdot 129204.7 / 3.14159)$$

$$= 4055.965 \text{ mm.}$$

MDMT Calculations in the Spherical Portion:

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature : 25 °C
 User entered Minimum Design Temperature : -10 °C
 The Curve for this Material is : A
 From Figure 3.7M- Non PWHT, the MDMT is : 34 °C
 Temperature reduction per Graph 3.12 : 9 °C
 The computed stress ratio is : 0.775

The computed thickness ratio [sr]:

$$= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext})$$

$$= (17.447 \cdot 1.0) / (24.0 - 1.5 - 0.0)$$

$$= 0.775$$

Warning: Impact Testing Is Mandatory!

MDMT Design Information:			
MDMT Curve	A	Press. at MDMT	51.000 bars
Governing Thk.	24.000 mm.	Min. Des. Mtl. Temp.	-10 °C
Stress/Thk. Ratio	0.775	Basic MDMT	34 °C
		Temperature Difference	9 °C
		Computed Min. Temp.	25 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 35.150 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 17.575 mm.

EXTERNAL PRESSURE RESULTS for: Top Dish

Spherical Head From 90 to 100 Ext. Chart: CS-2 at 286 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 0.075 \cdot E_y (t / R_o)$$

$$= 0.075 \cdot 0.18730 \text{E}+09 (22.5 / 1430.0)$$

$$= 221.040 \text{ N./mm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / S_y$$

$$= 221.0 / 247.7$$

$$= 0.892$$

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$

$$= 247.7 / 517.1$$

$$= 0.479$$

Ratio [AE]:

$$= F_{he} / E$$

$$= 221.0 / 1.8730E+09$$

$$= 0.001180$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / (\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.479) + (0.00002 - 0.002)) / (\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.160$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m_1$$

$$= 247.7(1 + 0.002) / \ln(1 + 0.002) 0.160318$$

$$= 672.232 \text{ N./mm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m_2 f(m_2 f_2 - R)$$

$$= 0.6(1.0 - 0.479)$$

$$= 0.313$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2)$$

$$= (517.1 \cdot \exp(0.313)) / (0.313 0.313)$$

$$= 1016.803 \text{ N./mm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R_{1.5} - 0.5 \cdot R_{2.5} - R_{3.5}$$

$$= 1.5 \cdot 0.4791.5 - 0.5 \cdot 0.4792.5 - 0.4793.5$$

$$= 0.342$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	258	-1.764	0.311E-05	0.262E-05	0.775E-05	-0.580E-05	0.122E+08	0.201E-01
2	129	-4.571	0.825E-07	0.825E-07	0.169E-05	-0.169E-05	0.143E+09	0.275E-03
3	193	-3.167	0.690E-06	0.684E-06	0.412E-05	-0.406E-05	0.510E+08	0.262E-02
4	161	-3.869	0.265E-06	0.265E-06	0.276E-05	-0.275E-05	0.933E+08	0.552E-03
5	145	-4.220	0.153E-06	0.153E-06	0.219E-05	-0.218E-05	0.119E+09	0.450E-04
6	137	-4.396	0.113E-06	0.113E-06	0.193E-05	-0.193E-05	0.131E+09	0.133E-03
7	141	-4.308	0.132E-06	0.132E-06	0.206E-05	-0.205E-05	0.125E+09	0.492E-04
8	143	-4.264	0.142E-06	0.142E-06	0.212E-05	-0.212E-05	0.122E+09	0.345E-05
9	144	-4.242	0.147E-06	0.147E-06	0.215E-05	-0.215E-05	0.120E+09	0.204E-04
10	143	-4.253	0.145E-06	0.145E-06	0.214E-05	-0.213E-05	0.121E+09	0.840E-05
11	143	-4.259	0.143E-06	0.143E-06	0.213E-05	-0.213E-05	0.121E+09	0.245E-05
12	143	-4.261	0.143E-06	0.143E-06	0.212E-05	-0.212E-05	0.122E+09	0.506E-06
13	143	-4.260	0.143E-06	0.143E-06	0.213E-05	-0.212E-05	0.122E+09	0.971E-06
14	143	-4.261	0.143E-06	0.143E-06	0.213E-05	-0.212E-05	0.122E+09	0.232E-06
15	143	-4.261	0.143E-06	0.143E-06	0.213E-05	-0.212E-05	0.122E+09	0.137E-06
16	143	-4.261	0.143E-06	0.143E-06	0.213E-05	-0.212E-05	0.122E+09	0.476E-07
17	143	-4.261	0.143E-06	0.143E-06	0.213E-05	-0.212E-05	0.122E+09	0.447E-07

Factor of Safety [FS]:

Since $F_{ic} > 0.55 F_y$ and $F_{ic} < F_y$

$$= 2.407 - 0.741 \cdot F_{ic} / F_y$$

$$= 2.407 - 0.741 \cdot 143.57 / 247.68$$

$$= 1.977$$

STEAM DRUM (WHB)
(A150NRJ10EQ001)

PV Elite 28 U2 Licensee: SL User

FileName : Steam drum (WHB) - division 2

Element: Top Dish

 PV Elite[®] v28



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Allowable Circumferential Compressive Stress [Fha]:

= Fc/FS

= 143.567/1.977

= 72.601 N./mm²

Allowable External Pressure at the given Thickness [Pa]:

= 2 • Fha[t/Ro]

= 2 • 72.601[22.5/1430.0]

= 22.845 bars

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Hydrostatic Test Pressure Results:

Test Pressure at high point in vessel per Equation 8.1
 $= \gamma_{St/s} \cdot MAWP(St/S)$
 $= 1.25 \cdot 47.3321(1.0)$
 $= 59.165 \text{ bars}$

Horizontal Test performed per: $\gamma_{St/s} \cdot MAWP(St/S)$

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
shell 6	178.9	294.5	0.608	59.40
shell 5	178.9	294.5	0.608	59.40
shell 4	178.9	294.5	0.608	59.40
cone transition	177.8	294.5	0.604	59.44
Shell-3	181.8	294.5	0.617	59.44
Shell-2	181.8	294.5	0.617	59.44
Shell-1	181.8	294.5	0.617	59.44
Top Dish	175.6	294.5	0.596	59.44

Theoretical Maximum Allowable test pressure is : 96.013 bars

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
A2-6"	Nozzle	138.00	138.00	1.000
L2-2"	Nozzle	138.00	138.00	1.000
L4-2"	Nozzle	138.00	138.00	1.000
L6-2"	Nozzle	138.00	138.00	1.000
L8-2"	Nozzle	138.00	138.00	1.000
L10-2"	Nozzle	138.00	138.00	1.000
L1-2"	Nozzle	138.00	138.00	1.000
L3-2"	Nozzle	138.00	138.00	1.000
L5-2"	Nozzle	138.00	138.00	1.000
L7-2"	Nozzle	138.00	138.00	1.000
L9-2"	Nozzle	138.00	138.00	1.000
M2-24"	Nozzle	138.00	132.00	1.045
B2-14"	Nozzle	138.00	138.00	1.000
P1-2"	Nozzle	138.00	138.00	1.000
P2-2"	Nozzle	138.00	138.00	1.000
V1-2"	Nozzle	138.00	138.00	1.000
S1-2"	Nozzle	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
shell 6	207.00	165.12	1.254
shell 5	207.00	165.12	1.254
shell 4	207.00	165.12	1.254
cone transition	207.00	165.12	1.254
Shell-3	207.00	165.12	1.254
Shell-2	207.00	165.12	1.254
Shell-1	207.00	165.12	1.254
Top Dish	207.00	165.12	1.254

Minimum

1.254

STEAM DRUM (WHB)
(A150NRJ10EQ001)



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FileName : Steam drum (WHB) - division 2
Shop Pressure Test Results:

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Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
A2-6"	28.26	247.50	0.114
L2-2"	11.73	247.50	0.047
L4-2"	11.73	247.50	0.047
L6-2"	11.73	247.50	0.047
L8-2"	11.73	247.50	0.047
L10-2"	11.73	247.50	0.047
L1-2"	11.73	247.50	0.047
L3-2"	11.73	247.50	0.047
L5-2"	11.73	247.50	0.047
L7-2"	11.73	247.50	0.047
L9-2"	11.73	247.50	0.047
M2-24"	87.58	247.50	0.354
B2-14"	40.42	247.50	0.163
P1-2"	11.73	247.50	0.047
P2-2"	11.73	247.50	0.047
V1-2"	11.73	247.50	0.047
S1-2"	13.77	247.50	0.056

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 2, 2025

External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
20	30	7000	2450	38.5	No Calc	No Calc
30	40	7000	2450	38.5	No Calc	No Calc
40	50	7000	2450	38.5	No Calc	No Calc
50	60	1200	2860	46.5	No Calc	No Calc
60	70	7468.67	2860	44.5	No Calc	No Calc
70	80	7468.67	2860	44.5	No Calc	No Calc
80	90	7468.67	2860	44.5	No Calc	No Calc
90	100	No Calc	2860	22.5	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
20	30	40	13.201	1.03419	18.0287
30	40	40	13.201	1.03419	18.0287
40	50	40	13.201	1.03419	18.0287
50	60	48	7.7694	1.03419	37.4083
60	70	46	14.6553	1.03419	18.6503
70	80	46	14.6553	1.03419	18.6503
80	90	46	14.6553	1.03419	18.6503
90	100	24	5.38107	1.03419	22.8452

Minimum

18

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
20	30	7000	25399974	No Calc	No Calc
30	40	7000	25399974	No Calc	No Calc
40	50	7000	25399974	No Calc	No Calc
50	60	1200	1200	No Calc	No Calc
60	70	7468.67	25399974	No Calc	No Calc
70	80	7468.67	25399974	No Calc	No Calc
80	90	7468.67	25399974	No Calc	No Calc
90	100	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
20	30	4694.48	8824587	4521.24	8846943	...
30	40	4694.48	8824587	4521.24	8846943	...
40	50	7041.71	13236881	6781.86	13270414	...
50	60	4532.61	6175766	4417.97	6190559	...
60	70	3151.82	6018661	3050.67	6031714	...
70	80	9455.46	18055984	9152.02	18095142	...
80	90	9455.46	18055984	9152.02	18095142	...
90	100	2749.54	5822269	2604.98	5840924	...
Total		45775	85014720.00	44202	85217768.00	0

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset, Dtl. Cent. mm.	Description
20	Ins1	354.283	...	1000	...	Ins: 20
20	Wght	55349	...	...	...	Boiler-fab weig
20	Wght	51247	...	...	...	Boiler -Hydro
20	Wght	1900	...	...	...	Boiler SF
30	Ins1	354.283	...	1000	...	Ins: 20
30	Lugs	3893.53	...	1750	...	Support Lugs
40	Liqd	13228.8	...	1500	...	Liquid: 40
40	Ins1	531.425	...	1500	...	Ins: 20
50	Liqd	6172	...	600	...	Liquid: 50
50	Ins1	283.443	...	600	...	Ins: 20
60	Liqd	6014.99	...	500	...	Liquid: 60
60	Ins1	205.512	...	500	...	Ins: 20
70	Ins1	616.536	...	1500	...	Ins: 20
70	Nozl	74.9367	-1484	1500	...	A2-6"
70	Nozl	11.7006	1482.42	1500	...	L2-2"
70	Nozl	11.7006	1283.82	1500	741.212	L4-2"
70	Nozl	11.7006	741.212	1500	1283.82	L6-2"
70	Nozl	11.7006	...	1500	1482.42	L8-2"
70	Nozl	11.7006	-741.212	1500	1283.82	L10-2"
80	Ins1	616.536	...	1500	...	Ins: 20
80	Nozl	11.7006	1482.42	1500	...	L1-2"
80	Nozl	11.7006	1283.82	1500	741.212	L3-2"
80	Nozl	11.7006	741.212	1500	1283.82	L5-2"
80	Nozl	11.7006	...	1500	1482.42	L7-2"
80	Nozl	11.7006	-741.212	1500	1283.82	L9-2"
80	Nozl	1793.16	-1482.42	1500	...	M2-24"
80	Wght	670	...	...	...	Demister
80	Wght	70	...	...	...	demister suppor
80	Wght	10000	...	...	...	10% misc
80	Wght	1550	...	...	...	Clip weight
90	Ins1	305.325	...	715	...	Ins: 20
90	Nozl	157.031	...	1544	...	B2-14"
90	Nozl	16.6962	600	1459.55	...	P1-2"
90	Nozl	16.6962	-600	1459.55	...	P2-2"
90	Nozl	16.6962	...	1459.55	600	V1-2"
90	Nozl	19.6728	...	1354.8	-600	S1-2"
90	Wght	3000	...	...	...	Trunioin
90	Wght	3000	...	...	...	SF weight

Total Weight of Each Detail Type:

Liquid 25415.8
Insulation 3267.3

STEAM DRUM (WHB)
(A150NRJ10EQ001)

Elite v28



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FileName : Steam drum (WHB) - division 2

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Element and Detail Weights: Step: 14 6:47pm Sep 16,2026

Nozzles 2211.9
Lugs 3893.5
Weights 126786.0

Sum of the Detail Weights 161574.6 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	45775.5	45775.5	45775.5	45775.5	45775.5	45775.5
Nozzles	2211.9	2211.9	2211.9	2211.9	2211.9	2211.9
Lugs	3893.5	3893.5	3893.5	3893.5	3893.5	3893.5
Wld Weights	74869.0	74869.0	74869.0	74869.0	74869.0	74869.0
Insulation	...	...	3267.3	3267.3	3267.3	3267.3
Empty Weights	...	...	...	670.0	670.0	...
Test Weights	...	51917.0	...	...	...	...
Ope Weights	...	...	...	...	...	670.0
Ope. Liquid	...	...	...	...	...	25415.8
Test Liquid	...	84962.8	...	...	...	...
Totals	126750.0	263629.8	130017.3	130687.3	130687.3	156103.1

Field Installation Options:

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	126750.0 kg.
Shop Test Wt.	- Fabricated weight + water (full)	263629.8 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	130017.3 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	130687.3 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	130687.3 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	156103.1 kg.
Field Test Wt.	- Empty weight + test medium (full)	266897.1 kg.

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	39910.2 kg.
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	70374.0 kg.
Mass of the Upper 1/3 of the Vertical Vessel (empty)	40567.2 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
20	30	153938	148912
30	40	153938	148912
40	50	230907	223367
50	60	102323	97787.6
60	70	89849.6	86959.3
70	80	269549	260878
80	90	269549	260878
90	100	129205	124208

Total	1399258.0	1351901.1 cm ²
Total	1506.1	1455.2 ft ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg.	Total. Ele. Oper. Wgt. kg.	Total. Ele. Hydro. Wgt. kg.	Total Dtl. Offset Mom. Kg-m.	Oper. Wgt. No Liquid kg.
20	30	62297.8	62297.8	122364	...	62297.8
30	Lugs	3786.57	3786.57	10401	...	3786.57
Lugs	40	1262.19	1262.19	3466.99	...	1262.19

STEAM DRUM (WHB)
(A150NRJ10EQ001)

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FileName : Steam drum (WHB) - division 2

Element and Detail Weights:

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40	50	7573.14	20801.9	20801.9	...	7573.14
50	60	4816.05	10988	10988	...	4816.05
60	70	3357.33	9372.32	9372.32	...	3357.33
70	80	10205.4	10205.4	28250.4	78.8409	10205.4
80	90	24213.7	24213.7	42258.6	2625.91	24213.7
90	100	9281.66	9281.66	15100.4	...	9281.66

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg.	Cumulative Oper. Wgt. kg.	Cumulative Hydro. Wgt. kg.
20	30	...	...	...
30	Lugs	-62297.8	-62297.8	-122364
Lugs	40	60709.5	86125.2	130239
40	50	59447.3	84863.1	126772
50	60	51874.1	64061.1	105970
60	70	47058.1	53073.1	94981.7
70	80	43700.8	43700.8	85609.4
80	90	33495.3	33495.3	57359
90	100	9281.66	9281.66	15100.4

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. Kg-m.	Cumulative Oper. Mom. Kg-m.	Cumulative Hydro. Mom. Kg-m.
20	30	...	...	...
30	Lugs	...	...	...
Lugs	40	2704.75	2704.75	2704.75
40	50	2704.75	2704.75	2704.75
50	60	2704.75	2704.75	2704.75
60	70	2704.75	2704.75	2704.75
70	80	2704.75	2704.75	2704.75
80	90	2625.91	2625.91	2625.91
90	100	...	...	...

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Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Ope.	Rating Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
A2-6"	80.95	102.00	286	600	GR 1.1	...	...	...	...
L2-2"	80.95	102.00	286	600	GR 1.1	...	...	...	...
L4-2"	80.95	102.00	286	600	GR 1.1	...	...	...	...
L6-2"	80.95	102.00	286	600	GR 1.1	...	...	...	...
L8-2"	80.95	102.00	286	600	GR 1.1	...	...	...	...
L10-2"	80.95	102.00	286	600	GR 1.1	...	...	...	...
L1-2"	80.95	102.00	286	600	GR 1.1	...	...	...	...
L3-2"	80.95	102.00	286	600	GR 1.1	...	...	...	...
L5-2"	80.95	102.00	286	600	GR 1.1	...	...	...	...
L7-2"	80.95	102.00	286	600	GR 1.1	...	...	...	...
L9-2"	80.95	102.00	286	600	GR 1.1	...	...	...	...
M2-24"	80.95	102.00	286	600	GR 1.1	...	...	...	...
Min Rating	80.949	102.000, [for Core Elements]					...	...	...

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-2, 4.16.12 for more information.

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STEAM DRUM (WHB)
(A150NRJ10EQ001)

PV Elite® v28

PV Elite 28 U2 Licensee: SL User

FileName : Steam drum (WHB) - division 2

Natural Frequency Calculation: Step: 16 6:47pm Sep 16,2026



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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 8.18744 Hz.

The Natural Frequency for the Vessel (Ope...) is 8.03963 Hz.

The Natural Frequency for the Vessel (Filled) is 6.30287 Hz.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 Km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	Moderately Smooth
Base Elevation	7500 mm.
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment	H or Hh 0 mm.
Distance Upwind of Crest	Lh 0 mm.
Distance from Crest to the Vessel	x 0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q)/(1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1+1.7 \cdot 3.4 \cdot 0.152 \cdot 0.937)/(1+1.7 \cdot 3.4 \cdot 0.152)))$
 $= \min(0.85, 0.898)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	8.040 Hz
Natural Frequency of Vessel (Empty)	8.187 Hz
Natural Frequency of Vessel (Test)	6.303 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.582
Structure Height to Diameter ratio	5.909
Height to top of Structure	16740.000 mm.

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of [α] and [Zg]:
 Exposure Category: D from Table 26.11-1
 α = 11.5: Zg = 589788 mm.

Effective Height [z]:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 1000.0 + 7500.0 = 8500.0 \text{ mm.}$
 $= 27.887 \text{ ft. Imperial Units}$

Velocity Pressure coefficient evaluated at height z [Kz]:
 Because z (27.887 ft.) > 15 ft.
 $= 2.41(z / Zg)^{2/\alpha}$
 $= 2.41(27.887/1935.0)^{2/11.5}$
 $= 1.153$

Type of Hill: No Hill

Wind Directionality Factor [Kd]:
 $= 1, \text{ per Table 26.6-1}$



As there is No Hill Present: [Kzt]:

$K1 = 0, K2 = 0, K3 = 0$

Topographical Factor [Kzt]:

$= (1 + K1 \cdot K2 \cdot K3)^2$
 $= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$
 $= 1.0$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$
 $= \max(16, 0.00256 \cdot 1.153 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 111.85^2)$
 $= 36.9 \text{ psf } [180.28] \text{ Kgs/m}^2$

Force on the first element [F]:

$= qz \cdot G \cdot Cf \cdot \text{WindArea}$
 $= 36.924 \cdot 0.85 \cdot 0.582 \cdot 68.975$
 $= 1259.5 \text{ lbs. } [571.3] \text{ Kg}$

Element	Hgt (z) mm.	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
shell 6	8500.0	0.000	0.000	0.000	1.153	1.000	180.280
shell 5	10500.0	0.000	0.000	0.000	1.196	1.000	187.029
shell 4	13000.0	0.000	0.000	0.000	1.241	1.000	194.106
cone transition	15116.0	0.000	0.000	0.000	1.274	1.000	199.264
Shell-3	16200.0	0.000	0.000	0.000	1.290	1.000	201.679
Shell-2	18200.0	0.000	0.000	0.000	1.316	1.000	205.804
Shell-1	21200.0	0.000	0.000	0.000	1.352	1.000	211.338
Top Dish	23353.6	0.000	0.000	0.000	1.374	1.000	214.924

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

Cf - Correction factor for natural frequency
D - Average outer insulated diameter of vessel, mm.
Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
Dr - Average outer insulated of top half of vessel, mm.
f - Natural frequency of vibration (Hertz)
f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
L - Total height of structure mm.
Lc - Total length of conical section(s) of vessel, mm.
tb - Uncorroded plate thickness at bottom of vessel, mm.
V30 - Design Wind Speed provided by user, Km/hr
Vc - Critical wind velocity, Km/hr
Vw - Maximum wind speed at top of structure, Km/hr
W - Total corroded weight of structure, Kg
Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. Kg
Z - Maximum amplitude of vibration at top of vessel, mm.
Dl - Logarithmic decrement (taken as 0.03 for Welded Structures)
Vp - Vib. Chance, $\leq 0.320\text{E-}06$ (High); $0.320\text{E-}06 < 0.400\text{E-}06$ (Probable)
P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5

:: $1200.0/15200.0 = 0.079$

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)

:: $[9.43/49.87^2] \cdot 10^4 = 37.926$ [Geometry Violation]

Compute the vibration possibility. If $Vp > 0.400\text{E-}06$ no chance. [Vp]:

$= W / (L \cdot Dr^2)$
 $= 205777 / (15200.0 \cdot 2991.892^2)$
 $= 0.15124\text{E-}05$



Since V_p is $> 0.400E-06$ no further vibration analysis is required !

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm.	Pressure Kgs/m ²	Force Kgf
Boiler-fab we	235458.00	7500.00	180.28	4244.85
Boiler -Hydro	0.00	7500.00	180.28	0.00
Boiler SF	0.00	7500.00	180.28	0.00
Demister	0.00	19700.00	208.57	0.00
demister supp	0.00	19700.00	208.57	0.00
10% misc	0.00	19700.00	208.57	0.00
Clip weight	0.00	19700.00	208.57	0.00
Trunioin	0.00	22699.97	213.84	0.00
SF weight	0.00	22699.97	213.84	0.00

The Natural Frequency for the Vessel (Ope...) is 8.03963 Hz.

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
20	30	8500	3204	64080	180.28	5297.79
30	40	10500	3204	64080	187.029	651.975
40	50	13000	3204	96120	194.106	1014.97
50	60	15116	3447.43	41369.1	199.264	448.442
60	70	16200	3696	36960	201.679	405.502
70	80	18200	3696	110880	205.804	1241.38
80	90	21200	3696	110880	211.338	1274.77
90	100	23353.6	3696	44704.7	214.924	522.683

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-22	
Importance Factor		1.500	
Force Modification Factor R		3.000	
Site Class		D	
Component Elevation Ratio	z/h	0.500	
Amplification Factor	Ap	1.000	
Force Factor		0.000	
Consider Vertical Acceleration		Yes	
Minimum Acceleration Multiplier		0.000	
User Value of Sds (used if > 0)		0.000	
User Value of Sd1 (used if > 0)		0.000	
Component resonance duct. fact.	[CAR]	1.000	
Component strength factor	[Rpo]	1.500	
Overstrength factor	[Omega0]	2.000	
Response Acceleration Factor	[SMS]	0.326	
Response Acceleration Factor	[SM1]	0.139	
Building Importance Factor	[Ie]	1.000	
Site Latitude		0.000	deg
Site Longitude		0.000	deg

Seismic Analysis Results:

$$SDS = 2/3 * Sms = 2/3 * 0.326 = 0.217$$

$$SD1 = 2/3 * Sm1 = 2/3 * 0.139 = 0.093$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
= $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
= $0.020(54.48160.75)$
= 0.401 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
= (1/Natural Frequency) = (1/8.04)
= 0.124

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
= Minimum Value of ($1.7 \cdot 0.401$, 0.124) per 12.8.2
= 0.124

Loads on Building/Structure Supported Equipment:

Factor [a1]:
= $\min(1/T_a, 2.5)$
= $\min(1/0.124, 2.5)$
= 2.500

Factor [a2]:
= $\max(0, 1 - (0.4/T_a)^2)$
= $\max(0, 1 - (0.4/0.124)^2)$
= 0.000

Factor for force amplification factor [Hf]:
= $1 + (a1 \cdot z/h) + (a2(z/h)^{10})$
= $1 + (2.5 \cdot 0.5) + (0.0(0.5)^{10})$
= 2.250

Structure ductility reduction factor [Rp]:
= $\max(1.3, \sqrt{(1.1 \cdot R / (I_e \cdot \omega_O))})$
= $\max(1.3, \sqrt{(1.1 \cdot 3.0 / (1.0 \cdot 2.0))})$
= 1.300

Seismic Force per equation 13.3-1, [Fp]:

$$= 0.4 \cdot SDS \cdot I_p \cdot W_p (H_f / R_\mu) (CAR / RPO)$$

$$= 0.4 \cdot 0.217 \cdot 1.5 \cdot 335559 (2.25/1.3) (1.0/1.5)$$

$$= 22901.689 \text{ Kgf}$$

Maximum value of Fp per equation 13.3-2:

$$= 1.6 \cdot Sds \cdot I \cdot W$$

$$= 1.6 \cdot 0.217 \cdot 1.5 \cdot 152210 = 79392.52 \text{ Kgf}$$

Minimum value of Fp per equation 13.3-3:

$$= 0.3 \cdot 0.22 \cdot 1.5 \cdot 152210$$

$$= 14886.101 \text{ Kgf}$$

Total Base Shear V = Fp, [V]:

$$= 22901.689 \text{ Kgf}$$

Vertical load per 12.4-4, [Ev]:

$$= 0.2 \cdot Sds \cdot W$$

$$= 0.2 \cdot 0.217 \cdot 152210$$

$$= 6616.0 \text{ Kgf}$$

Distribute the Base shear force to each element according to the equations

$$F_x = C_{vx} \cdot V \text{ (eqn. 12.8-11)}$$
 and the vertical distribution factor

$$C_{vx} = W_x \cdot h_x^k / (\text{Sum of } W_i \cdot h_i^k)$$
 and k is an exponent which is related to the period of Vibration.

In this case, the value of k was 1.0

The Natural Frequency for the Vessel (Ope...) is 8.03963 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm.	Earthquake Weight Kgf	Element Ope Load Kgf	Element Emp Load Kgf	Element Vertical Load Kgf
20	30	1000	62297.8	1514.97	1543.99	-2707.88
30	Lugs	3500	3786.57	322.29	328.463	-2872.47
Lugs	40	3750	1262.19	115.103	117.308	3743.58
40	50	5500	20801.9	2782.27	1032.31	3688.71
50	60	7600	10988	2030.8	907.144	2784.52
60	70	8700	9372.32	1982.89	723.911	2306.91
70	80	10700	10205.4	2655.51	2706.37	1899.53
80	90	13700	24213.7	8067.02	8221.53	1455.93
90	100	15200	9281.66	3430.84	3496.56	403.443



The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm.	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
20	30	2500	...	...	...	...
30	Lugs	750	5297.79	1514.97	5297.89	1515
Lugs	40	250	10857.5	22901.7	20804.7	163991
40	50	2000	4907.75	20949.3	31921.5	157517
50	60	4100	3892.78	18167.1	18720.5	98841
60	70	5200	3444.34	16136.3	14318.1	78258.6
70	80	7200	3038.83	14153.4	11076.5	63113.5
80	90	10200	1797.45	11497.9	3821.89	24635.9
90	100	12353.6	522.683	3430.84	341.638	2242.48

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
Wind: 1.100; Earthquake: 1.000.

Note:

Review the Vessel Design Summary for the cumulative shear force and bending moment on the support.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear Kgf	Centroid Deflection mm.	Elem. End Deflection mm.	Elem. Ang. Rotation
20	30	...	0.011485	0.025204	0.43596E-05
30	Lugs	5297.79	-0.0060324	...	...
Lugs	40	10857.5	-0.0053928	-0.0020533	0.60519E-05
40	50	4907.75	0.027685	0.071311	0.12513E-04
50	60	3892.78	0.091588	0.11282	0.15713E-04
60	70	3444.34	0.13112	0.14983	0.17413E-04
70	80	3038.83	0.20802	0.26834	0.20287E-04
80	90	1797.45	0.32984	0.39185	0.20997E-04
90	100	522.683	0.39185	0.39185	0.20997E-04

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed Km/hr	2nd Crit. Wind Speed Km/hr
20	30	462.413	2890.08
30	40	462.413	2890.08
40	50	462.413	2890.08
50	60	497.545	3109.66
60	70	533.42	3333.87
70	80	533.42	3333.87
80	90	533.42	3333.87
90	100	533.42	3333.87

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
Allowable deflection : 76.000 Actual deflection : 0.392 mm.

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STEAM DRUM (WHB)
(A150NRJ10EQ001)

PV Elite[®] v28

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FileName : Steam drum (WHB) - division 2

Longitudinal Stress Constants: Step: 21 6:47pm Sep 16,2026



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Longitudinal Stress Constants:

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm. ³	Section Modulus Corroded mm. ³
20	30	3028.5	2916.74	179537136	173123872
30	40	3028.5	2916.74	179537136	173123872
40	50	3028.5	2916.74	179537136	173123872
50	60	3701.4	3588.02	217810912	211403248
60	70	4066.6	3936.09	281559360	272808992
70	80	4066.6	3936.09	281559360	272808992
80	90	4066.6	3936.09	281559360	272808992
90	100	2138.3	2005.71	150343568	141169584

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Longitudinal Allowable Stresses:

From	To	Tensile N./mm ²	Hydrotest Tensile N./mm ²	Compressive N./mm ²	Hydrotest Compressive N./mm ²
20	30	165.119	294.495	-85.1877	-118.105
30	Lugs	165.119	294.495	-85.1877	-118.105
Lugs	40	165.119	294.495	-85.1877	-118.105
40	50	165.119	294.495	-85.1877	-118.105
50	60	165.119	294.495	-88.1456	-119.394
60	70	165.119	294.495	-85.0484	-118.044
70	80	165.119	294.495	-85.0484	-118.044
80	90	165.119	294.495	-85.0484	-118.044
90	100	165.119	294.495	-75.3565	-113.874

Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure N./mm ²	Longitudinal Stress External Pressure N./mm ²	Longitudinal Stress Hydrotest Pressure N./mm ²
20	30	77.3363	-1.67167	89.7179
30	40	77.3363	-1.67167	89.7179
40	50	77.3363	-1.67167	89.7179
50	60	63.2964	-1.38927	73.4302
60	70	78.1435	-1.68804	90.6544
70	80	78.1435	-1.68804	90.6544
80	90	78.1435	-1.68804	90.6544
90	100	158.261	-3.31268	183.599

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty N./mm ²	Wght. Str. Operating N./mm ²	Wght. Str. Hydrotest N./mm ²	Wght. Str. Emp. Mom. N./mm ²	Wght. Str. Opr. Mom. N./mm ²
20	30	...	...	...	...	...
30	Lugs	2.09461	2.09461	4.11419	...	...
Lugs	40	-2.04121	-2.04121	-2.04121	0.15321	0.15321
40	50	-1.99877	-1.99877	-1.99877	0.15321	0.15321
50	60	-1.41783	-1.41783	-1.41783	0.12547	0.12547
60	70	-1.17246	-1.17246	-1.17246	0.097227	0.097227
70	80	-1.08881	-1.08881	-1.08881	0.097227	0.097227
80	90	-0.83454	-0.83454	-0.83454	0.094393	0.094393
90	100	-0.45382	-0.45382	-0.45382	...	...

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. N./mm ²	Bend. Str. Oper. Wind N./mm ²	Bend. Str. Oper. Equ. N./mm ²	Bend. Str. Hyd. Wind N./mm ²	Bend. Str. Hyd. Equ. N./mm ²
20	30	...	...	...	...	...
30	Lugs	...	0.3001	0.085818	0.099033	...
Lugs	40	0.15321	1.17849	9.28931	0.3889	...
40	50	0.15321	1.8082	8.92257	0.59671	...
50	60	0.12547	0.86841	4.58506	0.28658	...
60	70	0.097227	0.51469	2.81315	0.16985	...
70	80	0.097227	0.39816	2.26873	0.13139	...
80	90	0.094393	0.13739	0.88558	0.045337	...
90	100	...	0.023733	0.15578	0.0078317	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case N./mm ²	Vortex Shedding Empty Case N./mm ²	Vortex Shedding Test Case N./mm ²	Earthquake Empty Case N./mm ²
20	30	...	...	...	...
30	Lugs	...	...	...	0.087461
Lugs	40	...	...	...	8.61091

40	50	...	...	...	8.35787
50	60	...	...	...	4.53825
60	70	...	...	...	2.84372
70	80	...	...	...	2.31219
80	90	...	...	...	0.90255
90	100	...	...	...	0.15876

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind N./mm ²	Longitudinal Stress Y Forces Seismic N./mm ²
20	30	...	-0.091046
30	Lugs	...	-0.09658
Lugs	40	...	0.12587
40	50	...	0.12402
50	60	...	0.076107
60	70	...	0.057477
70	80	...	0.047327
80	90	...	0.036275
90	100	...	0.019726

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded N./mm ²	Earthquake For/Mom Corroded N./mm ²	Wind For/Mom No Corrosion N./mm ²	Earthquake For/Mom No Corrosion N./mm ²
20	30	...	...	...	...
30	Lugs	...	...	...	...
Lugs	40	...	...	...	...
40	50	...	...	...	...
50	60	...	...	...	...
60	70	...	...	...	...
70	80	...	...	...	...
80	90	...	...	...	...
90	100	...	...	...	...

Bending Stresses due to Blast Load:

From	To	Bending Stress N./mm ²
20	30	...
30	Lugs	...
Lugs	40	...
40	50	...
50	60	...
60	70	...
70	80	...
80	90	...
90	100	...



Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

For Division 2, the overall stress intensity is computed in accordance with the Section VIII Division II Example problem manual PTB-3, example 4.3.6, stress due to combined loads.
 The stresses are combined per equation 4.3.44 to determine the overall stress intensity.
 The unity check is from Section 4.4.12 equations 4.4.85 and 4.4.90.

An asterisk (*) in the final column denotes overstress.

Note: Performing Combined Stress Analysis per VIII-2 paragraph 4.3.10.2.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
20	0.00	165.12	0.00	170.66	0.0000	-0.0000	0.0035
30	2.39	165.12		170.66	0.0145		0.0133
30	3.37	165.12	-3.37	170.66	0.0204	0.0198	0.0130
40	3.96	165.12	-3.96	170.66	0.0240	0.0232	0.0127
60	1.78	165.12	-1.78	170.19	0.0108	0.0105	0.0073
70	1.58	165.12	-1.58	170.19	0.0096	0.0093	0.0068
80	1.07	165.12	-1.07	170.19	0.0065	0.0063	0.0052
90	0.48	165.12	-0.48	72.60	0.0029	0.0066	0.0134

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
20	0.09	165.12	-0.09	170.66	0.0006	0.0005	...
30	2.09	165.12		170.66	0.0126		0.0127

30	10.68	165.12	-10.68	170.66	0.0647	0.0626	0.0514
40	10.39	165.12	-10.39	170.66	0.0629	0.0609	0.0499
60	4.06	165.12	-4.06	170.19	0.0246	0.0238	0.0173
70	3.45	165.12	-3.45	170.19	0.0209	0.0203	0.0142
80	1.80	165.12	-1.80	170.19	0.0109	0.0105	0.0059
90	0.59	165.12	-0.59	140.52	0.0036	0.0042	0.0121

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
20	0.00	165.12	0.00	170.66	0.0000	-0.0000	0.0035
30	2.39	165.12		170.66	0.0145		0.0133
30	3.37	165.12	-3.37	170.66	0.0204	0.0198	0.0130
40	3.96	165.12	-3.96	170.66	0.0240	0.0232	0.0127
60	1.78	165.12	-1.78	170.19	0.0108	0.0105	0.0073
70	1.58	165.12	-1.58	170.19	0.0096	0.0093	0.0068
80	1.07	165.12	-1.07	170.19	0.0065	0.0063	0.0052
90	0.48	165.12	-0.48	72.60	0.0029	0.0066	0.0134

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
20	0.09	165.12	-0.09	170.66	0.0006	0.0005	...
30	2.08	165.12		170.66	0.0126		0.0127
30	11.36	165.12	-11.36	170.66	0.0688	0.0666	0.0553
40	10.95	165.12	-10.95	170.66	0.0663	0.0642	0.0532
60	4.03	165.12	-4.03	170.19	0.0244	0.0237	0.0171
70	3.41	165.12	-3.41	170.19	0.0206	0.0200	0.0139
80	1.78	165.12	-1.78	170.19	0.0108	0.0104	0.0058
90	0.59	165.12	-0.59	140.52	0.0036	0.0042	0.0120

Analysis of Load Case 5 : NP+HW+HI

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
20	0.00	294.49	0.00	118.10	0.0000	-0.0000	0.0035
30	4.21	294.49		118.10	0.0143		0.0256
30	2.58	294.49	-2.58	118.10	0.0088	0.0219	0.0127
40	2.75	294.49	-2.75	118.10	0.0093	0.0233	0.0124
60	1.44	294.49	-1.44	118.04	0.0049	0.0122	0.0072
70	1.32	294.49	-1.32	118.04	0.0045	0.0112	0.0067
80	0.97	294.49	-0.97	118.04	0.0033	0.0083	0.0051
90	0.46	294.49	-0.46	113.87	0.0016	0.0041	0.0119

Analysis of Load Case 6 : NP+HW+HE

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
20	0.00	294.49	0.00	118.10	0.0000	-0.0000	...
30	4.11	294.49		118.10	0.0140		0.0256
30	2.19	294.49	-2.19	118.10	0.0075	0.0186	0.0127
40	2.15	294.49	-2.15	118.10	0.0073	0.0182	0.0124
60	1.27	294.49	-1.27	118.04	0.0043	0.0108	0.0072
70	1.19	294.49	-1.19	118.04	0.0040	0.0100	0.0067
80	0.93	294.49	-0.93	118.04	0.0032	0.0079	0.0051
90	0.45	294.49	-0.45	113.87	0.0015	0.0040	0.0113

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
20	137.50	165.12		170.66	0.8327		0.0035
30	137.48	165.12		170.66	0.8326		0.0133
30	137.59	165.12		170.66	0.8333		0.0130
40	137.61	165.12		170.66	0.8334		0.0127
60	137.82	165.12		170.19	0.8347		0.0073



70	137.55	165.12	170.19	0.8330	0.0068
80	137.54	165.12	170.19	0.8330	0.0052
90	158.05	165.12	75.36	0.9572	...

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
20	137.50	165.12		170.66	0.8327		...
30	137.48	165.12		170.66	0.8326		0.0127
30	138.14	165.12		170.66	0.8366		0.0553
40	138.10	165.12		170.66	0.8363		0.0532
60	137.89	165.12		170.19	0.8351		0.0171
70	137.60	165.12		170.19	0.8333		0.0139
80	137.56	165.12		170.19	0.8331		0.0058
90	158.12	165.12		75.36	0.9576		...

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
20	1.67	165.12	-1.67	170.66	0.0101	0.0098	0.0035
30	0.72	165.12		170.66	0.0044		0.0748
30	5.04	165.12	-5.04	170.66	0.0306	0.0296	0.0753
40	5.63	165.12	-5.63	170.66	0.0341	0.0330	0.0757
60	3.47	165.12	-3.47	170.19	0.0210	0.0204	0.0679
70	3.27	165.12	-3.27	170.19	0.0198	0.0192	0.0673
80	2.75	165.12	-2.75	170.19	0.0167	0.0162	0.0654
90	3.79	165.12	-3.79	72.60	0.0230	0.0522	0.0453

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
20	1.76	165.12	-1.76	170.66	0.0107	0.0103	0.0016
30	0.41	165.12		170.66	0.0025		0.0743
30	13.03	165.12	-13.03	170.66	0.0789	0.0763	0.0934
40	12.62	165.12	-12.62	170.66	0.0764	0.0740	0.0918
60	5.71	165.12	-5.71	170.19	0.0346	0.0336	0.0699
70	5.10	165.12	-5.10	170.19	0.0309	0.0299	0.0687
80	3.47	165.12	-3.47	170.19	0.0210	0.0204	0.0656
90	3.90	165.12	-3.90	72.60	0.0236	0.0538	0.0453

Analysis of Load Case 11 : HP+HW+HI

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
20	154.96	294.49		118.10	0.5262		0.0035
30	155.02	294.49		118.10	0.5264		0.0256
30	154.97	294.49		118.10	0.5262		0.0127
40	154.98	294.49		118.10	0.5262		0.0124
60	157.45	294.49		118.04	0.5346		0.0072
70	157.44	294.49		118.04	0.5346		0.0067
80	157.44	294.49		118.04	0.5346		0.0051
90	179.50	294.49		113.87	0.6095		...

Analysis of Load Case 12 : HP+HW+HE

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
20	154.96	294.49		118.10	0.5262		...
30	155.02	294.49		118.10	0.5264		0.0256
30	154.97	294.49		118.10	0.5262		0.0127
40	154.97	294.49		118.10	0.5262		0.0124
60	157.44	294.49		118.04	0.5346		0.0072
70	157.44	294.49		118.04	0.5346		0.0067
80	157.44	294.49		118.04	0.5346		0.0051
90	179.49	294.49		113.87	0.6095		...

Analysis of Load Case 13 : IP+WE+EW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
20	137.50	165.12		170.66	0.8327		...
30	137.48	165.12		170.66	0.8326		0.0133
30	137.55	165.12		170.66	0.8330		0.0130
40	137.55	165.12		170.66	0.8330		0.0127
60	137.81	165.12		170.19	0.8346		0.0073
70	137.54	165.12		170.19	0.8330		0.0068
80	137.54	165.12		170.19	0.8330		0.0052
90	158.03	165.12		75.36	0.9571		...

Analysis of Load Case 14 : IP+WF+CW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
20	137.50	165.12		170.66	0.8327		...
30	137.48	165.12		170.66	0.8326		0.0128
30	137.54	165.12		170.66	0.8330		0.0125
40	137.54	165.12		170.66	0.8330		0.0122
60	137.81	165.12		170.19	0.8346		0.0071
70	137.54	165.12		170.19	0.8330		0.0066
80	137.54	165.12		170.19	0.8330		0.0050
90	158.05	165.12		75.36	0.9572		...

Analysis of Load Case 15 : IP+VO+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
20	137.50	165.12		170.66	0.8327		...
30	137.48	165.12		170.66	0.8326		0.0133
30	137.55	165.12		170.66	0.8330		0.0130
40	137.55	165.12		170.66	0.8330		0.0127
60	137.81	165.12		170.19	0.8346		0.0073
70	137.54	165.12		170.19	0.8330		0.0068
80	137.54	165.12		170.19	0.8330		0.0052
90	158.03	165.12		75.36	0.9571		...

Analysis of Load Case 16 : IP+VE+EW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
20	137.50	165.12		170.66	0.8327		...
30	137.48	165.12		170.66	0.8326		0.0133
30	137.55	165.12		170.66	0.8330		0.0130
40	137.55	165.12		170.66	0.8330		0.0127
60	137.81	165.12		170.19	0.8346		0.0073
70	137.54	165.12		170.19	0.8330		0.0068
80	137.54	165.12		170.19	0.8330		0.0052
90	158.03	165.12		75.36	0.9571		...

Analysis of Load Case 17 : NP+VO+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
20	0.00	165.12	0.00	170.66	0.0000	-0.0000	...
30	2.09	165.12		170.66	0.0127		0.0133
30	2.19	165.12	-2.19	170.66	0.0133	0.0129	0.0130
40	2.15	165.12	-2.15	170.66	0.0130	0.0126	0.0127
60	1.27	165.12	-1.27	170.19	0.0077	0.0075	0.0073
70	1.19	165.12	-1.19	170.19	0.0072	0.0070	0.0068
80	0.93	165.12	-0.93	170.19	0.0056	0.0055	0.0052
90	0.45	165.12	-0.45	72.60	0.0027	0.0063	0.0115

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
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STEAM DRUM (WHB)
(A150NRJ10EQ001)



PV Elite 28 U2 Licensee: SL User

FileName : Steam drum (WHB) - division 2

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Stress due to Combined Loads: Step: 24 6:47pm Sep 16,2026

Node	Intensity	Intensity	Stress	Stress Fxa	Ratio	Ratio	Check
20	137.50	165.12		170.66	0.8327		...
30	137.48	165.12		170.66	0.8326		0.0127
30	137.55	165.12		170.66	0.8330		0.0122
40	137.54	165.12		170.66	0.8330		0.0119
60	137.81	165.12		170.19	0.8346		0.0070
70	137.54	165.12		170.19	0.8330		0.0065
80	137.54	165.12		170.19	0.8330		0.0050
90	158.04	165.12		75.36	0.9572		...

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
20	1.76	165.12	-1.76	170.66	0.0107	0.0103	0.0016
30	0.33	165.12		170.66	0.0020		0.0167
30	3.74	165.12	-3.74	170.66	0.0227	0.0219	0.0164
40	3.70	165.12	-3.70	170.66	0.0224	0.0217	0.0162
60	2.90	165.12	-2.90	170.19	0.0176	0.0170	0.0116
70	2.83	165.12	-2.83	170.19	0.0171	0.0166	0.0112
80	2.58	165.12	-2.58	170.19	0.0156	0.0152	0.0095
90	3.75	165.12	-3.75	72.60	0.0227	0.0516	0.0453

Analysis of Load Case 20 : BL+IP+OW

From Node	Stress Intensity	All. Stress Intensity	Compressive Stress	All. Comp. Stress Fxa	S. I. Ratio	Comp. Ratio	Unity Check
20	137.50	165.12		170.66	0.8327		...
30	137.48	165.12		170.66	0.8326		0.0133
30	137.55	165.12		170.66	0.8330		0.0130
40	137.55	165.12		170.66	0.8330		0.0127
60	137.81	165.12		170.19	0.8346		0.0073
70	137.54	165.12		170.19	0.8330		0.0068
80	137.54	165.12		170.19	0.8330		0.0052
90	158.03	165.12		75.36	0.9571		...

Absolute Maximum of the all of the Stress Ratio's 0.9576

Governing Element: Top Dish

Governing Load Case 8 : IP+OW+EQ+FS+BS

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
shell 6	77.3	-0.1	...	...	77.2	+ Tens
shell 6	77.3	-0.1	...	...	...	- Comp
shell 5	77.3	2.0	...	0.1	79.4	+ Tens
shell 5	77.3	2.0	...	-0.1	...	- Comp
shell 5	77.3	-1.9	0.2	9.3	84.9	+ Tens
shell 5	77.3	-1.9	-0.2	-9.3	...	- Comp
shell 4	77.3	-1.9	0.2	8.9	84.5	+ Tens
shell 4	77.3	-1.9	-0.2	-8.9	...	- Comp
cone transition	63.3	-1.3	0.1	4.6	66.7	+ Tens
cone transition	63.3	-1.3	-0.1	-4.6	...	- Comp
Shell-3	78.1	-1.1	0.1	2.8	79.9	+ Tens
Shell-3	78.1	-1.1	-0.1	-2.8	...	- Comp
Shell-2	78.1	-1.0	0.1	2.3	79.5	+ Tens
Shell-2	78.1	-1.0	-0.1	-2.3	...	- Comp
Shell-1	78.1	-0.8	0.1	0.9	78.3	+ Tens

Shell-1	78.1	-0.8	-0.1	-0.9	...	- Comp
Top Dish	158.3	-0.4	...	0.2	158.0	+ Tens
Top Dish	158.3	-0.4	...	-0.2	...	- Comp

Additional Results per Section 4.4, see 4.4.15

Nomenclature :

- L - Section Length mm.
- Fxa - Allowable comp mem stress due to compression with lamc ≤ 0.15 , N./mm²
- Fba - Allowable all comp mem stress due to bending moment, N./mm²
- Fva - Allowable shear stress when only shear is present, N./mm²
- Fha - Allowable hoop comp memb stress under ext pressure alone, N./mm²
- Fxha - Allowable axial comp w/hoop compression and lamc ≤ 0.15 , N./mm²
- Fhxa - Allowable hoop comp w/axial compression and lamc ≤ 0.15 , N./mm²
- Faha - Allowable axial comp w/hoop compression and lamc > 0.15 , N./mm²
- Fbha - Allowable axial comp from bending w/hoop compression, N./mm²
- Fhba - Allowable hoop comp w/long compression due to moment, N./mm²

Allowable Stress Results for Case 1 : NP+EW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
20	0.466	6999	170	170	102	57	6895	170	6895	6895	6895
30	0.466	6999	170	170	102	57	6895	170	6895	6895	6895
40	0.466	6999	170	170	102	57	6895	170	6895	6895	6895
60	0.399	7468	170	170	103	59	6895	170	6895	6895	6895
70	0.399	7468	170	170	103	59	6895	170	6895	6895	6895
80	0.399	7468	170	170	103	59	6895	170	6895	6895	6895
90						72	6895		6895	6895	6895

Allowable Stress Results for Case 2 : NP+EW+EE+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
20	0.466	6999	170	170	102	57	6895	170	6895	6895	6895
30	0.466	6999	170	170	102	57	6895	170	6895	6895	6895
40	0.466	6999	170	170	102	57	6895	170	6895	6895	6895
60	0.399	7468	170	170	103	59	6895	170	6895	6895	6895
70	0.399	7468	170	170	103	59	6895	170	6895	6895	6895
80	0.399	7468	170	170	103	59	6895	170	6895	6895	6895
90						140	6895		6895	6895	6895

Allowable Stress Results for Case 3 : NP+OW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
20	0.466	6999	170	170	102	57	6895	170	6895	6895	6895
30	0.466	6999	170	170	102	57	6895	170	6895	6895	6895
40	0.466	6999	170	170	102	57	6895	170	6895	6895	6895
60	0.399	7468	170	170	103	59	6895	170	6895	6895	6895
70	0.399	7468	170	170	103	59	6895	170	6895	6895	6895
80	0.399	7468	170	170	103	59	6895	170	6895	6895	6895
90						72	6895		6895	6895	6895

Allowable Stress Results for Case 4 : NP+OW+EQ+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
20	0.466	6999	170	170	102	57	6895	170	6895	6895	6895
30	0.466	6999	170	170	102	57	6895	170	6895	6895	6895
40	0.466	6999	170	170	102	57	6895	170	6895	6895	6895
60	0.399	7468	170	170	103	59	6895	170	6895	6895	6895
70	0.399	7468	170	170	103	59	6895	170	6895	6895	6895
80	0.399	7468	170	170	103	59	6895	170	6895	6895	6895
90						140	6895		6895	6895	6895

Allowable Stress Results for Case 5 : NP+HW+HI

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
20	0.451	6999	173	173	102	59	6895	173	6895	6895	6895
30	0.451	6999	173	173	102	59	6895	173	6895	6895	6895
40	0.451	6999	173	173	102	59	6895	173	6895	6895	6895
60	0.386	7468	173	173	103	61	6895	173	6895	6895	6895
70	0.386	7468	173	173	103	61	6895	173	6895	6895	6895
80	0.386	7468	173	173	103	61	6895	173	6895	6895	6895



90 74 6895 6895 6895 6895

Allowable Stress Results for Case 6 : NP+HW+HE

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
20	0.451	6999	173	173	102	59	6895	173	6895	6895	6895
30	0.451	6999	173	173	102	59	6895	173	6895	6895	6895
40	0.451	6999	173	173	102	59	6895	173	6895	6895	6895
60	0.386	7468	173	173	103	61	6895	173	6895	6895	6895
70	0.386	7468	173	173	103	61	6895	173	6895	6895	6895
80	0.386	7468	173	173	103	61	6895	173	6895	6895	6895
90						74	6895		6895	6895	6895

Allowable Stress Results for Case 7 : IP+OW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
20	0.466	6999	170	170	102	57	6895	170	6895	6895	6895
30	0.466	6999	170	170	102	57	6895	170	6895	6895	6895
40	0.466	6999	170	170	102	57	6895	170	6895	6895	6895
60	0.399	7468	170	170	103	59	6895	170	6895	6895	6895
70	0.399	7468	170	170	103	59	6895	170	6895	6895	6895
80	0.399	7468	170	170	103	59	6895	170	6895	6895	6895
90							6895		6895	6895	6895

Allowable Stress Results for Case 8 : IP+OW+EQ+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
20	0.466	6999	170	170	102	57	6895	170	6895	6895	6895
30	0.466	6999	170	170	102	57	6895	170	6895	6895	6895
40	0.466	6999	170	170	102	57	6895	170	6895	6895	6895
60	0.399	7468	170	170	103	59	6895	170	6895	6895	6895
70	0.399	7468	170	170	103	59	6895	170	6895	6895	6895
80	0.399	7468	170	170	103	59	6895	170	6895	6895	6895
90							6895		6895	6895	6895

Allowable Stress Results for Case 9 : EP+OW+1.1WI+FW+BW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
20	0.466	6999	170	170	102	57	28	59	26	6895	6895
30	0.466	6999	170	170	102	57	65	60	60	22	56
40	0.466	6999	170	170	102	57	65	60	59	33	55
60	0.399	7468	170	170	103	59	52	63	48	10	59
70	0.399	7468	170	170	103	59	50	63	47	8	59
80	0.399	7468	170	170	103	59	45	63	42	4	59
90						72	6895		6895	6895	6895

Allowable Stress Results for Case 10 : EP+OW+EQ+FS+BS

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
20	0.466	6999	170	170	102	57	30	59	27	6895	6895
30	0.466	6999	170	170	102	57	63	60	58	122	42
40	0.466	6999	170	170	102	57	62	60	57	120	43
60	0.399	7468	170	170	103	59	51	63	47	50	57
70	0.399	7468	170	170	103	59	49	63	46	41	57
80	0.399	7468	170	170	103	59	45	63	42	17	59
90						72	6895		6895	6895	6895

Allowable Stress Results for Case 11 : HP+HW+HI

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
20	0.451	6999	173	173	102	59	6895	173	6895	6895	6895
30	0.451	6999	173	173	102	59	6895	173	6895	6895	6895
40	0.451	6999	173	173	102	59	6895	173	6895	6895	6895
60	0.386	7468	173	173	103	61	6895	173	6895	6895	6895
70	0.386	7468	173	173	103	61	6895	173	6895	6895	6895
80	0.386	7468	173	173	103	61	6895	173	6895	6895	6895
90							6895		6895	6895	6895

Allowable Stress Results for Case 12 : HP+HW+HE

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
20	0.451	6999	173	173	102	59	6895	173	6895	6895	6895
30	0.451	6999	173	173	102	59	6895	173	6895	6895	6895
40	0.451	6999	173	173	102	59	6895	173	6895	6895	6895
60	0.386	7468	173	173	103	61	6895	173	6895	6895	6895

STEAM DRUM (WHB)

(A150NRJ10EQ001)

PV Elite 28 U2

Elite® v28

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FileName : Steam drum (WHB) - division 2

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70	0.386	7468	173	173	103	61	6895	173	6895	6895	6895
80	0.386	7468	173	173	103	61	6895	173	6895	6895	6895
90							6895		6895	6895	6895

Allowable Stress Results for Case 13 : IP+WE+EW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
20	0.466	6999	170	170	102	57	6895	170	6895	6895	6895
30	0.466	6999	170	170	102	57	6895	170	6895	6895	6895
40	0.466	6999	170	170	102	57	6895	170	6895	6895	6895
60	0.399	7468	170	170	103	59	6895	170	6895	6895	6895
70	0.399	7468	170	170	103	59	6895	170	6895	6895	6895
80	0.399	7468	170	170	103	59	6895	170	6895	6895	6895
90							6895		6895	6895	6895

Allowable Stress Results for Case 14 : IP+WF+CW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
20	0.466	6999	170	170	102	57	6895	170	6895	6895	6895
30	0.466	6999	170	170	102	57	6895	170	6895	6895	6895
40	0.466	6999	170	170	102	57	6895	170	6895	6895	6895
60	0.399	7468	170	170	103	59	6895	170	6895	6895	6895
70	0.399	7468	170	170	103	59	6895	170	6895	6895	6895
80	0.399	7468	170	170	103	59	6895	170	6895	6895	6895
90							6895		6895	6895	6895

Allowable Stress Results for Case 15 : IP+VO+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
20	0.466	6999	170	170	102	57	6895	170	6895	6895	6895
30	0.466	6999	170	170	102	57	6895	170	6895	6895	6895
40	0.466	6999	170	170	102	57	6895	170	6895	6895	6895
60	0.399	7468	170	170	103	59	6895	170	6895	6895	6895
70	0.399	7468	170	170	103	59	6895	170	6895	6895	6895
80	0.399	7468	170	170	103	59	6895	170	6895	6895	6895
90							6895		6895	6895	6895

Allowable Stress Results for Case 16 : IP+VE+EW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
20	0.466	6999	170	170	102	57	6895	170	6895	6895	6895
30	0.466	6999	170	170	102	57	6895	170	6895	6895	6895
40	0.466	6999	170	170	102	57	6895	170	6895	6895	6895
60	0.399	7468	170	170	103	59	6895	170	6895	6895	6895
70	0.399	7468	170	170	103	59	6895	170	6895	6895	6895
80	0.399	7468	170	170	103	59	6895	170	6895	6895	6895
90							6895		6895	6895	6895

Allowable Stress Results for Case 17 : NP+VO+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
20	0.466	6999	170	170	102	57	6895	170	6895	6895	6895
30	0.466	6999	170	170	102	57	6895	170	6895	6895	6895
40	0.466	6999	170	170	102	57	6895	170	6895	6895	6895
60	0.399	7468	170	170	103	59	6895	170	6895	6895	6895
70	0.399	7468	170	170	103	59	6895	170	6895	6895	6895
80	0.399	7468	170	170	103	59	6895	170	6895	6895	6895
90							72		6895	6895	6895

Allowable Stress Results for Case 18 : FS+BS+IP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
20	0.466	6999	170	170	102	57	6895	170	6895	6895	6895
30	0.466	6999	170	170	102	57	6895	170	6895	6895	6895
40	0.466	6999	170	170	102	57	6895	170	6895	6895	6895
60	0.399	7468	170	170	103	59	6895	170	6895	6895	6895
70	0.399	7468	170	170	103	59	6895	170	6895	6895	6895
80	0.399	7468	170	170	103	59	6895	170	6895	6895	6895
90							6895		6895	6895	6895

Allowable Stress Results for Case 19 : FS+BS+EP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
20	0.466	6999	170	170	102	57	30	59	27	6895	6895
30	0.466	6999	170	170	102	57	63	60	58	6895	6895

STEAM DRUM (WHB)
(A150NRJ10EQ001)



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40	0.466	6999	170	170	102	57	62	60	57	6895	6895
60	0.399	7468	170	170	103	59	51	63	47	6895	6895
70	0.399	7468	170	170	103	59	49	63	46	6895	6895
80	0.399	7468	170	170	103	59	45	63	42	6895	6895
90						72	6895		6895	6895	6895

Allowable Stress Results for Case 20 : BL+IP+OW

From	Lamc	L	Fxa	Fba	Fva	Fha	Fxha	Fhxa	Faha	Fbha	Fhba
20	0.466	6999	170	170	102	57	6895	170	6895	6895	6895
30	0.466	6999	170	170	102	57	6895	170	6895	6895	6895
40	0.466	6999	170	170	102	57	6895	170	6895	6895	6895
60	0.399	7468	170	170	103	59	6895	170	6895	6895	6895
70	0.399	7468	170	170	103	59	6895	170	6895	6895	6895
80	0.399	7468	170	170	103	59	6895	170	6895	6895	6895
90							6895		6895	6895	6895

Cone Junction Stresses and related Calculations are shown in the Conical Sections report.

Note:

Load cases involving Wind and Seismic are assumed to cause a net section bending moment. These load case components are: EQ, EE, WI, WE, VO, VE, VF, HE, HI, WF. Static bending moments due to the weights of nozzles etc. are considered to be local in nature and will not cause a net section bending moment. This is an important consideration when computing the unity check per section 4.4.12 equations 4.4.86, 4.4.87, 4.4.90, 4.4.91 and 4.4.92.

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Shop/Field Installation Options :

Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Liquid	6767.290 mm.
Center of Gravity of the Insulation	8626.215 mm.
Center of Gravity of the Nozzles	13827.527 mm.
Center of Gravity of the Lugs	3750.000 mm.
Center of Gravity of the Added Weights (Operating)	3192.230 mm.
Center of Gravity of the Added Weights (Empty)	3192.230 mm.
Center of Gravity of Bare Shell New and Cold	8605.125 mm.
Center of Gravity of Bare Shell Corroded	8604.257 mm.
Vessel CG in the Operating Condition	5609.476 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	5420.669 mm.
Vessel CG in the Test Condition	6717.600 mm.

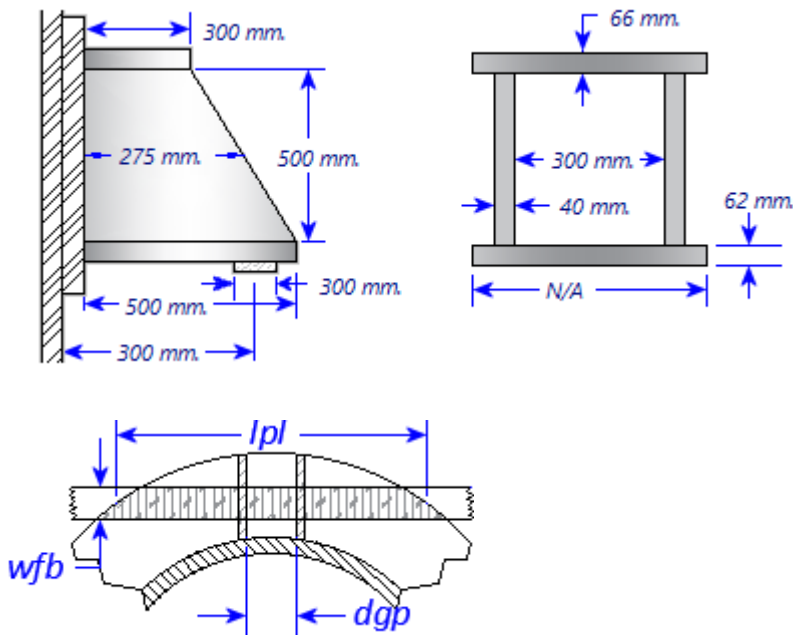
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Support Lug Calculations: Operating Case

Input Data: Support Lugs

Type of Geometry : Gussets with Continuous Top Ring
 Number of Support Lugs Nlug 4
 Distance from Vessel OD to Lug Contact Point Dlug 300.0000 mm.
 Lug Support Force Bearing Width Wfb 300.0000 mm.
 Lug Material SA-387 11
 Lug Yield Stress 247.68 N./mm²
 Radial Width of bottom Support Lug Plate Wpl 500.0000 mm.
 Effective Force Bearing Length Lpl 500.0000 mm.
 Thickness of bottom Support Lug Plate Tpl 62.0000 mm.
 Distance between Gussets Dgp 300.0000 mm.
 Mean Width of Gusset Plate Wgp 275.0000 mm.
 Height of Gusset Plate Hgp 500.0000 mm.
 Thickness of Gusset Plate Tgp 40.0000 mm.
 Radial Width of Reinforcing Ring Wtp 300.0000 mm.
 Thickness of Top Plate Ttp 66.0000 mm.

Bolt Material SA-193 B7
 Bolt Allowable Stress at Design Temperature 150.00 N./mm²
 Thread Series TEMA Metric
 Bolt Diameter 48.00 mm.



Results for Support Lugs:

Description: Support Lugs

Overturning Moment at Support Lug 163991.1 Kg-m.
 Weight Load at the top of one Lug 39706.4 Kgf

Force on one Lug, Operating Condition [Flug]:

$$= (W/Nlug + Mlug/(Rlug \cdot Nlug/2))$$

$$= (158826/4 + 163991/(1525.0 \cdot 4/2))$$

$$= 93472.92 \text{ Kgf}$$

Force on one lug in the Empty Condition [Flugemp]:

$$= (-Wemp/Nlug + Mlugemp/(Rlug \cdot Nlug/2))$$

$$= (-126794/4 + 152015/(1525.0 \cdot 4/2))$$

$$= 18141.47 \text{ Kgf}$$

Ring Girder Calculations Per Bednar pp.157-159:

Force P acting in the plane of the ring [P]:

$$\begin{aligned} &= \text{Flug} \cdot \text{Dlug} / (\text{Hgp} + \text{Tpl}) \\ &= 93473 \cdot 300.0 / (500.0 + 62.0) \\ &= 49896.58 \text{ Kgf} \end{aligned}$$

Angle [α]:

$$\begin{aligned} &= \pi / \text{Nlug} \\ &= \pi / 4 \\ &= 45.0 \text{ deg} \end{aligned}$$

Maximum bending moment (at the load points) [M1]:

$$\begin{aligned} &= 0.5 \cdot P(\text{OD}/2 + \text{Wtp}/2) \cdot (1/\alpha - \cos(\alpha)/\sin(\alpha)) \\ &= 0.5 \cdot 49896.58 \cdot 1375.0(1/0.79 - \cos(45.0)/\sin(45.0)) \\ &= 9373.36 \text{ Kg-m.} \end{aligned}$$

Tangential (axial) thrust induced in the ring [T1]:

$$\begin{aligned} &(\text{Top ring in compression, bottom ring in tension}) \\ &= 0.5 \cdot P \cdot \cos(\alpha) / \sin(\alpha) \\ &= 0.5 \cdot 49896.58 \cdot \cos(45.0) / \sin(45.0) \\ &= 24948.29 \text{ Kgf} \end{aligned}$$

Resulting combined stress at the load point [S1]:

$$\begin{aligned} &= T1 / (\text{Ttp} \cdot \text{Wtp}) + M1 / (\text{Ttp} \cdot \text{Wtp}^2 / 6) \\ &= 24948 / (66.0 \cdot 300.0) + 9373.36 / (66.0 \cdot 300.0^2 / 6) \\ &= 105.21 \text{ N./mm}^2 \end{aligned}$$

Bending moment at the midpoints between the loads [M2]:

$$\begin{aligned} &= 0.5 \cdot P(\text{OD}/2 + \text{Wtp}/2) \cdot (1/\sin(\alpha) - 1/\alpha) \\ &= 0.5 \cdot 49896.58 \cdot 1375.0(1/\sin(45.0) - 1/0.79) \\ &= 4836.05 \text{ Kg-m.} \end{aligned}$$

Axial thrust [T2]:

$$\begin{aligned} &= 0.5 \cdot P / \sin(\alpha) \\ &= 0.5 \cdot 49896.58 / \sin(45.0) \\ &= 35282.21 \text{ Kgf} \end{aligned}$$

Combined maximum stress [S2]:

$$\begin{aligned} &= T2 / (\text{Wtp} \cdot \text{Ttp}) + M2 / (\text{Ttp} \cdot \text{Wtp}^2 / 6) \\ &= 35282 / (300.0 \cdot 66.0) + 4836.05 / (66.0 \cdot 300.0^2 / 6) \\ &= 65.38 \text{ N./mm}^2 \end{aligned}$$

Ring Allowable Stress [Sa]:

$$\begin{aligned} &= 2/3 \cdot \text{Ylug} \\ &= 2/3 \cdot 248 \\ &= 165.12 \text{ N./mm}^2 \end{aligned}$$

Bottom Support Plate Bending Stress (Point Load) [Spl]:

$$\begin{aligned} &= (\text{Flugemp}/2 \cdot \text{Dgp}/2) / ((\text{Wpl}/6) \cdot \text{Tpl}^2) \text{ (Uplift Condition)} \\ &= (18141/2 \cdot 300.0/2) / ((500.0/6) \cdot 62.0^2) \\ &= 41.65 \text{ N./mm}^2 \end{aligned}$$

Bottom Plate Required Thickness (Point Load) 31.1403 mm.

Bottom Plate required thickness for Uplift per ADM S 3/4 [trUpl]:

$$\begin{aligned} &= 0.72 \cdot \sqrt{ (\text{Flugemp} \cdot \text{Dgp} / ((\text{Wpl} - \text{dt}) \cdot \text{Spa})) } \\ &= 0.72 \cdot \sqrt{ (18141.5 \cdot 300.0 / ((500.0 - 51.17) \cdot 165.119)) } \\ &= 19.32 \text{ mm.} \end{aligned}$$

Bearing Area [Ba]:

$$\begin{aligned} &= \text{Lpl} \cdot \text{Wfb} \\ &= 500.0 \cdot 300.0 \\ &= 1500.00 \text{ cm}^2 \end{aligned}$$

Bending Stress in bottom Plate (Unif. Load) Per Bednar p.156 [Spl2]:

$$= \beta_1 \cdot \text{Flug}/\text{Ba} \cdot \text{Wfb}^2 / \text{Tpl}^2 \text{ per Roark \& Young 5th Ed.}$$

$$= 0.691 \cdot 93472.9/1500.0 \cdot 300.0^2/62.0^2$$

$$= 98.82 \text{ N./mm}^2$$

Bottom Plate Required Thickness (Uniform Load) 47.9628 mm.

Bottom Plate Required Thickness based on ADM S 3/4 [trAD]:

$$= 0.71 \cdot D_{gp} \cdot \sqrt{(Flug / (L_{pl} \cdot W_{fb}) / Spa)}$$

$$= 0.71 \cdot 300.0 \cdot \sqrt{(93473 / (500.0 \cdot 300.0) / 165.119)}$$

$$= 40.977 \text{ mm.}$$

If using the AD Code recommendations, the force bearing width (Wfb) must be greater than or equal to 1/3 of the bottom plate radial width (Wpl) plus the pad thickness (Padthk), if there is a pad.

Bottom Support Plate Allowable Stress [Spa]:

$$= 2/3 \cdot Y_{lug}$$

$$= 2/3 \cdot 248$$

$$= 165.12 \text{ N./mm}^2$$

Gusset Plate Axial Stress (Force / Gusset Plate Area) [Sgp]:

$$= (Flug/2) / (W_{gp} \cdot T_{gp})$$

$$= (93473/2) / (275.0 \cdot 40.0)$$

$$= 41.67 \text{ N./mm}^2$$

Check required Gusset Plate Thickness:

Required Thickness of Gussets per AISC: 19.6749 mm.

Gusset Plate Allowable Stress [Sga]:

$$= (1 - (K_{lr})^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (K_{lr}) / (8 \cdot C_c) - (K_{lr})^3 / (8 \cdot C_c^3))$$

$$= (1 - (50.0)^2 / (2 \cdot 122.18^2)) 248 /$$

$$(5/3 + 3 \cdot (50.0) / (8 \cdot 122.18) - (50.0)^3 / (8 \cdot 122.18^3))$$

$$= 125.27 \text{ N./mm}^2$$

Dimension [Gtp]:

$$= 2 \cdot W_{gp} - W_{pl}$$

$$= 2 \cdot 275.0 - 500.0$$

$$= 50.0 \text{ mm.}$$

Angle [αG]:

$$= \pi / 2 - \text{atan}((W_{pl} - G_{tp}) / H_{gp})$$

$$= \pi / 2 - \text{atan}((500.0 - 50.0) / 500.0)$$

$$= 48.0 \text{ deg}$$

Maximum Compressive Gusset Plate Stress per Bednar [SgpB]:

$$= Flug(3 \cdot D_{lug} - W_{pl}) / (T_{gp} \cdot W_{pl}^2 \cdot (\sin(\alpha G))^2)$$

$$= 93473(3 \cdot 300.0 - 500.0) / (40.0 \cdot 500.0^2 \cdot (\sin(48.01))^2)$$

$$= 66.37 \text{ N./mm}^2$$

Gusset Plate Allowable Compressive Stress [SgaB]:

$$= 18000 / (1 + (1/18000) \cdot (H_{gp} / \sin(\alpha G) / (0.289 \cdot T_{gp}))^2)$$

$$= 18000 / (1 + (1/18000) \cdot (500.0 / \sin(48.01) / (0.289 \cdot 40.0))^2)$$

$$= 104.46 \text{ N./mm}^2$$

Determine the required bolt area due to Uplift [Abr]:

$$= Flug_{emp} / S_{bolt}$$

$$= 18141.469 / 150.0$$

$$= 11.86 \text{ cm}^2$$

The Area available in a single bolt : 13.430 cm² [OK]

Review notes about nozzle loadings and supports in the Warnings report.

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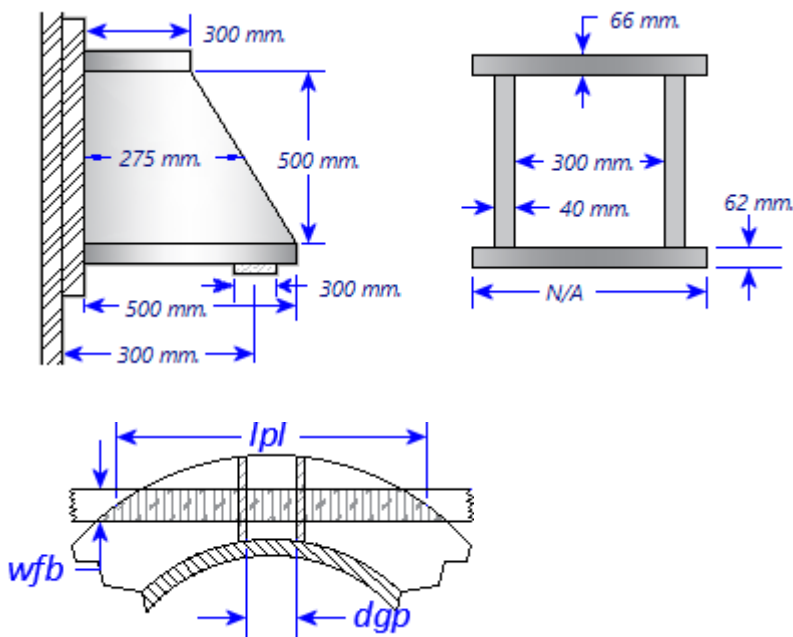
Support Lug Calculations: Test Case

Input Data: Support Lugs

Type of Geometry : Gussets with Continuous Top Ring

Number of Support Lugs	Nlug	4
Distance from Vessel OD to Lug Contact Point	Dlug	300.0000 mm.
Lug Support Force Bearing Width	Wfb	300.0000 mm.
Lug Material		SA-387 11
Lug Yield Stress		309.99 N./mm ²
Radial Width of bottom Support Lug Plate	Wpl	500.0000 mm.
Effective Force Bearing Length	Lpl	500.0000 mm.
Thickness of bottom Support Lug Plate	Tpl	62.0000 mm.
Distance between Gussets	Dgp	300.0000 mm.
Mean Width of Gusset Plate	Wgp	275.0000 mm.
Height of Gusset Plate	Hgp	500.0000 mm.
Thickness of Gusset Plate	Tgp	40.0000 mm.
Radial Width of Reinforcing Ring	Wtp	300.0000 mm.
Thickness of Top Plate	Ttp	66.0000 mm.

Bolt Material	SA-193 B7
Bolt Allowable Stress at Design Temperature	150.00 N./mm ²
Thread Series	TEMA Metric
Bolt Diameter	48.00 mm.



Results for Support Lugs:

Description: Support Lugs

Overturning Moment at Support Lug	6865.6 Kg-m.
Weight Load at the top of one Lug	65750.9 Kgf

Force on one Lug, Test Condition [Flug]:

$$= (W/Nlug + Mlug/(Rlug \cdot Nlug/2))$$

$$= (263004/4 + 6866/(1525.0 \cdot 4/2))$$

$$= 68001.87 \text{ Kgf}$$

Force on one lug in the Empty Condition [Flugemp]:

$$= (-Wemp/Nlug + Mlugemp/(Rlug \cdot Nlug/2))$$

$$= (-126794/4 + 152015/(1525.0 \cdot 4/2))$$

$$= 18141.47 \text{ Kgf}$$



Ring Girder Calculations Per Bednar pp.157-159:

Force P acting in the plane of the ring [P]:

$$\begin{aligned} &= \text{Flug} \cdot \text{Dlug} / (\text{Hgp} + \text{Tpl}) \\ &= 68002 \cdot 300.0 / (500.0 + 62.0) \\ &= 36299.93 \text{ Kgf} \end{aligned}$$

Angle [α]:

$$\begin{aligned} &= \pi / \text{Nlug} \\ &= \pi / 4 \\ &= 45.0 \text{ deg} \end{aligned}$$

Maximum bending moment (at the load points) [M1]:

$$\begin{aligned} &= 0.5 \cdot P(\text{OD}/2 + \text{Wtp}/2) \cdot (1/\alpha - \cos(\alpha)/\sin(\alpha)) \\ &= 0.5 \cdot 36299.93 \cdot 1375.0(1/0.79 - \cos(45.0)/\sin(45.0)) \\ &= 6819.15 \text{ Kg-m.} \end{aligned}$$

Tangential (axial) thrust induced in the ring [T1]:

$$\begin{aligned} &(\text{Top ring in compression, bottom ring in tension}) \\ &= 0.5 \cdot P \cdot \cos(\alpha) / \sin(\alpha) \\ &= 0.5 \cdot 36299.93 \cdot \cos(45.0) / \sin(45.0) \\ &= 18149.96 \text{ Kgf} \end{aligned}$$

Resulting combined stress at the load point [S1]:

$$\begin{aligned} &= T1 / (\text{Ttp} \cdot \text{Wtp}) + M1 / (\text{Ttp} \cdot \text{Wtp}^2 / 6) \\ &= 18150 / (66.0 \cdot 300.0) + 6819.15 / (66.0 \cdot 300.0^2 / 6) \\ &= 76.54 \text{ N./mm}^2 \end{aligned}$$

Bending moment at the midpoints between the loads [M2]:

$$\begin{aligned} &= 0.5 \cdot P(\text{OD}/2 + \text{Wtp}/2) \cdot (1/\sin(\alpha) - 1/\alpha) \\ &= 0.5 \cdot 36299.93 \cdot 1375.0(1/\sin(45.0) - 1/0.79) \\ &= 3518.25 \text{ Kg-m.} \end{aligned}$$

Axial thrust [T2]:

$$\begin{aligned} &= 0.5 \cdot P / \sin(\alpha) \\ &= 0.5 \cdot 36299.93 / \sin(45.0) \\ &= 25667.93 \text{ Kgf} \end{aligned}$$

Combined maximum stress [S2]:

$$\begin{aligned} &= T2 / (\text{Wtp} \cdot \text{Ttp}) + M2 / (\text{Ttp} \cdot \text{Wtp}^2 / 6) \\ &= 25668 / (300.0 \cdot 66.0) + 3518.25 / (66.0 \cdot 300.0^2 / 6) \\ &= 47.56 \text{ N./mm}^2 \end{aligned}$$

Ring Allowable Stress [Sa]:

$$\begin{aligned} &= 2/3 \cdot \text{Ylug} \\ &= 2/3 \cdot 310 \\ &= 206.66 \text{ N./mm}^2 \end{aligned}$$

Bottom Support Plate Bending Stress (Point Load) [Spl]:

$$\begin{aligned} &= (\text{Flugemp}/2 \cdot \text{Dgp}/2) / ((\text{Wpl}/6) \cdot \text{Tpl}^2) \text{ (Uplift Condition)} \\ &= (18141/2 \cdot 300.0/2) / ((500.0/6) \cdot 62.0^2) \\ &= 41.65 \text{ N./mm}^2 \end{aligned}$$

Bottom Plate Required Thickness (Point Load) 27.8350 mm.

Bottom Plate required thickness for Uplift per ADM S 3/4 [trUpl]:

$$\begin{aligned} &= 0.72 \cdot \sqrt{ (\text{Flugemp} \cdot \text{Dgp} / ((\text{Wpl} - \text{dt}) \cdot \text{Spa})) } \\ &= 0.72 \cdot \sqrt{ (18141.5 \cdot 300.0 / ((500.0 - 51.17) \cdot 206.663)) } \\ &= 17.27 \text{ mm.} \end{aligned}$$

Bearing Area [Ba]:

$$\begin{aligned} &= \text{Lpl} \cdot \text{Wfb} \\ &= 500.0 \cdot 300.0 \\ &= 1500.00 \text{ cm}^2 \end{aligned}$$

Bending Stress in bottom Plate (Unif. Load) Per Bednar p.156 [Spl2]:

$$= \beta_1 \cdot \text{Flug}/\text{Ba} \cdot \text{Wfb}^2 / \text{Tpl}^2 \text{ per Roark \& Young 5th Ed.}$$

$$= 0.691 \cdot 68001.9/1500.0 \cdot 300.0^2/62.0^2$$

$$= 71.89 \text{ N./mm}^2$$

Bottom Plate Required Thickness (Uniform Load) 36.5670 mm.

Bottom Plate Required Thickness based on ADM S 3/4 [trAD]:

$$= 0.71 \cdot D_{gp} \cdot \sqrt{(Flug / (L_{pl} \cdot W_{fb}) / Spa)}$$

$$= 0.71 \cdot 300.0 \cdot \sqrt{(68002 / (500.0 \cdot 300.0) / 206.663)}$$

$$= 31.241 \text{ mm.}$$

If using the AD Code recommendations, the force bearing width (Wfb) must be greater than or equal to 1/3 of the bottom plate radial width (Wpl) plus the pad thickness (Padthk), if there is a pad.

Bottom Support Plate Allowable Stress [Spa]:

$$= 2/3 \cdot Y_{lug}$$

$$= 2/3 \cdot 310$$

$$= 206.66 \text{ N./mm}^2$$

Gusset Plate Axial Stress (Force / Gusset Plate Area) [Sgp]:

$$= (Flug/2) / (W_{gp} \cdot T_{gp})$$

$$= (68002/2) / (275.0 \cdot 40.0)$$

$$= 30.31 \text{ N./mm}^2$$

Check required Gusset Plate Thickness:

Required Thickness of Gussets per AISC: 16.6490 mm.

Gusset Plate Allowable Stress [Sga]:

$$= (1 - (K_{lr})^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (K_{lr}) / (8 \cdot C_c) - (K_{lr})^3 / (8 \cdot C_c^3))$$

$$= (1 - (50.0)^2 / (2 \cdot 114.0^2)) 310 /$$

$$(5/3 + 3 \cdot (50.0) / (8 \cdot 114.0) - (50.0)^3 / (8 \cdot 114.0^3))$$

$$= 153.89 \text{ N./mm}^2$$

Dimension [Gtp]:

$$= 2 \cdot W_{gp} - W_{pl}$$

$$= 2 \cdot 275.0 - 500.0$$

$$= 50.0 \text{ mm.}$$

Angle [αG]:

$$= \pi / 2 - \text{atan}((W_{pl} - G_{tp}) / H_{gp})$$

$$= \pi / 2 - \text{atan}((500.0 - 50.0) / 500.0)$$

$$= 48.0 \text{ deg}$$

Maximum Compressive Gusset Plate Stress per Bednar [SgpB]:

$$= Flug(3 \cdot D_{lug} - W_{pl}) / (T_{gp} \cdot W_{pl}^2 \cdot (\sin(\alpha G))^2)$$

$$= 68002(3 \cdot 300.0 - 500.0) / (40.0 \cdot 500.0^2 \cdot (\sin(48.01))^2)$$

$$= 48.28 \text{ N./mm}^2$$

Gusset Plate Allowable Compressive Stress [SgaB]:

$$= 18000 / (1 + (1/18000) \cdot (H_{gp} / \sin(\alpha G) / (0.289 \cdot T_{gp}))^2)$$

$$= 18000 / (1 + (1/18000) \cdot (500.0 / \sin(48.01) / (0.289 \cdot 40.0))^2)$$

$$= 104.46 \text{ N./mm}^2$$

Determine the required bolt area due to Uplift [Abr]:

$$= Flug_{emp} / S_{bolt}$$

$$= 18141.469 / 150.0$$

$$= 11.86 \text{ cm}^2$$

The Area available in a single bolt : 13.430 cm² [OK]

Review notes about nozzle loadings and supports in the Warnings report.

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Conical Calculations per ASME VIII-2, 4.3.11

Input Values:

cone transition

Design Internal Pressure, Small End	51.216	bars
Design Internal Pressure, Large End	51.098	bars
Temperature for Internal Pressure	286.0	°C
Design External Pressure	1.034	bars
Temperature for External Pressure	286.0	°C

CONE Properties:

Cone Material		SA-387 11	
Cone Allowable Stress at Temperature	Ss	165.12	N./mm ²
Cone Allowable Stress At Ambient		207.00	N./mm ²
Longitudinal Joint Efficiency of Cone	Ec,E2	1.00	
Circumferential Joint Efficiency of Cone		1.00	
Actual Thickness of Cone	Tc	48.0000	mm.
Corrosion Allowance for Cone	Cac	1.5000	mm.
Diameter Basis for Cone and Cylinders		ID	
Diameter of Small End of Cone	Ds	2370.000	mm.
Diameter of Large End of Cone	Dl	2768.000	mm.
Half Apex Angle for Cone		12.00	degrees
Axial Length of Cone	Lc	1200.000	mm.

SMALL-END Properties:

Attached Small End Material Specification		SA-387 11	
Operating Allowable Stress	Ss	165.12	N./mm ²
Ambient Allowable Stress		207.00	N./mm ²
Longitudinal Joint Efficiency	Es,E1	1.00	
Actual Thickness	Ts	40.0000	mm.
Corrosion Allowance	Cas	1.5000	mm.
Axial Length of Attached Element	Ls	7000.000	mm.

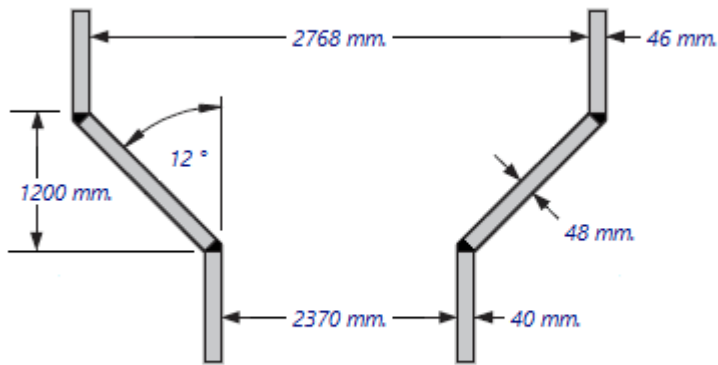
LARGE-END Properties:

Attached Large End Material Specification		SA-387 11	
Operating Allowable Stress	Sl	165.12	N./mm ²
Ambient Allowable Stress		207.00	N./mm ²
Longitudinal Joint Efficiency	El,E1	1.00	
Actual Thickness	Tl	46.0000	mm.
Corrosion Allowance	Cal	1.5000	mm.
Axial Length of Attached Element	Ll	7468.667	mm.

Type of Reinforcement at Large End of Cone: None

Type of Reinforcement at Small End of Cone: None

Warning: rtk -1.500mm., must be > Tl, 44.500mm., see 4.3.12



Cone Geometry

Calculated Values:

Elastic Modulus Data from ASME Section II Part D at 286 °C

Elastic Modulus of Cone Material	0.187E+09 KPa. at 286 °C
Elastic Modulus of Small Cylinder Material	0.187E+09 KPa. at 286 °C
Elastic Modulus of Large Cylinder Material	0.187E+09 KPa. at 286 °C
Elastic Modulus of Large End Reinforcement	0.187E+09 KPa. at 286 °C
Elastic Modulus of Small End Reinforcement	0.187E+09 KPa. at 286 °C

Elastic Modulus Data from ASME Section II Part D at 286 °C

Elastic Modulus of Cone Material	0.187E+09 KPa. at 286 °C
Elastic Modulus of Small Cylinder Material	0.187E+09 KPa. at 286 °C
Elastic Modulus of Large Cylinder Material	0.187E+09 KPa. at 286 °C
Elastic Modulus of Large End Reinforcement	0.187E+09 KPa. at 286 °C
Elastic Modulus of Small End Reinforcement	0.187E+09 KPa. at 286 °C

Forces and Moments on the ends of the Cone:

Axial Force on Small End of Cone	-51696.78 Kg
Axial Force on Large End of Cone	-44751.18 Kg
Moment on Small End of Cone	101545.74 Kg-m.
Moment on Large End of Cone	81813.81 Kg-m.

The moments above are computed from the Wind & Seismic loads as well as any user defined moments and include effects due to eccentric masses. The vertical forces include the weight of the elements and details above and below the cone as well as any user defined vertical forces. Vertical forces also include loads cause from vertical accelerations due to earthquake loading.

Maximum Centroid Reinforcement Distance Large End	25.4000 mm.
Maximum Centroid Reinforcement Distance Small End	25.4000 mm.

Note: No ring was found close enough to the large end to be considered.

Note: No ring was found close enough to the small end to be considered.

Junction Calculations per 4.3.11:

Calculations for the Large End Cone/Shell Junction - Without Knuckle

Fhe per 4.1.3 or 4.3.11.2 Step 6 [Fhe]:
 $= 0.4 \cdot E(tc/D)$
 $= 0.4 \cdot 0.18730E+09(46.5/2860.0)$
 $= 1218.2 \text{ N./mm}^2$

Factor of Safety [FS]:

Since $F_{ic} [247.68] \geq F_y [247.68]$, FS = 1.667

Allowable Stress per 4.3.11 Step 6 [Fha]:

= F_{ic}/FS
= $247.6785/1.667$
= 148.577 N./mm^2

Large End under Internal Pressure:

Cone Length along Cone surface per 4.3.46 [ConeMinLen]:

= $2 \cdot \sqrt{(RL \cdot Tc/\cos(\alpha)) + 1.4 \cdot \sqrt{(Rs \cdot tc/\cos(\alpha))}}$
= $2 \cdot \sqrt{(1385.534 \cdot 46.5/\cos(12.0)) + 1.4 \cdot \sqrt{(1186.534 \cdot 46.5/\cos(12.0))}}$
= 845.786 mm.

Results for the Large End Cylinder Without a Knuckle

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	81.410	247.678	Pass
Sigma sm-	75.431	247.678	Pass
Sigma thetam+	95.718	247.678	Pass
Sigma thetam-	100.120	247.678	Pass
Sigma sm+ + Sigma	-26.142	495.357	Pass
Sigma sm+ - Sigma	188.961	495.357	Pass
Sigma sm- + Sigma	-23.901	495.357	Pass
Sigma sm- - Sigma	174.763	495.357	Pass
Sigma thetam+ +	63.453	495.357	Pass
Sigma thetam+ -	127.984	495.357	Pass
Sigma thetam- +	70.320	495.357	Pass
Sigma thetam- -	129.919	495.357	Pass

Results for the Large End Cone Without a Knuckle

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	77.906	247.678	Pass
Sigma sm-	72.186	247.678	Pass
Sigma thetam+	93.616	247.678	Pass
Sigma thetam-	98.083	247.678	Pass
Sigma sm+ + Sigma	-20.592	495.357	Pass
Sigma sm+ - Sigma	176.405	495.357	Pass
Sigma sm- + Sigma	-18.785	495.357	Pass
Sigma sm- - Sigma	163.157	495.357	Pass
Sigma thetam+ +	64.066	495.357	Pass
Sigma thetam+ -	123.165	495.357	Pass
Sigma thetam- +	70.792	495.357	Pass
Sigma thetam- -	125.375	495.357	Pass

Cone Large end without a Knuckle Cylinder Section:

Net Axial Force ML +ve [XL+]:

= $FL / (2 \cdot \pi \cdot RL) + ML / (\pi \cdot RL^2)$
= $-44751.176 / (2 \cdot 3.141 \cdot 1385.534) + 81813.812 / (3.142 \cdot 1385.534^2)$
= 8.4249 Kg/mm.

Net Axial Force ML -ve [XL-]:

= $FL / (2 \cdot \pi \cdot RL) - ML / (\pi \cdot RL^2)$
= $-44751.176 / (2 \cdot 3.142 \cdot 1385.534) - 81813.812 / (3.141 \cdot 1385.534^2)$
= -18.706 Kg/mm.

Thickness Ratio [n]:

= $tc/tL = 46.5/44.5 = 1.0449$

Curve Fitting Geometry [H and B]:

$H = \sqrt{0.5 \cdot D/tL} = \sqrt{0.5 \cdot 2771.067/44.5} = 5.5799$
 $B = \tan(12.0) = 0.2126$

This analysis is preformed per Table 4.3.1 for the Large End:

Using Data from Table 4.3.3 and 4.3.4 knowing n

Table 4.3.3 - n: $1.0 < (n=1.0449) < 1.25$

Use n to construct a table: C1, C2, C3..... by interpolation

Table 4.3.3 [MsN1]:

C1 = -3.074157 C2 = 3.654484 C3 = 0.796858 C4 = -0.224421
C5 = -0.080729 C6 = 0.043703 C7 = 0.026073 C8 = -0.015457
C9 = 0.001903 C10 = -0.006971 C11 = 0.000000

Compute using Table 4.3.3 [MsN1]:

$$X = C1 + C2 \cdot \ln(H) + C3 \cdot \ln(B) + C4 \cdot (\ln(H))^2 + C5 \cdot (\ln(B))^2 + C6 \cdot \ln(H) \cdot \ln(B) + C7 \cdot (\ln(H))^3 + C8 \cdot (\ln(B))^3 + C9 \cdot \ln(H) \cdot (\ln(B))^2 + C10 \cdot (\ln(H))^2 \cdot \ln(B) + C1 \cdot C2 \cdot \ln(5.58) + C3 \cdot \ln(0.213) + C4 \cdot (\ln(5.58))^2 + C5 \cdot (\ln(0.213))^2 + C6 \cdot \ln(5.58) \cdot \ln(0.213) + C7 \cdot (\ln(5.58))^3 + C8 \cdot (\ln(0.213))^3 + C9 \cdot \ln(5.58) \cdot (\ln(0.213))^2 + C10 \cdot (\ln(5.58))^2 \cdot \ln(0.213)$$

$$X = 1.231, MsN1 = -\exp(X) = -3.4246$$

Table 4.3.3 [Qn1]:

C1 = -1.970822 C2 = 2.389375 C3 = 0.599762 C4 = -0.111963
C5 = -0.108437 C6 = 0.140297 C7 = 0.011277 C8 = -0.017818
C9 = 0.008270 C10 = -0.019738 C11 = 0.000000

Compute using Table 4.3.3 [Qn1]:

$$X = C1 + C2 \cdot \ln(H) + C3 \cdot \ln(B) + C4 \cdot (\ln(H))^2 + C5 \cdot (\ln(B))^2 + C6 \cdot \ln(H) \cdot \ln(B) + C7 \cdot (\ln(H))^3 + C8 \cdot (\ln(B))^3 + C9 \cdot \ln(H) \cdot (\ln(B))^2 + C10 \cdot (\ln(H))^2 \cdot \ln(B) + C1 \cdot C2 \cdot \ln(5.58) + C3 \cdot \ln(0.213) + C4 \cdot (\ln(5.58))^2 + C5 \cdot (\ln(0.213))^2 + C6 \cdot \ln(5.58) \cdot \ln(0.213) + C7 \cdot (\ln(5.58))^3 + C8 \cdot (\ln(0.213))^3 + C9 \cdot \ln(5.58) \cdot (\ln(0.213))^2 + C10 \cdot (\ln(5.58))^2 \cdot \ln(0.213)$$

$$X = 0.4916, Qn = -\exp(X) = -1.635$$

Table 4.3.4 MsN2:

C1 = -5.702604 C2 = 0.004364 C3 = 0.475511 C4 = -0.213129
C5 = 2.238168 C6 = 0.000028 C7 = 0.002659 C8 = -0.001738
C9 = -0.213621 C10 = 0.000094 C11 = -0.106215

Compute using Table 4.3.4 [MsN2]:

$$X1 = C1 + C3 \cdot \ln(H^2) + C5 \cdot \ln(A) + C7 \cdot (\ln(H^2))^2 + C9 \cdot (\ln(A))^2 + C11 \cdot \ln(H^2) \cdot \ln(A) + C1 + C3 \cdot \ln(5.58^2) + C5 \cdot \ln(0.209) + C7 \cdot (\ln(5.58^2))^2 + C9 \cdot (\ln(0.209))^2 + C11 \cdot \ln(5.58^2) \cdot \ln(0.209) = -0.7011$$

$$X2 = 1 + C2 \cdot \ln(H^2) + C4 \cdot \ln(AR) + C6 \cdot (\ln(H^2))^2 + C8 \cdot (\ln(AR))^2 + C10 \cdot \ln(H^2) \cdot \ln(AR) + 1 + C2 \cdot \ln(5.58) + C4 \cdot \ln(0.209) + C6 \cdot (\ln(5.58^2))^2 + C8 \cdot (\ln(0.209))^2 + C10 \cdot \ln(5.58^2) \cdot \ln(0.21) = 0.4758$$

$$MsN2 = \exp(X1/X2) = \exp(-0.701/0.476) = -0.2291$$

Table 4.3.4 QN2:

C1 = -4.837636 C2 = 0.004311 C3 = -0.012377 C4 = -0.197069
C5 = 2.013516 C6 = -0.000024 C7 = -0.000158 C8 = -0.003860
C9 = -0.206336 C10 = -0.000508 C11 = 0.003402

Compute using Table 4.3.4 [Qn2]:

$$X3 = C1 + C3 \cdot \ln(H^2) + C5 \cdot \ln(A) + C7 \cdot (\ln(H^2))^2 + C9 \cdot (\ln(A))^2 + C11 \cdot \ln(H^2) \cdot \ln(A) + C1 + C3 \cdot \ln(5.58^2) + C5 \cdot \ln(0.209) + C7 \cdot (\ln(5.58^2))^2 + C9 \cdot (\ln(0.209))^2 + C11 \cdot \ln(5.58^2) \cdot \ln(0.209) = -0.7011$$

$$X4 = 1 + C2 \cdot \ln(H^2) + C4 \cdot \ln(AR) + C6 \cdot (\ln(H^2))^2 + C8 \cdot (\ln(AR))^2 + C10 \cdot \ln(H^2) \cdot \ln(AR) + 1 + C2 \cdot \ln(5.58) + C4 \cdot \ln(0.209) + C6 \cdot (\ln(5.58^2))^2 + C8 \cdot (\ln(0.209))^2 + C10 \cdot \ln(5.58^2) \cdot \ln(0.209) = 0.4758$$

$$Qn2 = \exp(X3/X4) = \exp(-1.124/0.497) = -0.1041$$

Stresses for the Large End Cylinder per Table 4.3.1

Stress Resultant Value [Msp]:

$$= P \cdot tL^2 \cdot MsN1 = 51.098 \cdot 44.5^2 \cdot -3.425 = -3533.6631 \text{ Kgf}$$

Compute with XL+ [Msx+]:

$$= XL \cdot tL \cdot MsN2 = 8.425 \cdot 44.5 \cdot -0.229 = -85.8965 \text{ Kgf}$$

Compute wiht XL- [Msx-]:

$$= XL \cdot tL \cdot MsN2 = -18.706 \cdot 44.5 \cdot -0.229 = 190.7172 \text{ Kgf}$$

Compute with XL+ [Ms+]:
= MsP + Msx = -3533.663 + -85.896 = -3619.5596 Kgf

Compute with XL- [Ms-]:
= MsP + Msx = -3533.663 + 190.717 = -3342.946 Kgf

Compute [Qp]:
= P•tL•Qn1 = 51.098•44.5•-1.635 = -37.9114 Kgf/mm.

Compute with XL+ [Qx+]:
= XL•QN2 = 8.425•-0.002 = -0.877 Kgf/mm.

Compute with XL- [Qx-]:
= XL•QN2 = -18.706•-0.002 = 1.9472 Kgf/mm.

Compute with XL+ [Q+]:
= Qp + Qx = -37.911 + -0.877 = -38.7884 Kgf/mm.

Compute with XL- [Q-]:
= Qp + Qx = -37.911 + 1.947 = -35.9642 Kgf/mm.

Compute [βcy]:
= $3 \cdot (1 - v^2) / (RL^2 \cdot tL^2) \cdot 0.25$
= $(3 \cdot (1 - 0.3^2) / (1385.534^2 \cdot 44.5^2)) \cdot 0.25 = 0.0052 \text{ 1/mm.}$

Compute with XL+ [Ns+]:
= P • RL/2 + XL = 51.098•1385.534/2+8.425 = 369.4087 Kgf/mm.

Compute with XL- [Ns-]:
= P • RL/2 + XL = 51.098•1385.534/2+-18.706 = 342.2778 Kgf/mm.

Compute with XL+ [Ntheta+]:
= P • RL + 2 • βcy • RL • (-Ms • βcy + Q)
= 51.098 • 1385.534 + 2 • 0.0052 • 1385.534 •
(-3619.56 • 0.005 + -38.788) = 434.3365 Kgf/mm.

Compute with XL- [Ntheta-]:
= P • RL + 2 • βcy • RL • (-Ms • βcy + Q)
= 51.098 • 1385.534 + 2 • 0.0052 • 1385.534 •
(-3342.946 • 0.005 + -35.964) = 454.3077 Kgf/mm.

Note: Kpc has been set to 1.0

σSm with XL+ [σsm+]:
= Ns / tL
= 369.409/44.5
= 81.410 N./mm²

σSm with XL- [σsm-]:
= Ns / tL
= 342.278/44.5
= 75.431 N./mm²

σSb with XL+ [σsb+]:
= 6 • Ms / (tL² • Kpc)
= 6 • -3619.56/(44.5² • 1.0)
= -107.551 N./mm²

σSb with XL- [σsb-]:
= 6 • Ms / (tL² • Kpc)
= 6 • -3342.946/(44.5² • 1.0)
= -99.332 N./mm²

σThetam with XL+ [σthetam+]:
= Ntheta / tL
= 434.337/44.5 N./mm²
= 95.718 N./mm²

σ_{Thetam} with XL- [σ_{thetam}]:

$$= N_{\text{theta}} / tL$$

$$= 454.308/44.5$$

$$= 100.120 \text{ N./mm}^2$$

σ_{Thetab} with XL+ [σ_{thetab}]:

$$= 6 \cdot v \cdot Ms / (tL^2 \cdot K_{pc})$$

$$= 6 \cdot 0.3 \cdot -3619.56 / (44.5^2 \cdot 1.0)$$

$$= -32.265 \text{ N./mm}^2$$

σ_{Thetab} with XL- [σ_{thetab}]:

$$= 6 \cdot v \cdot Ms / (tL^2 \cdot K_{pc})$$

$$= 6 \cdot 0.3 \cdot -3342.946 / (44.5^2 \cdot 1.0)$$

$$= -29.800 \text{ N./mm}^2$$

For the Large End Cone Junction per Table 4.3.1

Compute [McsP]:

$$= M_{sp} = M_{sp} = -3533.6631 \text{ Kgf}$$

McsX with XL+ [McsX+]:

$$= M_{sX} = -85.8965 \text{ Kgf}$$

McsX with XL- [McsX-]:

$$= M_{sX} = 190.7172 \text{ Kgf}$$

Mcs with XL+ [Mcs+]:

$$= M_{csP} + M_{sX} = -3533.663 + -85.896$$

$$= -3619.560 \text{ Kgf}$$

Mcs with XL- [Mcs-]:

$$= M_{csP} + M_{sX} = -3533.663 + 190.717$$

$$= -3342.946 \text{ Kgf}$$

Compute with XL+ [McsX+]:

$$= M_{sX} = -85.8965 \text{ Kgf}$$

Compute with XL- [McsX-]:

$$= M_{sX} = 190.7172 \text{ Kgf}$$

Compute with XL+ [Qc+]:

$$= Q \cdot \cos(a) + N_s \cdot \sin(a)$$

$$= -38.788 \cdot \cos(12.0) + 369.409 \cdot \sin(12.0)$$

$$= 38.864 \text{ Kgf/mm.}$$

Compute with XL- [Qc-]:

$$= Q \cdot \cos(a) - N_s \cdot \sin(a)$$

$$= -35.964 \cdot \cos(12.0) + 342.278 \cdot \sin(12.0)$$

$$= 35.985 \text{ Kgf/mm.}$$

Compute [Rc]:

$$= RL/\cos(a) = 1385.534/\cos(12.0) = 1416.4872 \text{ mm.}$$

Compute [β -co]:

$$= (3 \cdot (1 - v^2) / (Rc^2 \cdot tC^2))^{1/4}$$

$$= (3 \cdot (1 - 0.3^2) / (1416.487^2 \cdot 46.5^2))^{1/4}$$

$$= 0.005 \text{ 1/mm.}$$

Compute with XL +ve [Ncs+]:

$$= N_s \cdot \cos(a) - Q \cdot \sin(a)$$

$$= 369.409 \cdot \cos(12.0) - -38.788 \cdot \sin(12.0)$$

$$= 369.401 \text{ Kgf/mm.}$$

Compute with XL -ve [Ncs-]:

$$= N_s \cdot \cos(a) - Q \cdot \sin(a)$$

$$= 342.278 \cdot \cos(12.0) - -35.964 \cdot \sin(12.0)$$

$$= 342.276 \text{ Kgf/mm.}$$

Compute with XL +ve [NcTheta+]:

$$= P \cdot RL / \cos(a) + 2 \cdot \beta_{co} \cdot Rc(-Mcs \cdot \beta_{co} - Qc)$$

$$= 51.098 \cdot 1385.534 / \cos(12.0) + 2 \cdot 0.005 \cdot 1416.487 \cdot (-3619.56 \cdot 0.005 - 38.864)$$

$$= 443.888 \text{ Kgf/mm.}$$

Compute with XL -ve [NcTheta-]:

$$= P \cdot RL / \cos(A) + 2 \cdot \beta_{co} \cdot Rc \cdot (-Mcs \cdot \beta_{co} - Qc)$$

$$= 51.098 \cdot 1385.534 / \cos(12.0) + 2 \cdot 0.005 \cdot 1416.487 \cdot (-3342.946 \cdot 0.005 - 35.985)$$

$$= 465.071 \text{ Kgf/mm.}$$

Note: Kcpc = 1.0

Compute with XL +ve [σ_{sm+}]:

$$= Ncs/tC = 369.401/46.5 = 77.9065 \text{ N./mm}^2$$

Compute with XL -ve [σ_{sm-}]:

$$= Ncs/tC = 342.276/46.5 = 721.8159 \text{ N./mm}^2$$

Compute with XL +ve [σ_{sb+}]:

$$= 6 \cdot Mcs / (tC^2 \cdot Kcpc)$$

$$= 6 \cdot -3619.56 / (46.5^2 \cdot 1.0)$$

$$= -98.499 \text{ N./mm}^2$$

Compute with XL -ve [σ_{sb-}]:

$$= 6 \cdot Mcs / (tC^2 \cdot Kcpc)$$

$$= 6 \cdot -3342.946 / (46.5^2 \cdot 1.0)$$

$$= -90.971 \text{ N./mm}^2$$

Compute with XL +ve [$\sigma_{Thetam+}$]:

$$= NCTheta/tC = 443.888/46.5 = 93.62 \text{ N./mm}^2$$

Compute with XL -ve [$\sigma_{Thetam-}$]:

$$= NCTheta/tC = 465.071/46.5 = 98.08 \text{ N./mm}^2$$

Compute with XL +ve [$\sigma_{thetab+}$]:

$$= 6 \cdot v \cdot Mcs / (tC^2 \cdot Kcpc)$$

$$= 6 \cdot 0.3 \cdot -3619.56 / (46.5^2 \cdot 1.0)$$

$$= -29.550 \text{ N./mm}^2$$

Compute with XL -ve [$\sigma_{thetab-}$]:

$$= 6 \cdot v \cdot Mcs / (tC^2 \cdot Kcpc)$$

$$= 6 \cdot 0.3 \cdot -3342.946 / (46.5^2 \cdot 1.0)$$

$$= -27.291 \text{ N./mm}^2$$

Large End under External Pressure:

Cone Length along Cone surface per 4.3.46 [ConeMinLen]:

$$= 2 \cdot \sqrt{(RL \cdot Tc / \cos(a))} + 1.4 \cdot \sqrt{(Rs \cdot tc / \cos(a))}$$

$$= 2 \cdot \sqrt{(1385.534 \cdot 46.5 / \cos(12.0))} + 1.4 \cdot \sqrt{(1186.534 \cdot 46.5 / \cos(12.0))}$$

$$= 845.786 \text{ mm.}$$

Results for the Large End Cylinder Without a Knuckle

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	0.247	247.678	Pass
Sigma sm-	5.732	247.678	Pass
Sigma thetam+	-3.332	247.678	Pass
Sigma thetam-	1.070	247.678	Pass
Sigma sm+ + Sigma	-0.181	495.357	Pass
Sigma sm+ - Sigma	0.674	495.357	Pass
Sigma sm- + Sigma	2.060	495.357	Pass
Sigma sm- - Sigma	-13.525	495.357	Pass
Sigma thetam+ +	-3.460	495.357	Pass
Sigma thetam+ -	-3.203	495.357	Pass

C1 = -1.970822 C2 = 2.389375 C3 = 0.599762 C4 = -0.111963
C5 = -0.108437 C6 = 0.140297 C7 = 0.011277 C8 = -0.017818
C9 = 0.008270 C10 = -0.019738 C11 = 0.000000

Compute using Table 4.3.3 [Qn1]:

$$\begin{aligned} X &= C1+C2*\ln(H)+C3*\ln(B)+C4*(\ln(H))^2+C5*(\ln(B))^2+C6*\ln(H)*\ln(B)+ \\ &C7*(\ln(H))^3+C8*(\ln(B))^3+C9*\ln(H)*(\ln(B))^2+C10*(\ln(H))^2*\ln(B) \\ &= C1+C2*\ln(5.58)+C3*\ln(0.213)+C4*(\ln(5.58))^2+C5*(\ln(0.213))^2+C6*\ln(5.58)*\ln(0.213)+ \\ &C7*(\ln(5.58))^3+C8*(\ln(0.213))^3+C9*\ln(5.58)*(\ln(0.213))^2+C10*(\ln(5.58))^2*\ln(0.213) \\ X &= 0.4916, Qn = -\exp(X) = -1.635 \end{aligned}$$

Table 4.3.4 MsN2:

$$\begin{aligned} C1 &= -5.702604 & C2 &= 0.004364 & C3 &= 0.475511 & C4 &= -0.213129 \\ C5 &= 2.238168 & C6 &= 0.000028 & C7 &= 0.002659 & C8 &= -0.001738 \\ C9 &= -0.213621 & C10 &= 0.000094 & C11 &= -0.106215 \end{aligned}$$

Compute using Table 4.3.4 [MsN2]:

$$\begin{aligned} X1 &= C1+C3*\ln(H^2)+C5*\ln(A)+C7*(\ln(H^2))^2+C9*(\ln(A))^2+C11*\ln(H^2)*\ln(A) \\ &= C1+C3*\ln(5.58^2)+C5*\ln(0.209)+C7*(\ln(5.58^2))^2+C9*(\ln(0.209))^2+ \\ &C11*\ln(5.58^2)*\ln(0.209) = -0.7011 \\ X2 &= 1+C2*\ln(H^2)+C4*\ln(AR)+C6*(\ln(H^2))^2+C8*(\ln(AR))^2+C10*\ln(H^2)*\ln(AR) \\ &= 1+C2*\ln(5.58)+C4*\ln(0.209)+C6*(\ln(5.58^2))^2+C8*(\ln(0.209))^2+ \\ &C10*\ln(5.58^2)*\ln(0.21) = 0.4758 \\ MsN2 &= \exp(X1/X2) = \exp(-0.701/0.476) = -0.2291 \end{aligned}$$

Table 4.3.4 QN2:

$$\begin{aligned} C1 &= -4.837636 & C2 &= 0.004311 & C3 &= -0.012377 & C4 &= -0.197069 \\ C5 &= 2.013516 & C6 &= -0.000024 & C7 &= -0.000158 & C8 &= -0.003860 \\ C9 &= -0.206336 & C10 &= -0.000508 & C11 &= 0.003402 \end{aligned}$$

Compute using Table 4.3.4 [Qn2]:

$$\begin{aligned} X3 &= C1+C3*\ln(H^2)+C5*\ln(A)+C7*(\ln(H^2))^2+C9*(\ln(A))^2+C11*\ln(H^2)*\ln(A) \\ &= C1+C3*\ln(5.58^2)+C5*\ln(0.209)+C7*(\ln(5.58^2))^2+C9*(\ln(0.209))^2+ \\ &C11*\ln(5.58^2)*\ln(0.209) = -0.7011 \\ X4 &= 1+C2*\ln(H^2)+C4*\ln(AR)+C6*(\ln(H^2))^2+C8*(\ln(AR))^2+C10*\ln(H^2)*\ln(AR) \\ &= 1+C2*\ln(5.58)+C4*\ln(0.209)+C6*(\ln(5.58^2))^2+C8*(\ln(0.209))^2+ \\ &C10*\ln(5.58^2)*\ln(0.209) = 0.4758 \\ Qn2 &= \exp(X3/X4) = \exp(-1.124/0.497) = -0.1041 \end{aligned}$$

Stresses for the Large End Cylinder per Table 4.3.1

Stress Resultant Value [Msp]:

$$= P \cdot t \cdot L^2 \cdot MsN1 = -1.034 \cdot 44.5^2 \cdot -3.425 = 71.519 \text{ Kgf}$$

Compute with XL+ [Msx+]:

$$= XL \cdot t \cdot L \cdot MsN2 = 8.425 \cdot 44.5 \cdot -0.229 = -85.8965 \text{ Kgf}$$

Compute with XL- [Msx-]:

$$= XL \cdot t \cdot L \cdot MsN2 = -18.706 \cdot 44.5 \cdot -0.229 = 190.7172 \text{ Kgf}$$

Compute with XL+ [Ms+]:

$$= MsP + Msx = 71.519 + -85.896 = -14.3775 \text{ Kgf}$$

Compute with XL- [Ms-]:

$$= MsP + Msx = 71.519 + 190.717 = 262.2362 \text{ Kgf}$$

Compute [Qp]:

$$= P \cdot t \cdot L \cdot Qn1 = -1.034 \cdot 44.5 \cdot -1.635 = 0.7673 \text{ Kgf/mm.}$$

Compute with XL+ [Qx+]:

$$= XL \cdot QN2 = 8.425 \cdot -0.002 = -0.877 \text{ Kgf/mm.}$$

Compute with XL- [Qx-]:

$$= XL \cdot QN2 = -18.706 \cdot -0.002 = 1.9472 \text{ Kgf/mm.}$$

Compute with XL+ [Q+]:

$$= Qp + Qx = 0.767 + -0.877 = -0.1097 \text{ Kgf/mm.}$$

Compute with XL- [Q-]:

$$= Qp + Qx = 0.767 + 1.947 = 2.7145 \text{ Kgf/mm.}$$



Compute [β_{cy}]:

$$= 3 \cdot (1 - v^2) / (RL^2 \cdot tL^2) \cdot 0.25$$

$$= (3 \cdot (1 - 0.3^2) / (1385.534^2 \cdot 44.5^2)) \cdot 0.25 = 0.0052 \text{ 1/mm.}$$

Compute with XL+ [N_{s+}]:

$$= P \cdot RL/2 + XL = -1.034 \cdot 1385.534/2 + 8.425 = 1.1188 \text{ Kgf/mm.}$$

Compute with XL- [N_{s-}]:

$$= P \cdot RL/2 + XL = -1.034 \cdot 1385.534/2 + -18.706 = -26.012 \text{ Kgf/mm.}$$

Compute with XL+ [$N_{\theta+}$]:

$$= P \cdot RL + 2 \cdot \beta_{cy} \cdot RL \cdot (-M_s \cdot \beta_{cy} + Q)$$

$$= -1.034 \cdot 1385.534 + 2 \cdot 0.0052 \cdot 1385.534 \cdot (-14.377 \cdot 0.005 + -0.11) = -15.1178 \text{ Kgf/mm.}$$

Compute with XL- [$N_{\theta-}$]:

$$= P \cdot RL + 2 \cdot \beta_{cy} \cdot RL \cdot (-M_s \cdot \beta_{cy} + Q)$$

$$= -1.034 \cdot 1385.534 + 2 \cdot 0.0052 \cdot 1385.534 \cdot (-262.236 \cdot 0.005 + 2.714) = 4.8533 \text{ Kgf/mm.}$$

Note: Kpc has been set to 1.0

σ_{Sm} with XL+ [σ_{sm+}]:

$$= N_s / tL$$

$$= 1.119/44.5$$

$$= 0.247 \text{ N./mm}^2$$

σ_{Sm} with XL- [σ_{sm-}]:

$$= N_s / tL$$

$$= -26.012/44.5$$

$$= -5.732 \text{ N./mm}^2$$

σ_{Sb} with XL+ [σ_{sb+}]:

$$= 6 \cdot M_s / (tL^2 \cdot Kpc)$$

$$= 6 \cdot -14.377 / (44.5^2 \cdot 1.0)$$

$$= -0.427 \text{ N./mm}^2$$

σ_{Sb} with XL- [σ_{sb-}]:

$$= 6 \cdot M_s / (tL^2 \cdot Kpc)$$

$$= 6 \cdot 262.236 / (44.5^2 \cdot 1.0)$$

$$= 7.792 \text{ N./mm}^2$$

$\sigma_{\theta_{tam}}$ with XL+ [$\sigma_{\theta_{tam+}}$]:

$$= N_{\theta} / tL$$

$$= -15.118/44.5 \text{ N./mm}^2$$

$$= -3.332 \text{ N./mm}^2$$

$\sigma_{\theta_{tam}}$ with XL- [$\sigma_{\theta_{tam-}}$]:

$$= N_{\theta} / tL$$

$$= 4.853/44.5$$

$$= 1.070 \text{ N./mm}^2$$

$\sigma_{\theta_{tab}}$ with XL+ [$\sigma_{\theta_{tab+}}$]:

$$= 6 \cdot v \cdot M_s / (tL^2 \cdot Kpc)$$

$$= 6 \cdot 0.3 \cdot -14.377 / (44.5^2 \cdot 1.0)$$

$$= -0.128 \text{ N./mm}^2$$

$\sigma_{\theta_{tab}}$ with XL- [$\sigma_{\theta_{tab-}}$]:

$$= 6 \cdot v \cdot M_s / (tL^2 \cdot Kpc)$$

$$= 6 \cdot 0.3 \cdot 262.236 / (44.5^2 \cdot 1.0)$$

$$= 2.338 \text{ N./mm}^2$$

For the Large End Cone Junction per Table 4.3.1

Compute [M_{csP}]:

$$= M_{sp} = M_{sp} = 71.519 \text{ Kgf}$$

M_{csX} with XL+ [M_{csX+}]:

$$= MsX = -85.8965 \text{ Kgf}$$

$$\text{McsX with XL- [McsX-]:}$$

$$= MsX = 190.7172 \text{ Kgf}$$

$$\text{Mcs with XL+ [Mcs+]:}$$

$$= McsP + MsX = 71.519 + -85.896$$

$$= -14.377 \text{ Kgf}$$

$$\text{Mcs with XL- [Mcs-]:}$$

$$= McsP + MsX = 71.519 + 190.717$$

$$= 262.236 \text{ Kgf}$$

$$\text{Compute with XL+ [McsX+]:}$$

$$= MsX = -85.8965 \text{ Kgf}$$

$$\text{Compute with XL- [McsX-]:}$$

$$= MsX = 190.7172 \text{ Kgf}$$

$$\text{Compute with XL+ [Qc+]:}$$

$$= Q \cdot \cos(a) + Ns \cdot \sin(a)$$

$$= -0.11 \cdot \cos(12.0) + 1.119 \cdot \sin(12.0)$$

$$= 0.125 \text{ Kgf/mm.}$$

$$\text{Compute with XL- [Qc-]:}$$

$$= Q \cdot \cos(a) + Ns \cdot \sin(a)$$

$$= 2.714 \cdot \cos(12.0) + -26.012 \cdot \sin(12.0)$$

$$= -2.753 \text{ Kgf/mm.}$$

$$\text{Compute [Rc]:}$$

$$= RL/\cos(a) = 1385.534/\cos(12.0) = 1416.4872 \text{ mm.}$$

$$\text{Compute } [\beta\text{-co}]:$$

$$= (3 \cdot (1 - v^2) / (Rc^2 \cdot tC^2))^{1/4}$$

$$= (3 \cdot (1 - 0.3^2) / (1416.487^2 \cdot 46.5^2))^{1/4}$$

$$= 0.005 \text{ 1/mm.}$$

$$\text{Compute with XL +ve [Ncs+]:}$$

$$= Ns \cdot \cos(a) - Q \cdot \sin(a)$$

$$= 1.119 \cdot \cos(12.0) - -0.11 \cdot \sin(12.0)$$

$$= 1.117 \text{ Kgf/mm.}$$

$$\text{Compute with XL -ve [Ncs-]:}$$

$$= Ns \cdot \cos(a) - Q \cdot \sin(a)$$

$$= -26.012 \cdot \cos(12.0) - 2.714 \cdot \sin(12.0)$$

$$= -26.008 \text{ Kgf/mm.}$$

$$\text{Compute with XL +ve [NcTheta+]:}$$

$$= P \cdot RL/\cos(a) + 2 \cdot \beta_{co} \cdot Rc(-Mcs \cdot \beta_{co} - Qc)$$

$$= -1.034 \cdot 1385.534/\cos(12.0) + 2 \cdot 0.005 \cdot 1416.487 \cdot$$

$$(-14.377 \cdot 0.005 - 0.125)$$

$$= -15.695 \text{ Kgf/mm.}$$

$$\text{Compute with XL -ve [NcTheta-]:}$$

$$= P \cdot RL/\cos(A) + 2 \cdot \beta_{co} \cdot Rc \cdot (-Mcs \cdot \beta_{co} - Qc)$$

$$= -1.034 \cdot 1385.534/\cos(12.0) + 2 \cdot 0.005 \cdot 1416.487 \cdot$$

$$(-262.236 \cdot 0.005 - -2.753)$$

$$= 5.488 \text{ Kgf/mm.}$$

Note: $Kcpc = 1.0$

$$\text{Compute with XL +ve } [\sigma_{sm}+]:$$

$$= Ncs/tC = 1.117/46.5 = 0.2356 \text{ N./mm}^2$$

$$\text{Compute with XL -ve } [\sigma_{sm}-]:$$

$$= Ncs/tC = -26.008/46.5 = -54.8476 \text{ N./mm}^2$$

$$\text{Compute with XL +ve } [\sigma_{sb}+]$$



$$= 6 \cdot \text{Mcs} / (tC^2 \cdot Kcpc)$$

$$= 6 \cdot -14.377 / (46.5^2 \cdot 1.0)$$

$$= -0.391 \text{ N./mm}^2$$

Compute with XL -ve [σ_{sb}]:

$$= 6 \cdot \text{Mcs} / (tC^2 \cdot Kcpc)$$

$$= 6 \cdot 262.236 / (46.5^2 \cdot 1.0)$$

$$= 7.136 \text{ N./mm}^2$$

Compute with XL +ve [$\sigma_{\text{Thetam+}}$]:

$$= \text{NCTheta}/tC = -15.695/46.5 = -3.31 \text{ N./mm}^2$$

Compute with XL -ve [$\sigma_{\text{Thetam-}}$]:

$$= \text{NCTheta}/tC = 5.488/46.5 = 1.16 \text{ N./mm}^2$$

Compute with XL +ve [$\sigma_{\text{thetab+}}$]:

$$= 6 \cdot v \cdot \text{Mcs} / (tC^2 \cdot Kcpc)$$

$$= 6 \cdot 0.3 \cdot -14.377 / (46.5^2 \cdot 1.0)$$

$$= -0.117 \text{ N./mm}^2$$

Compute with XL -ve [$\sigma_{\text{thetab-}}$]:

$$= 6 \cdot v \cdot \text{Mcs} / (tC^2 \cdot Kcpc)$$

$$= 6 \cdot 0.3 \cdot 262.236 / (46.5^2 \cdot 1.0)$$

$$= 2.141 \text{ N./mm}^2$$

Calculations for the Small End Cone/Shell Junction

Fhe per 4.1.3 or 4.3.11.2 Step 6 [Fhe]:

$$= 0.4 \cdot E(tc/D)$$

$$= 0.4 \cdot 0.18730E+09(46.5/2450.0)$$

$$= 1422.0 \text{ N./mm}^2$$

Factor of Safety [FS]:

Since $Fic [247.68] \geq Fy [247.68]$, FS = 1.667

Allowable Stress per 4.3.11 Step 6 [Fha]:

$$= Fic/FS$$

$$= 247.6785/1.667$$

$$= 148.577 \text{ N./mm}^2$$

Small End under Internal Pressure:

Results for Cylinder Small End Without a Flare:

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	83.007	247.678	Pass
Sigma sm-	71.311	247.678	Pass
Sigma thetam+	203.129	247.678	Pass
Sigma thetam-	194.948	247.678	Pass
Sigma sm+ + Sigm	193.011	495.357	Pass
Sigma sm- + Sigm	164.903	495.357	Pass
Sigma sm+ - Sigm	-26.996	495.357	Pass
Sigma sm- - Sigm	-22.282	495.357	Pass
Sigma thetam+ +	236.130	495.357	Pass
Sigma thetam- +	223.026	495.357	Pass
Sigma thetam+ -	170.128	495.357	Pass
Sigma thetam- -	166.871	495.357	Pass

Results for the Cone Small End Without a Flare:

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	68.502	247.678	Pass
Sigma sm-	58.821	247.678	Pass
Sigma thetam+	200.826	247.678	Pass
Sigma thetam-	193.413	247.678	Pass

Sigma sm+ + Sigm	143.910	495.357	Pass
Sigma sm- + Sigm	122.980	495.357	Pass
Sigma sm+ - Sigm	-6.907	495.357	Pass
Sigma sm- - Sigm	-5.338	495.357	Pass
Sigma thetam+ +	223.449	495.357	Pass
Sigma thetam- +	212.661	495.357	Pass
Sigma thetam+ -	178.203	495.357	Pass
Sigma thetam- -	174.166	495.357	Pass

Small End Cylinder Without a Flare

Compute the ratio equation 4.3.60 [n]:
= tc/ts = 46.5/38.5 = 1.2078

Compute the ratio equation 4.3.61 [H]:
= $\sqrt{(Rs/ts)} = \sqrt{(1186.534/38.5)} = 5.5515$

Compute the ratio equation 4.3.62 [B]:
= tan(a) = tan(12.0) = 0.2126

Compute Ms+ from equation 4.3.59 [Xs+]:
= $F_s / (2 \cdot \pi \cdot R_s) + M_s / (\pi \cdot R_s^2)$
= $-51696.777 / (2 \cdot 3.142 \cdot 1186.534) + 0.10154E+09 / (3.141 \cdot 1186.534^2)$
= 16.0253 Kgf/mm.

Compute Ms- from equation 4.3.59 [Xs-]:
= $F_s / (2 \cdot \pi \cdot R_s) - M_s / (\pi \cdot R_s^2) + X_s'$
= $-51696.777 / (2 \cdot 3.142 \cdot 1186.534) - 0.10154E+09 / (3.141 \cdot 1186.534^2) + 0.0$
= -29.8943 Kgf/mm.

From Table 4.3.5 MsN:

C1 = -10.030961 C2 = 1.840670 C3 = 4.174480 C4 = -0.051804
C5 = -1.089546 C6 = -0.029715 C7 = 0.004320 C8 = 0.120527
C9 = 0.015006 C10 = -0.006230 C11 = 0.000000

Compute from Table 4.3.5 [MsN]:
= $\exp[C1 + C2 \cdot \ln(H^2) + C3 \cdot \ln(a) + C4 \cdot (\ln(H^2))^2 + C5 \cdot \ln(a)^2 + C6 \cdot \ln(H^2) \cdot \ln(a) + C7 \cdot \ln(H^2)^3 + C8 \cdot \ln(a)^3 + C9 \cdot \ln(H^2) \cdot \ln(a)^2 + C10 \cdot \ln(H^2)^2 \cdot \ln(a)]$
= $\exp[-10.031 + 1.841 \cdot \ln(5.551^2) + 4.174 \cdot \ln(12.0) + -0.052 \cdot (\ln(5.551^2))^2 + -1.09 \cdot \ln(12.0)^2 + -0.03 \cdot \ln(5.551^2) \cdot \ln(12.0) + 0.004 \cdot \ln(5.551^2)^3 + 0.121 \cdot \ln(12.0)^3 + 0.015 \cdot \ln(5.551^2) \cdot \ln(12.0)^2 + -0.006 \cdot \ln(5.551^2)^2 \cdot \ln(12.0)] = 3.3932$

Compute from Table 4.3.2 [Msp]:
= $p \cdot tL^2 \cdot Ms_n = 51.216 \cdot 38.5^2 \cdot 3.39318 = 2626.7996 \text{ Kgf}$

From table 4.3.6 MsN:

C1 = -0.000099 C2 = -0.000378 C3 = -0.001194 C4 = 0.134080
C5 = 0.009899 C6 = 0.000007 C7 = -0.000002 C8 = 0.008980
C9 = 0.005621 C10 = 0.000001 C11 = 0.207273

Compute from Table 4.3.6 [MsN]:
= $(C1 + C3 \cdot H + C5 \cdot B + C7 \cdot H^2 + C9 \cdot B^2 + C11 \cdot H \cdot B) / (1 + C2 \cdot H + C4 \cdot B + C6 \cdot H^2 + C8 \cdot B^2 + C10 \cdot H \cdot B)$
= $(-0.0001 + -0.00119 \cdot 5.551 + 0.0099 \cdot 12.0 + -0.00551^2 + 0.00562 \cdot 12.0^2 + 0.20727 \cdot 5.551 \cdot 12.0) / (1 + -0.00038 \cdot 5.551 + 0.13408 \cdot 12.0 + 0.00001 \cdot 5.551^2 + 0.00898 \cdot 12.0^2 + 0.5551 \cdot 12.0)$
= 0.2338

Compute from Table 4.3.2 [Msx+]:
= $X_s \cdot t_s \cdot Ms_N = 16.025 \cdot 38.5 \cdot 0.234 = 144.2712 \text{ Kgf}$

Compute from Table 4.3.2 [Msx-]:
= $X_s \cdot t_s \cdot Ms_N = -29.894 \cdot 38.5 \cdot 0.234 = -269.1302 \text{ Kgf}$

Compute from Table 4.3.2 [Ms+]:
= Ms + Msx+ = 2626.8 + 144.271 = 2771.0708 Kgf

Compute from Table 4.3.2 [Ms-]:
= Ms + Msx- = 2626.8 + -269.13 = 2357.6692 Kgf

From table 4.3.5 Qn:

C1 = 0.177110 C2 = 0.000009 C3 = 0.003487 C4 = 0.002902
C5 = -0.000084 C6 = 0.000000 C7 = -0.000001 C8 = -0.000120
C9 = -0.000146 C10 = 0.000001 C11 = -0.004469

Compute from Table 4.3.5 [Qn]:
= (C1 + C3•H² + C5•α + C7•H⁴ + C9•α² + C11•H²•α) /
(1 + C2•H² + C4•α + C6•H⁴ + C8•α² + C10•H²•α)
= (0.17711 + 0.00349•5.551² + -0.00008•12.0
-0.5.551⁴ + -0.00015•12.0² + -0.00447•5.551²
* 12.0) / (1 + 0.00001•5.551² + 0.0029•12.0 +
0.5.551⁴ + -0.00012•12.0² + 0.000001•5.551²•12.0)
= -1.366

Compute from Table 4.3.2 [Qp]:
= P • ts • Qn = 51.216 • 38.5 • -1.366 = -27.4674 Kgf/mm.

From table 4.3.6 Qn:

C1 = -0.447437 C2 = 0.005176 C3 = -0.380296 C4 = -0.000297
C5 = -0.131252 C6 = 0.001755 C7 = -0.000056 C8 = -0.017447
C9 = 0.000578 C10 = 0.000022 C11 = 0.000000

Compute from Table 4.3.6 [Qn]:
= C1+C2•ln(H)+C3•ln(B)+C4•ln(H)²+C5•ln(B)²+C6•ln(H)•ln(B)+
C7•ln(H)³+C8•ln(B)³+C9•ln(H)•ln(B)²+C10•ln(H)²•ln(B)
= -0.44744 + 0.00518•ln(5.551) + -0.3803•ln(0.213) +
-0.0003•ln(5.551)² + -0.13125•(ln(0.213))² +
0.00175•ln(5.551)•ln(0.213) + -0.00006•ln(5.551)³ +
-0.01745•ln(0.213)³ + 0.00058•ln(5.551) * ln(0.213)² +
0.00002•ln(5.551)²•ln(0.213)
= -0.1031

Compute with Xs+ [Qx+]:
= Xs+ • Qn = 16.025 • -0.103 = -1.653 Kgf/mm.

Compute with Xs- [Qx-]:
= Xs- • Qn = -29.894 • -0.103 = 3.0835 Kgf/mm.

Compute with Xs+ [Q+]:
= Qp + Qx+ = -27.467 + -1.653 = -29.1204 Kgf/mm.

Compute with Xs- [Q-]:
= Qp + Qx- = -27.467 + 3.084 = -24.3839 Kgf/mm.

Compute angle [βcy]:
= (3•(1-ν²)/(Rs²•ts²))^{0.25} = (3•(1- 0.3²)/(1186.534²•38.5²))^{0.25}
= 0.0060 1/mm.

Compute with Xs+ [Ns+]:
= P•Rs/2 + Xs+ = 51.216•1186.534/2 + 16.025+ = 325.8736 Kgf/mm.

Compute with Xs- [Ns-]:
= P•Rs/2 + Xs- = 51.216•1186.534/2 + -29.894- = 279.9541 Kgf/mm.

Compute with Xs+ [NTheta+]:
= P • Rs + 2 • βcy • Rs • (-Ms • βcy - Q)
= 51.216 • 1186.534 + 2 • 0.006 • 1186.534 •
(-2771.071 • 0.006 - -0.11) = 797.4515 Kgf/mm.

Compute with Xs- [NTheta-]:
= P • Rs + 2 • βcy • Rs • (-Ms • βcy - Q)
= 51.216 • 1186.534 + 2 • 0.006 • 1186.534 •



$$(-2357.669 \cdot 0.006 - 2.714) = 765.3356 \text{ Kg/mm.}$$

Compute Stress with Xs+ [σ_{sm+}]:
 $= Ns(+)/ts = 325.874/38.5 = 83.0075 \text{ N./mm}^2$

Compute Stress with Xs- [σ_{sm-}]:
 $= Ns(-)/ts = 279.954/38.5 = 71.3107 \text{ N./mm}^2$

Compute stress with Xs+ [σ_{sb+}]:
 $= 6 \cdot Ms(+)/(ts^2 \cdot Kpc) = 6 \cdot 2771.071/(38.5^2 \cdot 1.0) = 110.0034 \text{ N./mm}^2$

Compute stress with Xs- [σ_{sb-}]:
 $= 6 \cdot Ms(-)/(ts^2 \cdot Kpc) = 6 \cdot 2357.669/(38.5^2 \cdot 1.0) = 93.5926 \text{ N./mm}^2$

Compute stress with Xs+ [$\sigma_{\theta sm+}$]:
 $= N\theta(+)/ts = 797.452/38.5 = 203.1292 \text{ N./mm}^2$

Compute stress with Xs- [$\sigma_{\theta sm-}$]:
 $= N\theta(-)/ts = 765.336/38.5 = 194.9485 \text{ N./mm}^2$

Compute stress with Xs+ [$\sigma_{\theta sb+}$]:
 $= 6 \cdot v \cdot Ms(+)/(ts^2 \cdot Kpc) = 6 \cdot 0.3 \cdot 2771.071/(38.5^2 \cdot 1.0) = 33.001 \text{ N./mm}^2$

Compute stress with Xs- [$\sigma_{\theta sb-}$]:
 $= 6 \cdot v \cdot Ms(-)/(ts^2 \cdot Kpc) = 6 \cdot 0.3 \cdot 2357.669/(38.5^2 \cdot 1.0) = 28.0778 \text{ N./mm}^2$

Small End Cone Without a Flare

Compute with Xs+ [$Qc+$]:
 $= Q \cdot \cos(a) + Ns \cdot \sin(a) = -29.1 \cdot \cos(12.0) + 325.9 \cdot \sin(12.0)$
 $= 39.2689 \text{ Kg/mm.}$

Compute with Xs- [$Qc-$]:
 $= Q \cdot \cos(a) + Ns \cdot \sin(a) = -24.4 \cdot \cos(12.0) + 280.0 \cdot \sin(12.0)$
 $= 34.3547 \text{ Kg/mm.}$

Compute effective radius [Rc]:
 $= Rs/\cos(a) = 1186.534/\cos(12.0) = 1213.0415 \text{ mm.}$

Compute the angle [β_{co}]:
 $= (3 \cdot (1 - v^2)/(Rc^2 \cdot t_c^2))^{0.25} = (3 \cdot (1 - 0.3^2)/(1213.042^2 \cdot 46.5^2))^{0.25}$
 $= 0.0054 \text{ 1/mm.}$

Compute with Ms+ [$Ncs+$]:
 $= Ns \cdot \cos(a) - Q \cdot \sin(a)$
 $= 325.874 \cdot \cos(12.0) - 29.12 \cdot \sin(12.0) = 324.807 \text{ Kg/mm.}$

Compute with Ms- [$Ncs-$]:
 $= Ns \cdot \cos(a) - Q \cdot \sin(a)$
 $= 279.954 \cdot \cos(12.0) - 24.384 \cdot \sin(12.0) = 278.9061 \text{ Kg/mm.}$

Compute with Ms+ [$N\theta_{sb+}$]:
 $= P \cdot Rs/\cos(a) + 2 \cdot \beta_{co} \cdot Rc \cdot (-Mcs \cdot \beta_{co} + Qc)$
 $= 51.216 \cdot 1186.534/\cos(12.0) + 2 \cdot 0.0054 \cdot 1213.042 \cdot (-2771.071 \cdot 0.0054 + 39.269) = 952.2348 \text{ Kg/mm.}$

Compute with Ms- [$N\theta_{sb-}$]:
 $= P \cdot Rs/\cos(a) + 2 \cdot \beta_{co} \cdot Rc \cdot (-Mcs \cdot \beta_{co} + Qc)$
 $= 51.216 \cdot 1186.534/\cos(12.0) + 2 \cdot 0.0054 \cdot (-2357.669 \cdot 0.0054 + 34.355) = 917.0875 \text{ Kg/mm.}$

Compute the stress with Ms+ [σ_{sm+}]:
 $= Ncs/tc = 324.807/46.5 = 68.5017 \text{ N./mm}^2$

Compute the stress with Ms- [σ_{sm-}]:
 $= Ncs/tc = 278.906/46.5 = 58.8212 \text{ N./mm}^2$

Compute the stress with Ms+ [σ_{sb+}]:

$$= 6 \cdot M_{cs} / (t_c^2 \cdot K_{cpc}) = 6 \cdot 2771.071 / (46.5^2 \cdot 1.0) = 75.4087 \text{ N./mm}^2$$

Compute the stress with M_{s-} [σ_{sb-}]:

$$= 6 \cdot M_{cs} / (t_c^2 \cdot K_{cpc}) = 6 \cdot 2357.669 / (46.5^2 \cdot 1.0) = 64.1589 \text{ N./mm}^2$$

Compute stress with M_{s+} [$\sigma_{\theta\theta am+}$]:

$$= N_{\theta\theta} / t_c = 952.235 / 46.5 = 200.826 \text{ N./mm}^2$$

Compute stress with M_{s-} [$\sigma_{\theta\theta am-}$]:

$$= N_{\theta\theta} / t_c = 917.087 / 46.5 = 193.413 \text{ N./mm}^2$$

Compute stress with M_{s+} [$\sigma_{\theta\theta ab+}$]:

$$= 6 \cdot v \cdot M_{cs} / (t_c^2 \cdot K_{cpc}) = 6 \cdot 0.3 \cdot 2771.071 / (46.5^2 \cdot 1.0) = 22.6226 \text{ N./mm}^2$$

Compute stress with M_{s-} [$\sigma_{\theta\theta ab-}$]:

$$= 6 \cdot v \cdot M_{cs} / (t_c^2 \cdot K_{cpc}) = 6 \cdot 0.3 \cdot 2357.669 / (46.5^2 \cdot 1.0) = 19.2477 \text{ N./mm}^2$$

Small End under External Pressure:

Results for Cylinder Small End Without a Flare:

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	2.488	247.678	Pass
Sigma sm-	-9.208	247.678	Pass
Sigma thetam+	-1.189	247.678	Pass
Sigma thetam-	-9.370	247.678	Pass
Sigma sm+ + Sigma	6.110	495.357	Pass
Sigma sm- + Sigma	-21.998	495.357	Pass
Sigma sm+ - Sigma	-1.133	495.357	Pass
Sigma sm- - Sigma	3.581	495.357	Pass
Sigma thetam+ +	-0.103	495.357	Pass
Sigma thetam- +	-13.207	495.357	Pass
Sigma thetam+ -	-2.276	495.357	Pass
Sigma thetam- -	-5.533	495.357	Pass

Results for the Cone Small End Without a Flare:

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	2.063	247.678	Pass
Sigma sm-	-7.617	247.678	Pass
Sigma thetam+	-1.416	247.678	Pass
Sigma thetam-	-8.829	247.678	Pass
Sigma sm+ + Sigma	4.546	495.357	Pass
Sigma sm- + Sigma	-16.384	495.357	Pass
Sigma sm+ - Sigma	-0.419	495.357	Pass
Sigma sm- - Sigma	1.150	495.357	Pass
Sigma thetam+ +	-0.671	495.357	Pass
Sigma thetam- +	-11.459	495.357	Pass
Sigma thetam+ -	-2.161	495.357	Pass
Sigma thetam- -	-6.199	495.357	Pass

Average Compressive Stresses and comparisons with F_{ha} and F_{xa} :

Sigma smCyl+ + S	2.276	-170.656	Pass
Sigma smCyl- + S	-8.413	-170.656	Pass
Sigma thetamCyl+	-1.303	-148.577	Pass
Sigma thetamCyl-	-1.303	-148.577	Pass

Small End Cylinder Without a Flare

Compute the ratio equation 4.3.60 [n]:

$$= t_c / t_s = 46.5 / 38.5 = 1.2078$$

Compute the ratio equation 4.3.61 [H]:

$$= \sqrt{R_s / t_s} = \sqrt{1186.534 / 38.5} = 5.5515$$



Compute the ratio equation 4.3.62 [B]:
 $= \tan(a) = \tan(12.0) = 0.2126$

Compute Ms+ from equation 4.3.59 [Xs+]:
 $= F_s / (2 \cdot \pi \cdot R_s) + M_s / (\pi \cdot R_s^2)$
 $= -51696.777 / (2 \cdot 3.142 \cdot 1186.534) + 0.10154E+09 / (3.141 \cdot 1186.534^2)$
 $= 16.0253 \text{ Kg/mm.}$

Compute Ms- from equation 4.3.59 [Xs-]:
 $= F_s / (2 \cdot \pi \cdot R_s) - M_s / (\pi \cdot R_s^2) + X_s'$
 $= -51696.777 / (2 \cdot 3.142 \cdot 1186.534) - 0.10154E+09 / (3.141 \cdot 1186.534^2) + 0.0$
 $= -29.8943 \text{ Kg/mm.}$

From Table 4.3.5 MsN:

C1 = -10.030961 C2 = 1.840670 C3 = 4.174480 C4 = -0.051804
 C5 = -1.089546 C6 = -0.029715 C7 = 0.004320 C8 = 0.120527
 C9 = 0.015006 C10 = -0.006230 C11 = 0.000000

Compute from Table 4.3.5 [MsN]:
 $= \exp[C1 + C2 \cdot \ln(H^2) + C3 \cdot \ln(a) + C4 \cdot (\ln(H^2))^2 + C5 \cdot \ln(a)^2 + C6 \cdot \ln(H^2) \cdot \ln(a) +$
 $C7 \cdot \ln(H^2)^3 + C8 \cdot \ln(a)^3 + C9 \cdot \ln(H^2) \cdot \ln(a)^2 + C10 \cdot \ln(H^2)^2 \cdot \ln(a)]$
 $= \exp[-10.031 + 1.841 \cdot \ln(5.551^2) + 4.174 \cdot \ln(12.0) +$
 $-0.052 \cdot (\ln(5.551^2))^2 + -1.09 \cdot \ln(12.0)^2 +$
 $-0.03 \cdot \ln(5.551^2) \cdot \ln(12.0) + 0.004 \cdot \ln(5.551^2)^3$
 $0.121 \cdot \ln(12.0)^3 + 0.015 \cdot \ln(5.551^2) \cdot \ln(12.0)^2 +$
 $-0.006 \cdot \ln(5.551^2)^2 \cdot \ln(12.0)] = 3.3932$

Compute from Table 4.3.2 [Msp]:
 $= p \cdot tL^2 \cdot M_{sn} = -1.034 \cdot 38.5^2 \cdot 3.39318 = -53.0426 \text{ Kg}$

From table 4.3.6 MsN:

C1 = -0.000099 C2 = -0.000378 C3 = -0.001194 C4 = 0.134080
 C5 = 0.009899 C6 = 0.000007 C7 = -0.000002 C8 = 0.008980
 C9 = 0.005621 C10 = 0.000001 C11 = 0.207273

Compute from Table 4.3.6 [MsN]:
 $= (C1 + C3 \cdot H + C5 \cdot B + C7 \cdot H^2 + C9 \cdot B^2 + C11 \cdot H \cdot B) /$
 $(1 + C2 \cdot H + C4 \cdot B + C6 \cdot H^2 + C8 \cdot B^2 + C10 \cdot H \cdot B)$
 $= (-0.0001 + -0.00119 \cdot 5.551 + 0.0099 \cdot 12.0 +$
 $-0.5 \cdot 5.551^2 + 0.00562 \cdot 12.0^2 +$
 $0.20727 \cdot 5.551 \cdot 12.0) / (1 + -0.00038 \cdot 5.551 +$
 $0.13408 \cdot 12.0 + 0.00001 \cdot 5.551^2 + 0.00898 \cdot 12.0^2 +$
 $0.5 \cdot 5.551 \cdot 12.0)$
 $= 0.2338$

Compute from Table 4.3.2 [Msx+]:
 $= X_s \cdot t_s \cdot M_{sN} = 16.025 \cdot 38.5 \cdot 0.234 = 144.2712 \text{ Kg}$

Compute from Table 4.3.2 [Msx-]:
 $= X_s \cdot t_s \cdot M_{sN} = -29.894 \cdot 38.5 \cdot 0.234 = -269.1302 \text{ Kg}$

Compute from Table 4.3.2 [Ms+]:
 $= M_s + M_{sx+} = -53.043 + 144.271 = 91.2286 \text{ Kg}$

Compute from Table 4.3.2 [Ms-]:
 $= M_s + M_{sx-} = -53.043 + -269.13 = -322.1728 \text{ Kg}$

From table 4.3.5 Qn:

C1 = 0.177110 C2 = 0.000009 C3 = 0.003487 C4 = 0.002902
 C5 = -0.000084 C6 = 0.000000 C7 = -0.000001 C8 = -0.000120
 C9 = -0.000146 C10 = 0.000001 C11 = -0.004469

Compute from Table 4.3.5 [Qn]:
 $= (C1 + C3 \cdot H^2 + C5 \cdot \alpha + C7 \cdot H^4 + C9 \cdot \alpha^2 + C11 \cdot H^2 \cdot \alpha) /$
 $(1 + C2 \cdot H^2 + C4 \cdot \alpha + C6 \cdot H^4 + C8 \cdot \alpha^2 + C10 \cdot H^2 \cdot \alpha)$
 $= (0.17711 + 0.00349 \cdot 5.551^2 + -0.00008 \cdot 12.0$
 $-0.5 \cdot 5.551^4 + -0.00015 \cdot 12.0^2 + -0.00447 \cdot 5.551^2$
 $\cdot 12.0) / (1 + 0.00001 \cdot 5.551^2 + 0.0029 \cdot 12.0 +$



$$0.5551^4 + -0.00012 \cdot 12.0^2 + 0.000001 \cdot 5.551^2 \cdot 12.0) \\ = -1.366$$

Compute from Table 4.3.2 [Qp]:

$$= P \cdot ts \cdot Q_n = -1.034 \cdot 38.5 \cdot -1.366 = 0.5546 \text{ Kgf/mm.}$$

From table 4.3.6 Qn:

$$\begin{aligned} C1 &= -0.447437 & C2 &= 0.005176 & C3 &= -0.380296 & C4 &= -0.000297 \\ C5 &= -0.131252 & C6 &= 0.001755 & C7 &= -0.000056 & C8 &= -0.017447 \\ C9 &= 0.000578 & C10 &= 0.000022 & C11 &= 0.000000 \end{aligned}$$

Compute from Table 4.3.6 [Qn]:

$$\begin{aligned} &= C1 + C2 \cdot \ln(H) + C3 \cdot \ln(B) + C4 \cdot \ln(H)^2 + C5 \cdot \ln(B)^2 + C6 \cdot \ln(H) \cdot \ln(B) + \\ &C7 \cdot \ln(H)^3 + C8 \cdot \ln(B)^3 + C9 \cdot \ln(H) \cdot \ln(B)^2 + C10 \cdot \ln(H)^2 \cdot \ln(B) \\ &= -0.44744 + 0.00518 \cdot \ln(5.551) + -0.3803 \cdot \ln(0.213) + \\ &-0.0003 \cdot \ln(5.551)^2 + -0.13125 \cdot (\ln(0.213))^2 + \\ &0.00175 \cdot \ln(5.551) \cdot \ln(0.213) + -0.00006 \cdot \ln(5.551)^3 + \\ &-0.01745 \cdot \ln(0.213)^3 + 0.00058 \cdot \ln(5.551) \cdot \ln(0.213)^2 + \\ &0.00002 \cdot \ln(5.551)^2 \cdot \ln(0.213) \\ &= -0.1031 \end{aligned}$$

Compute with Xs+ [Qx+]:

$$= X_{s+} \cdot Q_n = 16.025 \cdot -0.103 = -1.653 \text{ Kgf/mm.}$$

Compute with Xs- [Qx-]:

$$= X_{s-} \cdot Q_n = -29.894 \cdot -0.103 = 3.0835 \text{ Kgf/mm.}$$

Compute with Xs+ [Q+]:

$$= Q_p + Q_{x+} = 0.555 + -1.653 = -1.0983 \text{ Kgf/mm.}$$

Compute with Xs- [Q-]:

$$= Q_p + Q_{x-} = 0.555 + 3.084 = 3.6382 \text{ Kgf/mm.}$$

Compute angle [β_{cy}]:

$$\begin{aligned} &= (3 \cdot (1 - v^2) / (R_s^2 \cdot t_s^2))^{0.25} = (3 \cdot (1 - 0.3^2) / (1186.534^2 \cdot 38.5^2))^{0.25} \\ &= 0.0060 \text{ 1/mm.} \end{aligned}$$

Compute with Xs+ [Ns+]:

$$= P \cdot R_s / 2 + X_{s+} = -1.034 \cdot 1186.534 / 2 + 16.025 = 9.7685 \text{ Kgf/mm.}$$

Compute with Xs- [Ns-]:

$$= P \cdot R_s / 2 + X_{s-} = -1.034 \cdot 1186.534 / 2 + -29.894 = -36.151 \text{ Kgf/mm.}$$

Compute with Xs+ [NTheta+]:

$$\begin{aligned} &= P \cdot R_s + 2 \cdot \beta_{cy} \cdot R_s \cdot (-M_s \cdot \beta_{cy} - Q) \\ &= -1.034 \cdot 1186.534 + 2 \cdot 0.006 \cdot 1186.534 \cdot \\ &(-91.229 \cdot 0.006 - -0.11) = -4.6685 \text{ Kgf/mm.} \end{aligned}$$

Compute with Xs- [NTheta-]:

$$\begin{aligned} &= P \cdot R_s + 2 \cdot \beta_{cy} \cdot R_s \cdot (-M_s \cdot \beta_{cy} - Q) \\ &= -1.034 \cdot 1186.534 + 2 \cdot 0.006 \cdot 1186.534 \cdot \\ &(-322.173 \cdot 0.006 - 2.714) = -36.7844 \text{ Kgf/mm.} \end{aligned}$$

Compute Stress with Xs+ [σ_{sm+}]:

$$= N_{s+} / t_s = 9.769 / 38.5 = 2.4883 \text{ N./mm}^2$$

Compute Stress with Xs- [σ_{sm-}]:

$$= N_{s-} / t_s = -36.151 / 38.5 = -9.2085 \text{ N./mm}^2$$

Compute stress with Xs+ [σ_{sb+}]:

$$= 6 \cdot M_{s+} / (t_s^2 \cdot K_{pc}) = 6 \cdot 91.229 / (38.5^2 \cdot 1.0) = 3.6215 \text{ N./mm}^2$$

Compute stress with Xs- [σ_{sb-}]:

$$= 6 \cdot M_{s-} / (t_s^2 \cdot K_{pc}) = 6 \cdot -322.173 / (38.5^2 \cdot 1.0) = -12.7893 \text{ N./mm}^2$$

Compute stress with Xs+ [$\sigma_{\theta\theta\theta\theta+}$]:

$$= N_{\theta\theta\theta\theta+} / t_s = -4.669 / 38.5 = -1.1892 \text{ N./mm}^2$$

Compute stress with Xs- [$\sigma_{\theta\theta am-}$]:
= $N_{\theta\theta}/t_s = -36.784/38.5 = -9.3698 \text{ N./mm}^2$

Compute stress with Xs+ [$\sigma_{\theta\theta ab+}$]:
= $6 \cdot v \cdot M_s(+)/(t_s^2 \cdot K_{pc}) = 6 \cdot 0.3 \cdot 91.229/(38.5^2 \cdot 1.0) = 1.0865 \text{ N./mm}^2$

Compute stress with Xs- [$\sigma_{\theta\theta ab-}$]:
= $6 \cdot v \cdot M_s(-)/(t_s^2 \cdot K_{pc}) = 6 \cdot 0.3 \cdot -322.173/(38.5^2 \cdot 1.0) = -3.8368 \text{ N./mm}^2$

Small End Cone Without a Flare

Compute with Xs+ [Q_c+]:
= $Q \cdot \cos(a) + N_s \cdot \sin(a) = -1.1 \cdot \cos(12.0) + 9.8 \cdot \sin(12.0)$
= 0.9567 Kg/mm .

Compute with Xs- [Q_c-]:
= $Q \cdot \cos(a) + N_s \cdot \sin(a) = 3.6 \cdot \cos(12.0) + -36.2 \cdot \sin(12.0)$
= -3.9575 Kg/mm .

Compute effective radius [R_c]:
= $R_s/\cos(a) = 1186.534/\cos(12.0) = 1213.0415 \text{ mm}$.

Compute the angle [β_{co}]:
= $(3 \cdot (1-v^2)/(R_c^2 \cdot t_c^2))^{0.25} = (3 \cdot (1-0.3^2)/(1213.042^2 \cdot 46.5^2))^{0.25}$
= 0.0054 1/mm .

Compute with Ms+ [N_{cs+}]:
= $N_s \cdot \cos(a) - Q \cdot \sin(a)$
= $9.769 \cdot \cos(12.0) - -1.098 \cdot \sin(12.0) = 9.7834 \text{ Kg/mm}$.

Compute with Ms- [N_{cs-}]:
= $N_s \cdot \cos(a) - Q \cdot \sin(a)$
= $-36.151 \cdot \cos(12.0) - 3.638 \cdot \sin(12.0) = -36.1174 \text{ Kg/mm}$.

Compute with Ms+ [$N_{\theta\theta ab+}$]:
= $P \cdot R_s/\cos(a) + 2 \cdot \beta_{co} \cdot R_c \cdot (-M_{cs} \cdot \beta_{co} + Q_c)$
= $-1.034 \cdot 1186.534/\cos(12.0) + 2 \cdot 0.0054 \cdot 1213.042 \cdot (-91.229 \cdot 0.0054 + 0.957) = -6.7148 \text{ Kg/mm}$.

Compute with Ms- [$N_{\theta\theta ab-}$]:
= $P \cdot R_s/\cos(a) + 2 \cdot \beta_{co} \cdot R_c \cdot (-M_{cs} \cdot \beta_{co} + Q_c)$
= $-1.034 \cdot 1186.534/\cos(12.0) + 2 \cdot 0.0054 \cdot (-322.173 \cdot 0.0054 + -3.958) = -41.862 \text{ Kg/mm}$.

Compute the stress with Ms+ [σ_{sm+}]:
= $N_{cs}/t_c = 9.783/46.5 = 2.0633 \text{ N./mm}^2$

Compute the stress with Ms- [σ_{sm-}]:
= $N_{cs}/t_c = -36.117/46.5 = -7.6172 \text{ N./mm}^2$

Compute the stress with Ms+ [σ_{sb+}]:
= $6 \cdot M_{cs}/(t_c^2 \cdot K_{cpc}) = 6 \cdot 91.229/(46.5^2 \cdot 1.0) = 2.4826 \text{ N./mm}^2$

Compute the stress with Ms- [σ_{sb-}]:
= $6 \cdot M_{cs}/(t_c^2 \cdot K_{cpc}) = 6 \cdot -322.173/(46.5^2 \cdot 1.0) = -8.7672 \text{ N./mm}^2$

Compute stress with Ms+ [$\sigma_{\theta\theta am+}$]:
= $N_{\theta\theta}/t_c = -6.715/46.5 = -1.416 \text{ N./mm}^2$

Compute stress with Ms- [$\sigma_{\theta\theta am-}$]:
= $N_{\theta\theta}/t_c = -41.862/46.5 = -8.829 \text{ N./mm}^2$

Compute stress with Ms+ [$\sigma_{\theta\theta ab+}$]:
= $6 \cdot v \cdot M_{cs}/(t_c^2 \cdot K_{cpc}) = 6 \cdot 0.3 \cdot 91.229/(46.5^2 \cdot 1.0) = 0.7448 \text{ N./mm}^2$

Compute stress with Ms- [$\sigma_{\theta\theta ab-}$]:
= $6 \cdot v \cdot M_{cs}/(t_c^2 \cdot K_{cpc}) = 6 \cdot 0.3 \cdot -322.173/(46.5^2 \cdot 1.0) = -2.6302 \text{ N./mm}^2$

STEAM DRUM (WHB)
(A150NRJ10EQ001)



PV Elite 28 U2 Licensee: SL User

FileName : Steam drum (WHB) - division 2

Conical Section: cone transition Cone: 3 6:47pm Sep 16,2026



HEXAGON

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Input, Nozzle Desc: A2-6"

From: 70

Pressure for Reinforcement Calculations	P	47.332	bars
Temperature for Internal Pressure	Temp	286	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	286	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	165.12	N./mm ²
Parent Allowable Stress At Ambient	Sva	207.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2768.00	mm.
Design Length of Section	L	7468.6665	mm.
Shell Finished (Minimum) Thickness	t	46.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		10700.00	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

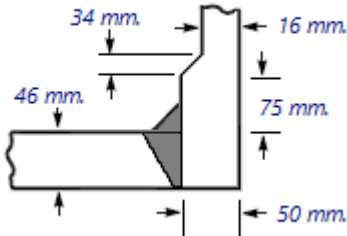
Material		SA-182 F11	
Material UNS Number		K11597	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	75.0000	mm.
Height of Beveled Transition	L'	34.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	52.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: A2-6"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	5.368	in.
Actual Thickness Used in Calculation	0.628	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 139.36 \cdot (\exp(47.332/(138.0 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 2.4315 + 1.5 + 0.0 = 3.9315 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 138.0 \cdot 1.0 \cdot \ln((2 \cdot 14.458 + 139.36) / 139.36) - 0.0$$

$$= 260.17 - 0.0 = 260.1699 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	7.720 mm
Min. required shell thickness, internal p + c	41.796 mm.
Min. required shell thickness, external p + c	16.155 mm.

Nozzle neck minimum required thickness	7.720 mm.
--	-----------

Minimum nozzle neck thickness 15.951 mm., is greater than required thickness 7.720 mm.

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 30.3,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 314.7 N./mm ²	Passed
Occasional	: 10.3,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 13.6,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[138.0/165.1, 1]$$

$$= 0.836$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 50.0 - 1.5$$

$$= 48.500 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 15.958 - 1.5$$

$$= 14.458 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2771.0/44.5$$

$$= 62.270 \text{ must be less than 400.}$$



Effective Pressure Radius [Reff]:

$$\begin{aligned}
 &= Di/2 + \text{corrosion allowance} \\
 &= 2768.0/2 + 1.5 \\
 &= 1385.500 \text{ mm.}
 \end{aligned}$$

Effective Length of Vessel Wall [LR]:

$$\begin{aligned}
 &= \min(\sqrt{Reff \cdot t}, 2 \cdot Rn) \\
 &= \min(\sqrt{1385.5 \cdot 44.5}, 2 \cdot 69.68) \\
 &= 139.360 \text{ mm.}
 \end{aligned}$$

Thickness Limit Candidate [LH1]:

$$\begin{aligned}
 &= \min[1.5t, te] + \sqrt{Rn \cdot tn} \\
 &= \min[1.5 \cdot 44.5, 0.0] + \sqrt{69.68 \cdot 48.5} \\
 &= 58.133 \text{ mm.}
 \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned}
 &= Lpr1 \\
 &= 200.0 \\
 &= 200.000 \text{ mm.}
 \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned}
 &= 8(t + te) \\
 &= 8(44.5 + 0.0) \\
 &= 356.000 \text{ mm.}
 \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned}
 &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\
 &= \min[58.133, 200.0, 356.0] + 44.5 \\
 &= 102.633 \text{ mm.}
 \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned}
 &= t + ((A5 \cdot frp) / LR) \\
 &= 44.5 + ((0.0 \cdot 1.0) / 139.36) \\
 &= 44.500 \text{ mm.}
 \end{aligned}$$

Parameter [λ]:

$$\begin{aligned}
 &= \min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff})) \\
 &= \min(12, (2 \cdot 69.68 + 48.5) / (\sqrt{(2771.0 + 44.5) \cdot 44.5})) \\
 &= 0.531
 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned}
 &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\
 &= 44.5 \cdot 139.36 \cdot \max((0.531/5)^{0.85}, 1) \\
 &= 62.015 \text{ cm}^2
 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned}
 &= tn \cdot LH \\
 &= 48.5 \cdot 102.633 \\
 &= 49.777 \text{ cm}^2
 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned}
 &= 0.5 \cdot Leg41^2 \\
 &= 0.5 \cdot 10.0^2 \\
 &= 0.500 \text{ cm}^2
 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned}
 &= A1 + frn(A2 + A3) + A41 + A42 + A43 \\
 &= 62.015 + 0.836(49.777 + 0.0) + 0.5 + 0.0 + 0.0 \\
 &= 104.117 \text{ cm}^2
 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned}
 &= tn / \ln[1 + tn/Rn] \\
 &= 48.5 / \ln[1 + 48.5/69.68] \\
 &= 91.805 \text{ mm.}
 \end{aligned}$$



Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 44.5 / \ln[1 + 44.5/1385.5]$$

$$= 1407.635 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 165.119 \cdot 1.0$$

$$= 247.7 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 47.332 \cdot 91.805 \cdot 102.633$$

$$= 4547.8 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 47.332 \cdot 1407.635 (139.36 + 48.5)$$

$$= 127637.0 \text{ Kg}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 47.332 \cdot 1407.635 \cdot 69.68$$

$$= 47342.4 \text{ Kg}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (127637.0 + 47342.4 + 4547.8) / 47.332$$

$$= 3719.4 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 247.678 / (2 \cdot 3719.443 / 104.117 - 1407.635 / 44.5)$$

$$= 62.2 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / [t / R_{xs}]$$

$$= 165.119 [44.5 / 1407.635]$$

$$= 52.2 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(62.203, 52.197)$$

$$= 52.197 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (4547.839 + 127637.016 + 47342.406) / 104.117$$

$$= 169.098 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 47.332 \cdot 1407.635 / 44.5$$

$$= 149.7 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 169.098 - 149.731, 149.731)$$

$$= 188.5 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	188.46	N./mm ²
Maximum Allowable Working Pressure	Pmax	52.20	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:



$$= (R_{nc} + t_n) / R_{nc}$$

$$= (69.68 + 48.5) / 69.68$$

$$= 1.696 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [L_T]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (69.68 + 48.5)$$

$$= 185.637 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L_{41T}, L_{42T}, L_{43T}]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$t_c, L_{41T} = 7.071, \text{ must be } \geq \min(0.7 \cdot 48.500, 6.000) \text{ Passed}$$

Weld Load Value [f_{welds}]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(47342 \cdot 1.7, 1.5 \cdot 138.0(49.777 + 0.0), \pi/4 \cdot 47.3 \cdot 69.68^2 \cdot 1.7^2)$$

$$= 5294.562 \text{ Kgf}$$

Weld Stress Value [τ]:

$$= f_{welds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$

$$= 5294.562 / (185.637(0.49 \cdot 7.071 + 0.6 \cdot 44.5 + 0.49 \cdot 0.0))$$

$$= 9.272 < \text{ or } = \text{ to } 165.119 \text{ Weld Stress Passed}$$

Nozzle Calculations per Section 4.5: External Pressure Case:Nozzle Radius for Force Calculation [R_{xn}]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 48.5 / \ln[1 + 48.5/69.68]$$

$$= 91.805 \text{ mm.}$$

Shell Radius for Force Calculation [R_{xs}]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 44.5 / \ln[1 + 44.5/1385.5]$$

$$= 1407.635 \text{ mm.}$$

Allowable Local Primary Membrane Stress [S_{allow}]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 165.119$$

$$= 247.7 \text{ N./mm}^2$$

Force acting on the Nozzle [f_N]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 1.034 \cdot 91.805 \cdot 102.633$$

$$= 99.4 \text{ Kgf}$$

Force acting on the Shell [f_S]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 1.034 \cdot 1407.635 (139.36 + 48.5)$$

$$= 2788.8 \text{ Kgf}$$

Discontinuity Force from External Pressure [f_Y]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 1.034 \cdot 1407.635 \cdot 69.68$$

$$= 1034.4 \text{ Kgf}$$

Area Resisting External Pressure [A_p]:

$$= (f_S + f_Y + f_N) / P$$

$$= (2788.8 + 1034.4 + 99.4) / 1.034$$

$$= 3719.4 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [P_{max1}]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 247.678 / (2 \cdot 3719.442 / 104.117 - 1407.635 / 44.5)$$

$$= 62.2 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [P_{max2}]:

= S[t/Rxs]
= 165.119[44.5/1407.635]
= 52.2 bars

Maximum Allowable Working Pressure [Pmax]:
= min(Pmax1, Pmax2)
= min(62.203, 52.197)
= 52.197 bars

Average Primary Membrane Stress [σAvg]:
= (fN + fS + fY) / AT
= (99.369 + 2788.823 + 1034.415)/104.117
= 3.695 N./mm²

General Primary Membrane Stress [σCirc]:
= P • Rxs / teff
= 1.034 • 1407.635/44.5
= 3.3 N./mm²

Maximum Local Primary Membrane Stress [PL]:
= max(2 • σAvg - σCirc, σCirc)
= max(2 • 3.695 - 3.272, 3.272)
= 4.1 N./mm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	4.12	N./mm ²
Maximum Allowable External Pressure	Pmax	52.20	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

This element is Not Post Weld Heat Treated
Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : -10 °C
The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.168

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (2.432 • 1.0)/(15.958 - 1.5 - 0.0)
= 0.168

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

This element is Not Post Weld Heat Treated
Computed Minimum Design Metal Temperature : 36 °C
User entered Minimum Design Temperature : -10 °C
The Curve for this Material is : A
From Figure 3.7M- Non PWHT, the MDMT is : 48 °C
Temperature reduction per Graph 3.12 : 11 °C
The computed stress ratio is : 0.720

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (32.048 • 1.0)/(46.0 - 1.5 - 0.0)
= 0.720

Warning: Impact Testing Is Mandatory!

Gov. MDMT of the nozzle to shell joint welded assembly : 36 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction -48 °C



The Drop for this Nozzle is : 5.0549 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 251.0549 mm.

Input Echo, WRC297 Item 1, Description: A2-6" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	2768.000	mm.
Shell Thickness	46.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-387 11	
Vessel Cold S.I. Allowable	Smc 207.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 165.12	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.5000	mm.
Nozzle Diameter	136.360	mm.
Nozzle Thickness	50.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-182 F11	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUstained loads.*

Design Internal Pressure	Dp	47.33	bars
Radial Load	P	-978.99	Kgf
Circumferential Shear	Vc	978.99	Kgf
Longitudinal Shear	Vl	734.25	Kgf
Circumferential Moment	Mc	381.25	Kg-m.
Longitudinal Moment	Ml	381.25	Kg-m.
Torsional Moment	Mt	440.38	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (4.727) is < 10
check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
= NozzleOD + 2 • 1.65 • $\sqrt{(R_{mean}(t - ca))}$
= 236.36 + 2 • 1.65 • $\sqrt{(1407.75(46.0 - 1.5))}$
= 1062.316 mm.

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2815.500	mm.
Nozzle Outside Diameter	(d) =	236.360	mm.
Vessel Thickness used	(T) =	44.500	mm.
Nozzle Thickness used	(t) =	48.500	mm.
T / t	=	0.918	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.668$$



Nr/P	=	0.151
Mr/P	=	0.247
M0/P	=	0.083
N0/P	=	0.150
MrD/Mc	=	0.292
NrDL/Mc	=	0.135
M0d/Mc	=	0.112
N0DL/Mc	=	-0.072
MrD/Ml	=	0.240
NrDL/Ml	=	0.124
M0D/Ml	=	0.091
N0DL/Ml	=	0.061

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-4	4	4	-4
Normal Pressure Stress	145	149	145	149
Outplane Stress Summary	143	151	153	144

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	7	-7	7	-7
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-11	11	11	-11
Inplane Pressure Stress	72	72	72	72
Inplane Stress Summary	67	76	92	55

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	1	1	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	143	151	153	144

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	2	-2	2	-2



Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-5	5	5	-5
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	72	72	72	72

Outplane Stress Summary	70	76	80	64
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	7	-7	7	-7
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-13	13	13	-13
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	145	149	145	149

Inplane Stress Summary	137	156	168	130
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	0	0	1	1
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	137	156	168	130

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	11	6	11

Hoop Stress Summary	6	11	7	12
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	4	-4	4	-4
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-3	-3	3	3
Axial Bending (ML)	-7	7	7	-7
Axial Pressure Stress	3	3	3	3



Axial Stress Summary	-2	3	17	-4
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary	1	1	0	0
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside

Two * Max Shear Stress	9	12	18	17
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	11	6	11

Hoop Stress Summary	7	13	6	11
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	4	-4	4	-4
Axial Membrane (Mc)	-3	-3	3	3
Axial Bending (Mc)	-9	9	9	-9
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3

Axial Stress Summary	-4	4	19	-6
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary	0	0	1	1
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Outside	Inside	Outside	Inside

Two * Max Shear Stress	12	13	20	18
------------------------	----	----	----	----

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	145	149	145	149	145	149	145	149
Circ. Pl (SUS)	0	0	0	0	-1	-1	1	1
Circ. Q (SUS)	-2	2	6	-6	-6	6	20	-20
Long. Pm (SUS)	72	72	72	72	72	72	72	72
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	-4	4	18	-18	-3	3	7	-7
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	145.0	149.0	145.0	149.0	145.0	149.0	145.0	149.0
Pm+Pl (SUS)	145.0	149.0	145.0	149.0	144.0	148.0	146.0	150.0
Pm+Pl+Q (Total)	143.0	151.0	151.0	143.0	138.0	154.0	166.0	130.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	149.00	165.12	Passed
Pm+Pl (SUS)	150.00	247.68	Passed
Pm+Pl+Q (TOTAL)	166.01	495.36	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	6	11	6	11	6	11	6	11
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-3	-3	3	3	-3	-3	3	3
Long. Q (SUS)	-3	3	11	-11	-5	5	13	-13
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	6.0	11.0	6.0	11.0	6.0	11.0	6.0	11.0
Pm+Pl (SUS)	6.0	11.0	6.0	11.0	6.0	11.0	6.0	11.0
Pm+Pl+Q (Total)	9.2	11.1	17.1	16.1	11.2	11.2	19.1	18.1

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	11.00	138.00	Passed
Pm+Pl (SUS)	11.00	207.00	Passed

STEAM DRUM (WHB)
(A150NRJ10EQ001)



PV Elite 28 U2 Licensee: SL User

FileName : Steam drum (WHB) - division 2

Nozzle Calcs.: A2-6"

Nozl: 35 6:47pm Sep 16,2026

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HEXAGON

Pm+Pl+Q (TOTAL)	19.08	414.00	Passed
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*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: L2-2"

From: 70

Pressure for Reinforcement Calculations	P	47.332	bars
Temperature for Internal Pressure	Temp	286	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	286	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	165.12	N./mm ²
Parent Allowable Stress At Ambient	Sva	207.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2768.00	mm.
Design Length of Section	L	7468.6665	mm.
Shell Finished (Minimum) Thickness	t	46.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		10700.00	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

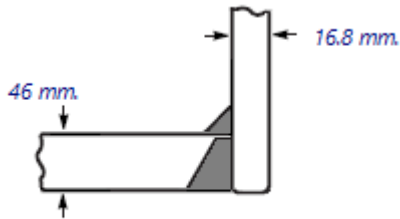
Material		SA-182 F11	
Material UNS Number		K11597	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Outside Projection	ho	196.8500	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		52.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: L2-2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 53.8 \cdot (\exp(47.332/(138.0 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 0.9387 + 1.5 + 0.0 = 2.4387 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 138.0 \cdot 1.0 \cdot \ln((2 \cdot 15.264 + 53.8) / 53.8) - 0.0$$

$$= 620.192 - 0.0 = 620.1918 \text{ bars}$$

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 95.96 mm.

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	6.300 mm
Min. required shell thickness, internal p + c :	41.796 mm.
Min. required shell thickness, external p + c :	16.155 mm.

Nozzle neck minimum required thickness : 6.300 mm.

Minimum nozzle neck thickness 16.764 mm., is greater than required thickness 6.300 mm.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[138.0/165.1, 1]$$

$$= 0.836$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.764 - 1.5$$

$$= 15.264 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2771.0/44.5$$

$$= 62.270 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2768.0/2 + 1.5$$

$$= 1385.500 \text{ mm.}$$



Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{(\text{Reff} \cdot t)}, 2 \cdot R_n)$$

$$= \min(\sqrt{1385.5 \cdot 44.5}, 2 \cdot 26.9)$$

$$= 53.800 \text{ mm.}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 44.5, 0.0] + \sqrt{26.9 \cdot 15.264}$$

$$= 20.263 \text{ mm.}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 196.85$$

$$= 196.850 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(44.5 + 0.0)$$

$$= 356.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles}$$

$$= \min[20.263, 196.85, 356.0] + 44.5$$

$$= 64.763 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot \text{frp}) / LR)$$

$$= 44.5 + ((0.0 \cdot 1.0) / 53.8)$$

$$= 44.500 \text{ mm.}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + \text{teff}) \cdot \text{teff}}))$$

$$= \min(12, (2 \cdot 26.9 + 15.264) / (\sqrt{(2771.0 + 44.5) \cdot 44.5}))$$

$$= 0.195$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 44.5 \cdot 53.8 \cdot \max((0.195/5)^{0.85}, 1)$$

$$= 23.941 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 15.264 \cdot 64.763$$

$$= 9.885 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 23.941 + 0.836(9.885 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 32.703 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 15.264 / \ln[1 + 15.264/26.9]$$

$$= 33.962 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= \text{teff} / \ln[1 + \text{teff}/\text{Reff}]$$

$$= 44.5 / \ln[1 + 44.5/1385.5]$$

$$= 1407.635 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:



$$\begin{aligned}
 &= 1.5 \cdot S \cdot E \\
 &= 1.5 \cdot 165.119 \cdot 1.0 \\
 &= 247.7 \text{ N./mm}^2
 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned}
 &= P \cdot R_{xn} \cdot L_H \\
 &= 47.332 \cdot 33.962 \cdot 64.763 \\
 &= 1061.6 \text{ Kg}
 \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned}
 &= P \cdot R_{xs} (L_R + t_n) \\
 &= 47.332 \cdot 1407.635 (53.8 + 15.264) \\
 &= 46923.9 \text{ Kg}
 \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned}
 &= P \cdot R_{xs} \cdot R_{nc} \\
 &= 47.332 \cdot 1407.635 \cdot 26.9 \\
 &= 18276.6 \text{ Kg}
 \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned}
 &= (f_S + f_Y + f_N) / P \\
 &= (46923.9 + 18276.6 + 1061.6) / 47.332 \\
 &= 1372.8 \text{ cm}^2
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned}
 &= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff}) \\
 &= 247.678 / (2 \cdot 1372.818 / 32.703 - 1407.635 / 44.5) \\
 &= 47.3 \text{ bars}
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned}
 &= S / [t / R_{xs}] \\
 &= 165.119 [44.5 / 1407.635] \\
 &= 52.2 \text{ bars}
 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned}
 &= \min(P_{max1}, P_{max2}) \\
 &= \min(47.332, 52.197) \\
 &= 47.332 \text{ bars}
 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned}
 &= (f_N + f_S + f_Y) / A_T \\
 &= (1061.641 + 46923.918 + 18276.574) / 32.703 \\
 &= 198.705 \text{ N./mm}^2
 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned}
 &= P \cdot R_{xs} / t_{eff} \\
 &= 47.332 \cdot 1407.635 / 44.5 \\
 &= 149.7 \text{ N./mm}^2
 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned}
 &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\
 &= \max(2 \cdot 198.705 - 149.731, 149.731) \\
 &= 247.7 \text{ N./mm}^2
 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	247.68	N./mm ²
Maximum Allowable Working Pressure	Pmax	47.33	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned}
 &= (R_{nc} + t_n) / R_{nc} \\
 &= (26.9 + 15.264) / 26.9 \\
 &= 1.567 \text{ For set-in Nozzles}
 \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (R_n + t_n)$$



$$= \pi/2 \cdot (26.9 + 15.264)$$
$$= 66.231 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:
= 7.071, 0.000, 0.000, mm.

Required Weld Throat Dimensions :
tc, L41T = 7.071, must be $\geq \min(0.7 \cdot 15.264, 6.000)$ Passed

Weld Load Value [fwelds]:
= $\min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$
= $\min(18277 \cdot 1.57, 1.5 \cdot 138.0(9.885 + 0.0), \pi/4 \cdot 47.3 \cdot 26.9^2 \cdot 1.57^2)$
= 673.947 Kgf

Weld Stress Value [τ]:
= $fwelds / (L_t(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
= $673.947 / (66.231(0.49 \cdot 7.071 + 0.6 \cdot 44.5 + 0.49 \cdot 0.0))$
= 3.308 < or = to 165.119 Weld Stress Passed

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:
= $t_n / \ln[1 + t_n/R_n]$
= $15.264 / \ln[1 + 15.264/26.9]$
= 33.962 mm.

Shell Radius for Force Calculation [Rxs]:
= $t_{eff} / \ln[1 + t_{eff}/R_{eff}]$
= $44.5 / \ln[1 + 44.5/1385.5]$
= 1407.635 mm.

Allowable Local Primary Membrane Stress [Sallow]:
= $1.5 \cdot S$
= $1.5 \cdot 165.119$
= 247.7 N./mm²

Force acting on the Nozzle [fN]:
= $P \cdot R_{xn} \cdot L_H$
= $1.034 \cdot 33.962 \cdot 64.763$
= 23.2 Kgf

Force acting on the Shell [fS]:
= $P \cdot R_{xs}(L_R + t_n)$
= $1.034 \cdot 1407.635(53.8 + 15.264)$
= 1025.3 Kgf

Discontinuity Force from External Pressure [fY]:
= $P \cdot R_{xs} \cdot R_{nc}$
= $1.034 \cdot 1407.635 \cdot 26.9$
= 399.3 Kgf

Area Resisting External Pressure [Ap]:
= $(f_S + f_Y + f_N)/P$
= $(1025.3 + 399.3 + 23.2)/1.034$
= 1372.8 cm²

Maximum Allowable Working Pressure Candidate [Pmax1]:
= $Sallow / (2 \cdot A_p/AT - R_{xs}/t_{eff})$
= $247.678 / (2 \cdot 1372.818/32.703 - 1407.635/44.5)$
= 47.3 bars

Maximum Allowable Working Pressure Candidate [Pmax2]:
= $S[t/R_{xs}]$
= $165.119[44.5/1407.635]$
= 52.2 bars

Maximum Allowable Working Pressure [Pmax]:
= $\min(P_{max1}, P_{max2})$



= min(47.332, 52.197)
= 47.332 bars

Average Primary Membrane Stress [σ_{Avg}]:
= ($f_N + f_S + f_Y$) / AT
= (23.196 + 1025.271 + 399.337)/32.703
= 4.342 N./mm²

General Primary Membrane Stress [σ_{Circ}]:
= P • Rxs / teff
= 1.034 • 1407.635/44.5
= 3.3 N./mm²

Maximum Local Primary Membrane Stress [PL]:
= max(2 • σ_{Avg} - σ_{Circ} , σ_{Circ})
= max(2 • 4.342 - 3.272, 3.272)
= 5.4 N./mm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	5.41	N./mm ²
Maximum Allowable External Pressure	Pmax	47.33	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

This element is Not Post Weld Heat Treated
Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : -10 °C
The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.061

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (0.939 • 1.0)/(16.764 - 1.5 - 0.0)
= 0.061

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 0.6424 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 243.4924 mm.

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Input, Nozzle Desc: L4-2"

From: 70

Pressure for Reinforcement Calculations	P	47.332	bars
Temperature for Internal Pressure	Temp	286	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	286	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	165.12	N./mm ²
Parent Allowable Stress At Ambient	Sva	207.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2768.00	mm.
Design Length of Section	L	7468.6665	mm.
Shell Finished (Minimum) Thickness	t	46.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		10700.00	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

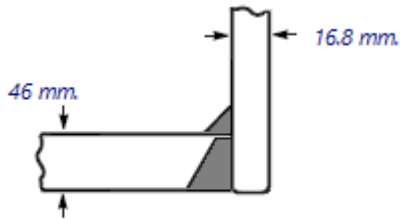
Material		SA-182 F11	
Material UNS Number		K11597	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		120.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Outside Projection	ho	196.8500	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	52.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: L4-2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 53.8 \cdot (\exp(47.332/(138.0 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 0.9387 + 1.5 + 0.0 = 2.4387 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 138.0 \cdot 1.0 \cdot \ln((2 \cdot 15.264 + 53.8) / 53.8) - 0.0$$

$$= 620.192 - 0.0 = 620.1918 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	6.300 mm
Min. required shell thickness, internal p + c :	41.796 mm.
Min. required shell thickness, external p + c :	16.155 mm.
Nozzle neck minimum required thickness :	6.300 mm.

Minimum nozzle neck thickness 16.764 mm., is greater than required thickness 6.300 mm.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[138.0/165.1, 1]$$

$$= 0.836$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.764 - 1.5$$

$$= 15.264 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2771.0/44.5$$

$$= 62.270 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2768.0/2 + 1.5$$

$$= 1385.500 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1385.5 \cdot 44.5}, 2 \cdot 26.9)$$

$$= 53.800 \text{ mm.}$$



Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\ &= \min[1.5 \cdot 44.5, 0.0] + \sqrt{(26.9 \cdot 15.264)} \\ &= 20.263 \text{ mm.} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 196.85 \\ &= 196.850 \text{ mm.} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(44.5 + 0.0) \\ &= 356.000 \text{ mm.} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ &= \min[20.263, 196.85, 356.0] + 44.5 \\ &= 64.763 \text{ mm.} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / LR) \\ &= 44.5 + ((0.0 \cdot 1.0) / 53.8) \\ &= 44.500 \text{ mm.} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}})) \\ &= \min(12, (2 \cdot 26.9 + 15.264) / (\sqrt{(2771.0 + 44.5) \cdot 44.5})) \\ &= 0.195 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 44.5 \cdot 53.8 \cdot \max((0.195/5)^{0.85}, 1) \\ &= 23.941 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 15.264 \cdot 64.763 \\ &= 9.885 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 \\ &= 23.941 + 0.836(9.885 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 32.703 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 15.264 / \ln[1 + 15.264/26.9] \\ &= 33.962 \text{ mm.} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 44.5 / \ln[1 + 44.5/1385.5] \\ &= 1407.635 \text{ mm.} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 165.119 \cdot 1.0 \\ &= 247.7 \text{ N./mm}^2 \end{aligned}$$



Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 47.332 \cdot 33.962 \cdot 64.763$$

$$= 1061.6 \text{ Kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 47.332 \cdot 1407.635 (53.8 + 15.264)$$

$$= 46923.9 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 47.332 \cdot 1407.635 \cdot 26.9$$

$$= 18276.6 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (46923.9 + 18276.6 + 1061.6) / 47.332$$

$$= 1372.8 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 247.678 / (2 \cdot 1372.818 / 32.703 - 1407.635 / 44.5)$$

$$= 47.3 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / R_{xs}$$

$$= 165.119 [44.5 / 1407.635]$$

$$= 52.2 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(47.332, 52.197)$$

$$= 47.332 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (1061.641 + 46923.918 + 18276.574) / 32.703$$

$$= 198.705 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 47.332 \cdot 1407.635 / 44.5$$

$$= 149.7 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 198.705 - 149.731, 149.731)$$

$$= 247.7 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	247.68	N./mm ²
Maximum Allowable Working Pressure	Pmax	47.33	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (26.9 + 15.264) / 26.9$$

$$= 1.567 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (R_n + t_n)$$

$$= \pi / 2 \cdot (26.9 + 15.264)$$

$$= 66.231 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:



= 7.071, 0.000, 0.000, mm.

Required Weld Throat Dimensions :

tc, L41T = 7.071, must be $\geq \min(0.7 * 15.264, 6.000)$ Passed

Weld Load Value [fwelds]:

= $\min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$
= $\min(18277 \cdot 1.57, 1.5 \cdot 138.0(9.885 + 0.0), \pi/4 \cdot 47.3 \cdot 26.9^2 \cdot 1.57^2)$
= 673.947 Kgf

Weld Stress Value [τ]:

= $\text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
= $673.947 / (66.231(0.49 \cdot 7.071 + 0.6 \cdot 44.5 + 0.49 \cdot 0.0))$
= 3.308 < or = to 165.119 Weld Stress Passed

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

= $t_n / \ln[1 + t_n/R_n]$
= $15.264 / \ln[1 + 15.264/26.9]$
= 33.962 mm.

Shell Radius for Force Calculation [Rxs]:

= $t_{eff} / \ln[1 + t_{eff}/R_{eff}]$
= $44.5 / \ln[1 + 44.5/1385.5]$
= 1407.635 mm.

Allowable Local Primary Membrane Stress [Sallow]:

= $1.5 \cdot S$
= $1.5 \cdot 165.119$
= 247.7 N/mm²

Force acting on the Nozzle [fN]:

= $P \cdot R_{xn} \cdot L_H$
= $1.034 \cdot 33.962 \cdot 64.763$
= 23.2 Kgf

Force acting on the Shell [fS]:

= $P \cdot R_{xs} (L_R + t_n)$
= $1.034 \cdot 1407.635 (53.8 + 15.264)$
= 1025.3 Kgf

Discontinuity Force from External Pressure [fY]:

= $P \cdot R_{xs} \cdot R_{nc}$
= $1.034 \cdot 1407.635 \cdot 26.9$
= 399.3 Kgf

Area Resisting External Pressure [Ap]:

= $(f_S + f_Y + f_N) / P$
= $(1025.3 + 399.3 + 23.2) / 1.034$
= 1372.8 cm²

Maximum Allowable Working Pressure Candidate [Pmax1]:

= $S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$
= $247.678 / (2 \cdot 1372.818 / 32.703 - 1407.635 / 44.5)$
= 47.3 bars

Maximum Allowable Working Pressure Candidate [Pmax2]:

= $S[t/R_{xs}]$
= $165.119[44.5/1407.635]$
= 52.2 bars

Maximum Allowable Working Pressure [Pmax]:

= $\min(P_{max1}, P_{max2})$
= $\min(47.332, 52.197)$
= 47.332 bars

Average Primary Membrane Stress [σ_{Avg}]:



$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (23.196 + 1025.271 + 399.337) / 32.703 \\ &= 4.342 \text{ N./mm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 1.034 \cdot 1407.635 / 44.5 \\ &= 3.3 \text{ N./mm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 4.342 - 3.272, 3.272) \\ &= 5.4 \text{ N./mm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	5.41	N./mm ²
Maximum Allowable External Pressure	Pmax	47.33	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	:	-104 °C
User entered Minimum Design Temperature	:	-10 °C
The Curve for this Material is	:	A

As the Stress Ratio < 0.240, the MDMT is	:	-104 °C
The computed stress ratio is	:	0.061

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (0.939 \cdot 1.0) / (16.764 - 1.5 - 0.0) \\ &= 0.061 \end{aligned}$$

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -48 °C

The Drop for this Nozzle is : 0.6424 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 243.4924 mm.

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Input, Nozzle Desc: L6-2"

From: 70

Pressure for Reinforcement Calculations	P	47.332	bars
Temperature for Internal Pressure	Temp	286	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	286	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	165.12	N./mm ²
Parent Allowable Stress At Ambient	Sva	207.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2768.00	mm.
Design Length of Section	L	7468.6665	mm.
Shell Finished (Minimum) Thickness	t	46.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		10700.00	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

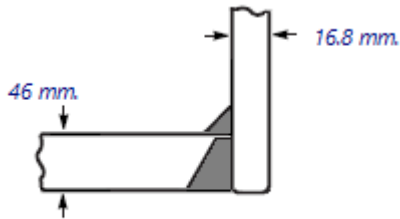
Material		SA-182 F11	
Material UNS Number		K11597	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		150.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Outside Projection	ho	196.8500	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	52.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: L6-2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 53.8 \cdot (\exp(47.332/(138.0 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 0.9387 + 1.5 + 0.0 = 2.4387 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 138.0 \cdot 1.0 \cdot \ln((2 \cdot 15.264 + 53.8) / 53.8) - 0.0$$

$$= 620.192 - 0.0 = 620.1918 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	6.300 mm
Min. required shell thickness, internal p + c :	41.796 mm.
Min. required shell thickness, external p + c :	16.155 mm.
Nozzle neck minimum required thickness :	6.300 mm.

Minimum nozzle neck thickness 16.764 mm., is greater than required thickness 6.300 mm.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[138.0/165.1, 1]$$

$$= 0.836$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.764 - 1.5$$

$$= 15.264 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2771.0/44.5$$

$$= 62.270 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2768.0/2 + 1.5$$

$$= 1385.500 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1385.5 \cdot 44.5}, 2 \cdot 26.9)$$

$$= 53.800 \text{ mm.}$$



Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\ &= \min[1.5 \cdot 44.5, 0.0] + \sqrt{(26.9 \cdot 15.264)} \\ &= 20.263 \text{ mm.} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 196.85 \\ &= 196.850 \text{ mm.} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(44.5 + 0.0) \\ &= 356.000 \text{ mm.} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ &= \min[20.263, 196.85, 356.0] + 44.5 \\ &= 64.763 \text{ mm.} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / LR) \\ &= 44.5 + ((0.0 \cdot 1.0) / 53.8) \\ &= 44.500 \text{ mm.} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}})) \\ &= \min(12, (2 \cdot 26.9 + 15.264) / (\sqrt{(2771.0 + 44.5) \cdot 44.5})) \\ &= 0.195 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 44.5 \cdot 53.8 \cdot \max((0.195/5)^{0.85}, 1) \\ &= 23.941 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 15.264 \cdot 64.763 \\ &= 9.885 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 \\ &= 23.941 + 0.836(9.885 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 32.703 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 15.264 / \ln[1 + 15.264/26.9] \\ &= 33.962 \text{ mm.} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 44.5 / \ln[1 + 44.5/1385.5] \\ &= 1407.635 \text{ mm.} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 165.119 \cdot 1.0 \\ &= 247.7 \text{ N./mm}^2 \end{aligned}$$



Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 47.332 \cdot 33.962 \cdot 64.763$$

$$= 1061.6 \text{ Kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 47.332 \cdot 1407.635 (53.8 + 15.264)$$

$$= 46923.9 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 47.332 \cdot 1407.635 \cdot 26.9$$

$$= 18276.6 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (46923.9 + 18276.6 + 1061.6) / 47.332$$

$$= 1372.8 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 247.678 / (2 \cdot 1372.818 / 32.703 - 1407.635 / 44.5)$$

$$= 47.3 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / R_{xs}$$

$$= 165.119 [44.5 / 1407.635]$$

$$= 52.2 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(47.332, 52.197)$$

$$= 47.332 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (1061.641 + 46923.918 + 18276.574) / 32.703$$

$$= 198.705 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 47.332 \cdot 1407.635 / 44.5$$

$$= 149.7 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 198.705 - 149.731, 149.731)$$

$$= 247.7 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	247.68	N./mm ²
Maximum Allowable Working Pressure	Pmax	47.33	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (26.9 + 15.264) / 26.9$$

$$= 1.567 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (R_n + t_n)$$

$$= \pi / 2 \cdot (26.9 + 15.264)$$

$$= 66.231 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:



= 7.071, 0.000, 0.000, mm.

Required Weld Throat Dimensions :

tc, L41T = 7.071, must be $\geq \min(0.7 * 15.264, 6.000)$ Passed

Weld Load Value [fwelds]:

= $\min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$
= $\min(18277 \cdot 1.57, 1.5 \cdot 138.0(9.885 + 0.0), \pi/4 \cdot 47.3 \cdot 26.9^2 \cdot 1.57^2)$
= 673.947 Kgf

Weld Stress Value [τ]:

= $\text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
= $673.947 / (66.231(0.49 \cdot 7.071 + 0.6 \cdot 44.5 + 0.49 \cdot 0.0))$
= 3.308 < or = to 165.119 Weld Stress Passed

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

= $t_n / \ln[1 + t_n/R_n]$
= $15.264 / \ln[1 + 15.264/26.9]$
= 33.962 mm.

Shell Radius for Force Calculation [Rxs]:

= $t_{eff} / \ln[1 + t_{eff}/R_{eff}]$
= $44.5 / \ln[1 + 44.5/1385.5]$
= 1407.635 mm.

Allowable Local Primary Membrane Stress [Sallow]:

= $1.5 \cdot S$
= $1.5 \cdot 165.119$
= 247.7 N/mm²

Force acting on the Nozzle [fN]:

= $P \cdot R_{xn} \cdot L_H$
= $1.034 \cdot 33.962 \cdot 64.763$
= 23.2 Kgf

Force acting on the Shell [fS]:

= $P \cdot R_{xs} (L_R + t_n)$
= $1.034 \cdot 1407.635 (53.8 + 15.264)$
= 1025.3 Kgf

Discontinuity Force from External Pressure [fY]:

= $P \cdot R_{xs} \cdot R_{nc}$
= $1.034 \cdot 1407.635 \cdot 26.9$
= 399.3 Kgf

Area Resisting External Pressure [Ap]:

= $(f_S + f_Y + f_N) / P$
= $(1025.3 + 399.3 + 23.2) / 1.034$
= 1372.8 cm²

Maximum Allowable Working Pressure Candidate [Pmax1]:

= $S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$
= $247.678 / (2 \cdot 1372.818 / 32.703 - 1407.635 / 44.5)$
= 47.3 bars

Maximum Allowable Working Pressure Candidate [Pmax2]:

= $S[t/R_{xs}]$
= $165.119[44.5/1407.635]$
= 52.2 bars

Maximum Allowable Working Pressure [Pmax]:

= $\min(P_{max1}, P_{max2})$
= $\min(47.332, 52.197)$
= 47.332 bars

Average Primary Membrane Stress [σ_{Avg}]:



$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (23.196 + 1025.271 + 399.337) / 32.703 \\ &= 4.342 \text{ N./mm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 1.034 \cdot 1407.635 / 44.5 \\ &= 3.3 \text{ N./mm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 4.342 - 3.272, 3.272) \\ &= 5.4 \text{ N./mm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	5.41	N./mm ²
Maximum Allowable External Pressure	Pmax	47.33	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	:	-104 °C
User entered Minimum Design Temperature	:	-10 °C
The Curve for this Material is	:	A

As the Stress Ratio < 0.240, the MDMT is	:	-104 °C
The computed stress ratio is	:	0.061

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (0.939 \cdot 1.0) / (16.764 - 1.5 - 0.0) \\ &= 0.061 \end{aligned}$$

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -48 °C

The Drop for this Nozzle is : 0.6424 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 243.4924 mm.

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Input, Nozzle Desc: L8-2"

From: 70

Pressure for Reinforcement Calculations	P	47.332	bars
Temperature for Internal Pressure	Temp	286	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	286	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	165.12	N./mm ²
Parent Allowable Stress At Ambient	Sva	207.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2768.00	mm.
Design Length of Section	L	7468.6665	mm.
Shell Finished (Minimum) Thickness	t	46.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		10700.00	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

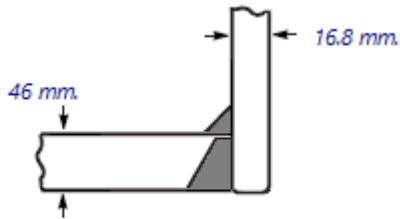
Material		SA-182 F11	
Material UNS Number		K11597	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Outside Projection	ho	196.8500	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	52.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: L8-2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 53.8 \cdot (\exp(47.332/(138.0 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 0.9387 + 1.5 + 0.0 = 2.4387 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 138.0 \cdot 1.0 \cdot \ln((2 \cdot 15.264 + 53.8) / 53.8) - 0.0$$

$$= 620.192 - 0.0 = 620.1918 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	6.300 mm
Min. required shell thickness, internal p + c :	41.796 mm.
Min. required shell thickness, external p + c :	16.155 mm.

Nozzle neck minimum required thickness :	6.300 mm.
--	-----------

Minimum nozzle neck thickness 16.764 mm., is greater than required thickness 6.300 mm.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[138.0/165.1, 1]$$

$$= 0.836$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.764 - 1.5$$

$$= 15.264 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2771.0/44.5$$

$$= 62.270 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2768.0/2 + 1.5$$

$$= 1385.500 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1385.5 \cdot 44.5}, 2 \cdot 26.9)$$

$$= 53.800 \text{ mm.}$$



Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\ &= \min[1.5 \cdot 44.5, 0.0] + \sqrt{(26.9 \cdot 15.264)} \\ &= 20.263 \text{ mm.} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 196.85 \\ &= 196.850 \text{ mm.} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(44.5 + 0.0) \\ &= 356.000 \text{ mm.} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ &= \min[20.263, 196.85, 356.0] + 44.5 \\ &= 64.763 \text{ mm.} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / LR) \\ &= 44.5 + ((0.0 \cdot 1.0) / 53.8) \\ &= 44.500 \text{ mm.} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}})) \\ &= \min(12, (2 \cdot 26.9 + 15.264) / (\sqrt{(2771.0 + 44.5) \cdot 44.5})) \\ &= 0.195 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 44.5 \cdot 53.8 \cdot \max((0.195/5)^{0.85}, 1) \\ &= 23.941 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 15.264 \cdot 64.763 \\ &= 9.885 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 \\ &= 23.941 + 0.836(9.885 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 32.703 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 15.264 / \ln[1 + 15.264/26.9] \\ &= 33.962 \text{ mm.} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 44.5 / \ln[1 + 44.5/1385.5] \\ &= 1407.635 \text{ mm.} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 165.119 \cdot 1.0 \\ &= 247.7 \text{ N./mm}^2 \end{aligned}$$



Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 47.332 \cdot 33.962 \cdot 64.763$$

$$= 1061.6 \text{ Kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 47.332 \cdot 1407.635 (53.8 + 15.264)$$

$$= 46923.9 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 47.332 \cdot 1407.635 \cdot 26.9$$

$$= 18276.6 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (46923.9 + 18276.6 + 1061.6) / 47.332$$

$$= 1372.8 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 247.678 / (2 \cdot 1372.818 / 32.703 - 1407.635 / 44.5)$$

$$= 47.3 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / R_{xs}$$

$$= 165.119 [44.5 / 1407.635]$$

$$= 52.2 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(47.332, 52.197)$$

$$= 47.332 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (1061.641 + 46923.918 + 18276.574) / 32.703$$

$$= 198.705 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 47.332 \cdot 1407.635 / 44.5$$

$$= 149.7 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 198.705 - 149.731, 149.731)$$

$$= 247.7 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	247.68	N./mm ²
Maximum Allowable Working Pressure	Pmax	47.33	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (26.9 + 15.264) / 26.9$$

$$= 1.567 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (R_n + t_n)$$

$$= \pi / 2 \cdot (26.9 + 15.264)$$

$$= 66.231 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:



= 7.071, 0.000, 0.000, mm.

Required Weld Throat Dimensions :

tc, L41T = 7.071, must be $\geq \min(0.7 * 15.264, 6.000)$ Passed

Weld Load Value [fwelds]:

= $\min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$
= $\min(18277 \cdot 1.57, 1.5 \cdot 138.0(9.885 + 0.0), \pi/4 \cdot 47.3 \cdot 26.9^2 \cdot 1.57^2)$
= 673.947 Kgf

Weld Stress Value [τ]:

= $fwelds / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
= $673.947 / (66.231(0.49 \cdot 7.071 + 0.6 \cdot 44.5 + 0.49 \cdot 0.0))$
= 3.308 < or = to 165.119 Weld Stress Passed

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

= $t_n / \ln[1 + t_n/R_n]$
= $15.264 / \ln[1 + 15.264/26.9]$
= 33.962 mm.

Shell Radius for Force Calculation [Rxs]:

= $t_{eff} / \ln[1 + t_{eff}/R_{eff}]$
= $44.5 / \ln[1 + 44.5/1385.5]$
= 1407.635 mm.

Allowable Local Primary Membrane Stress [Sallow]:

= $1.5 \cdot S$
= $1.5 \cdot 165.119$
= 247.7 N/mm²

Force acting on the Nozzle [fN]:

= $P \cdot R_{xn} \cdot L_H$
= $1.034 \cdot 33.962 \cdot 64.763$
= 23.2 Kgf

Force acting on the Shell [fS]:

= $P \cdot R_{xs} (L_R + t_n)$
= $1.034 \cdot 1407.635 (53.8 + 15.264)$
= 1025.3 Kgf

Discontinuity Force from External Pressure [fY]:

= $P \cdot R_{xs} \cdot R_{nc}$
= $1.034 \cdot 1407.635 \cdot 26.9$
= 399.3 Kgf

Area Resisting External Pressure [Ap]:

= $(f_S + f_Y + f_N) / P$
= $(1025.3 + 399.3 + 23.2) / 1.034$
= 1372.8 cm²

Maximum Allowable Working Pressure Candidate [Pmax1]:

= $S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$
= $247.678 / (2 \cdot 1372.818 / 32.703 - 1407.635 / 44.5)$
= 47.3 bars

Maximum Allowable Working Pressure Candidate [Pmax2]:

= $S[t/R_{xs}]$
= $165.119[44.5/1407.635]$
= 52.2 bars

Maximum Allowable Working Pressure [Pmax]:

= $\min(P_{max1}, P_{max2})$
= $\min(47.332, 52.197)$
= 47.332 bars

Average Primary Membrane Stress [σ_{Avg}]:



$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (23.196 + 1025.271 + 399.337) / 32.703 \\ &= 4.342 \text{ N./mm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 1.034 \cdot 1407.635 / 44.5 \\ &= 3.3 \text{ N./mm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 4.342 - 3.272, 3.272) \\ &= 5.4 \text{ N./mm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	5.41	N./mm ²
Maximum Allowable External Pressure	Pmax	47.33	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	:	-104 °C
User entered Minimum Design Temperature	:	-10 °C
The Curve for this Material is	:	A

As the Stress Ratio < 0.240, the MDMT is	:	-104 °C
The computed stress ratio is	:	0.061

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (0.939 \cdot 1.0) / (16.764 - 1.5 - 0.0) \\ &= 0.061 \end{aligned}$$

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -48 °C

The Drop for this Nozzle is : 0.6424 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 243.4924 mm.

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Input, Nozzle Desc: L10-2"

From: 70

Pressure for Reinforcement Calculations	P	47.332	bars
Temperature for Internal Pressure	Temp	286	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	286	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	165.12	N./mm ²
Parent Allowable Stress At Ambient	Sva	207.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2768.00	mm.
Design Length of Section	L	7468.6665	mm.
Shell Finished (Minimum) Thickness	t	46.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		10700.00	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

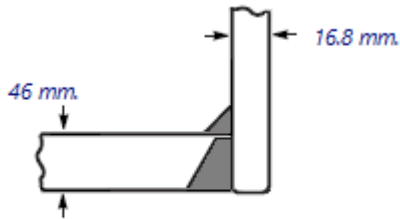
Material		SA-182 F11	
Material UNS Number		K11597	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		210.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Outside Projection	ho	196.8500	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		52.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: L10-2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 53.8 \cdot (\exp(47.332/(138.0 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 0.9387 + 1.5 + 0.0 = 2.4387 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 138.0 \cdot 1.0 \cdot \ln((2 \cdot 15.264 + 53.8) / 53.8) - 0.0$$

$$= 620.192 - 0.0 = 620.1918 \text{ bars}$$

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 51.37 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 95.96 mm.

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	6.300 mm
Min. required shell thickness, internal p + c :	41.796 mm.
Min. required shell thickness, external p + c :	16.155 mm.

Nozzle neck minimum required thickness : 6.300 mm.

Minimum nozzle neck thickness 16.764 mm., is greater than required thickness 6.300 mm.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[138.0/165.1, 1]$$

$$= 0.836$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.764 - 1.5$$

$$= 15.264 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2771.0/44.5$$

$$= 62.270 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2768.0/2 + 1.5$$

$$= 1385.500 \text{ mm.}$$



Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{(\text{Reff} \cdot t)}, 2 \cdot R_n)$$

$$= \min(\sqrt{1385.5 \cdot 44.5}, 2 \cdot 26.9)$$

$$= 53.800 \text{ mm.}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 44.5, 0.0] + \sqrt{26.9 \cdot 15.264}$$

$$= 20.263 \text{ mm.}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 196.85$$

$$= 196.850 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(44.5 + 0.0)$$

$$= 356.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles}$$

$$= \min[20.263, 196.85, 356.0] + 44.5$$

$$= 64.763 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot \text{frp}) / LR)$$

$$= 44.5 + ((0.0 \cdot 1.0) / 53.8)$$

$$= 44.500 \text{ mm.}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + \text{teff}) \cdot \text{teff}}))$$

$$= \min(12, (2 \cdot 26.9 + 15.264) / (\sqrt{(2771.0 + 44.5) \cdot 44.5}))$$

$$= 0.195$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 44.5 \cdot 53.8 \cdot \max((0.195/5)^{0.85}, 1)$$

$$= 23.941 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 15.264 \cdot 64.763$$

$$= 9.885 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 23.941 + 0.836(9.885 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 32.703 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 15.264 / \ln[1 + 15.264/26.9]$$

$$= 33.962 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= \text{teff} / \ln[1 + \text{teff}/\text{Reff}]$$

$$= 44.5 / \ln[1 + 44.5/1385.5]$$

$$= 1407.635 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 165.119 \cdot 1.0$$

$$= 247.7 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 47.332 \cdot 33.962 \cdot 64.763$$

$$= 1061.6 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 47.332 \cdot 1407.635 (53.8 + 15.264)$$

$$= 46923.9 \text{ Kg}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 47.332 \cdot 1407.635 \cdot 26.9$$

$$= 18276.6 \text{ Kg}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (46923.9 + 18276.6 + 1061.6) / 47.332$$

$$= 1372.8 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 247.678 / (2 \cdot 1372.818 / 32.703 - 1407.635 / 44.5)$$

$$= 47.3 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / [t / R_{xs}]$$

$$= 165.119 [44.5 / 1407.635]$$

$$= 52.2 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(47.332, 52.197)$$

$$= 47.332 \text{ bars}$$

Average Primary Membrane Stress [σAvg]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (1061.641 + 46923.918 + 18276.574) / 32.703$$

$$= 198.705 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 47.332 \cdot 1407.635 / 44.5$$

$$= 149.7 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 198.705 - 149.731, 149.731)$$

$$= 247.7 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	247.68	N./mm ²
Maximum Allowable Working Pressure	Pmax	47.33	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (26.9 + 15.264) / 26.9$$

$$= 1.567 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (R_n + t_n)$$



$$= \pi/2 \cdot (26.9 + 15.264)$$

$$= 66.231 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:
 = 7.071, 0.000, 0.000, mm.

Required Weld Throat Dimensions :
 tc, L41T = 7.071, must be $\geq \min(0.7 \cdot 15.264, 6.000)$ Passed

Weld Load Value [fwelds]:
 = $\min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$
 = $\min(18277 \cdot 1.57, 1.5 \cdot 138.0(9.885 + 0.0), \pi/4 \cdot 47.3 \cdot 26.9^2 \cdot 1.57^2)$
 = 673.947 Kgf

Weld Stress Value [τ]:
 = $\text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
 = $673.947 / (66.231(0.49 \cdot 7.071 + 0.6 \cdot 44.5 + 0.49 \cdot 0.0))$
 = 3.308 < or = to 165.119 Weld Stress Passed

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:
 = $t_n / \ln[1 + t_n/R_n]$
 = $15.264 / \ln[1 + 15.264/26.9]$
 = 33.962 mm.

Shell Radius for Force Calculation [Rxs]:
 = $t_{eff} / \ln[1 + t_{eff}/R_{eff}]$
 = $44.5 / \ln[1 + 44.5/1385.5]$
 = 1407.635 mm.

Allowable Local Primary Membrane Stress [Sallow]:
 = $1.5 \cdot S$
 = $1.5 \cdot 165.119$
 = 247.7 N./mm²

Force acting on the Nozzle [fN]:
 = $P \cdot R_{xn} \cdot L_H$
 = $1.034 \cdot 33.962 \cdot 64.763$
 = 23.2 Kgf

Force acting on the Shell [fS]:
 = $P \cdot R_{xs}(L_R + t_n)$
 = $1.034 \cdot 1407.635(53.8 + 15.264)$
 = 1025.3 Kgf

Discontinuity Force from External Pressure [fY]:
 = $P \cdot R_{xs} \cdot R_{nc}$
 = $1.034 \cdot 1407.635 \cdot 26.9$
 = 399.3 Kgf

Area Resisting External Pressure [Ap]:
 = $(f_S + f_Y + f_N)/P$
 = $(1025.3 + 399.3 + 23.2)/1.034$
 = 1372.8 cm²

Maximum Allowable Working Pressure Candidate [Pmax1]:
 = $S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$
 = $247.678 / (2 \cdot 1372.818 / 32.703 - 1407.635 / 44.5)$
 = 47.3 bars

Maximum Allowable Working Pressure Candidate [Pmax2]:
 = $S[t/R_{xs}]$
 = $165.119[44.5/1407.635]$
 = 52.2 bars

Maximum Allowable Working Pressure [Pmax]:
 = $\min(P_{max1}, P_{max2})$



= min(47.332, 52.197)
= 47.332 bars

Average Primary Membrane Stress [σ_{Avg}]:
= ($f_N + f_S + f_Y$) / AT
= (23.196 + 1025.271 + 399.337)/32.703
= 4.342 N./mm²

General Primary Membrane Stress [σ_{Circ}]:
= P • Rxs / teff
= 1.034 • 1407.635/44.5
= 3.3 N./mm²

Maximum Local Primary Membrane Stress [PL]:
= max(2 • σ_{Avg} - σ_{Circ} , σ_{Circ})
= max(2 • 4.342 - 3.272, 3.272)
= 5.4 N./mm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	5.41	N./mm ²
Maximum Allowable External Pressure	Pmax	47.33	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

This element is Not Post Weld Heat Treated
Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : -10 °C
The Curve for this Material is : A

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.061

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (0.939 • 1.0)/(16.764 - 1.5 - 0.0)
= 0.061

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 0.6424 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 243.4924 mm.

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Input, Nozzle Desc: L1-2"

From: 80

Pressure for Reinforcement Calculations	P	47.332	bars
Temperature for Internal Pressure	Temp	286	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	286	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	165.12	N./mm ²
Parent Allowable Stress At Ambient	Sva	207.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2768.00	mm.
Design Length of Section	L	7468.6665	mm.
Shell Finished (Minimum) Thickness	t	46.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		13700.00	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

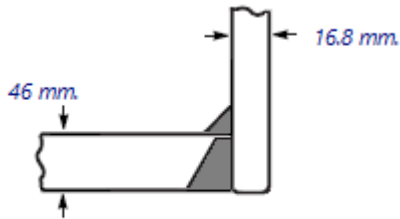
Material		SA-182 F11	
Material UNS Number		K11597	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Outside Projection	ho	196.8500	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	52.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: L1-2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 53.8 \cdot (\exp(47.332/(138.0 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 0.9387 + 1.5 + 0.0 = 2.4387 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 138.0 \cdot 1.0 \cdot \ln((2 \cdot 15.264 + 53.8) / 53.8) - 0.0$$

$$= 620.192 - 0.0 = 620.1918 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	6.300 mm
Min. required shell thickness, internal p + c :	41.796 mm.
Min. required shell thickness, external p + c :	16.155 mm.
Nozzle neck minimum required thickness :	6.300 mm.

Minimum nozzle neck thickness 16.764 mm., is greater than required thickness 6.300 mm.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[138.0/165.1, 1]$$

$$= 0.836$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.764 - 1.5$$

$$= 15.264 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2771.0/44.5$$

$$= 62.270 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2768.0/2 + 1.5$$

$$= 1385.500 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1385.5 \cdot 44.5}, 2 \cdot 26.9)$$

$$= 53.800 \text{ mm.}$$



Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)}$$

$$= \min[1.5 \cdot 44.5, 0.0] + \sqrt{(26.9 \cdot 15.264)}$$

$$= 20.263 \text{ mm.}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 196.85$$

$$= 196.850 \text{ mm.}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(44.5 + 0.0)$$

$$= 356.000 \text{ mm.}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[20.263, 196.85, 356.0] + 44.5$$

$$= 64.763 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 44.5 + ((0.0 \cdot 1.0) / 53.8)$$

$$= 44.500 \text{ mm.}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + teff) \cdot teff}))$$

$$= \min(12, (2 \cdot 26.9 + 15.264) / (\sqrt{(2771.0 + 44.5) \cdot 44.5}))$$

$$= 0.195$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 44.5 \cdot 53.8 \cdot \max((0.195/5)^{0.85}, 1)$$

$$= 23.941 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 15.264 \cdot 64.763$$

$$= 9.885 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + frn(A2 + A3) + A41 + A42 + A43$$

$$= 23.941 + 0.836(9.885 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 32.703 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 15.264 / \ln[1 + 15.264/26.9]$$

$$= 33.962 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= teff / \ln[1 + teff/Reff]$$

$$= 44.5 / \ln[1 + 44.5/1385.5]$$

$$= 1407.635 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 165.119 \cdot 1.0$$

$$= 247.7 \text{ N./mm}^2$$



Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 47.332 \cdot 33.962 \cdot 64.763$$

$$= 1061.6 \text{ Kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 47.332 \cdot 1407.635 (53.8 + 15.264)$$

$$= 46923.9 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 47.332 \cdot 1407.635 \cdot 26.9$$

$$= 18276.6 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (46923.9 + 18276.6 + 1061.6) / 47.332$$

$$= 1372.8 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 247.678 / (2 \cdot 1372.818 / 32.703 - 1407.635 / 44.5)$$

$$= 47.3 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / R_{xs}$$

$$= 165.119 [44.5 / 1407.635]$$

$$= 52.2 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(47.332, 52.197)$$

$$= 47.332 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (1061.641 + 46923.918 + 18276.574) / 32.703$$

$$= 198.705 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 47.332 \cdot 1407.635 / 44.5$$

$$= 149.7 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 198.705 - 149.731, 149.731)$$

$$= 247.7 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	247.68	N./mm ²
Maximum Allowable Working Pressure	Pmax	47.33	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (26.9 + 15.264) / 26.9$$

$$= 1.567 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lr]:

$$= \pi / 2 \cdot (R_n + t_n)$$

$$= \pi / 2 \cdot (26.9 + 15.264)$$

$$= 66.231 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:



= 7.071, 0.000, 0.000, mm.

Required Weld Throat Dimensions :

tc, L41T = 7.071, must be $\geq \min(0.7 * 15.264, 6.000)$ Passed

Weld Load Value [fwelds]:

= $\min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$
= $\min(18277 \cdot 1.57, 1.5 \cdot 138.0(9.885 + 0.0), \pi/4 \cdot 47.3 \cdot 26.9^2 \cdot 1.57^2)$
= 673.947 Kgf

Weld Stress Value [τ]:

= $\text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
= $673.947 / (66.231(0.49 \cdot 7.071 + 0.6 \cdot 44.5 + 0.49 \cdot 0.0))$
= 3.308 < or = to 165.119 Weld Stress Passed

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

= $t_n / \ln[1 + t_n/R_n]$
= $15.264 / \ln[1 + 15.264/26.9]$
= 33.962 mm.

Shell Radius for Force Calculation [Rxs]:

= $t_{eff} / \ln[1 + t_{eff}/R_{eff}]$
= $44.5 / \ln[1 + 44.5/1385.5]$
= 1407.635 mm.

Allowable Local Primary Membrane Stress [Sallow]:

= $1.5 \cdot S$
= $1.5 \cdot 165.119$
= 247.7 N/mm²

Force acting on the Nozzle [fN]:

= $P \cdot R_{xn} \cdot L_H$
= $1.034 \cdot 33.962 \cdot 64.763$
= 23.2 Kgf

Force acting on the Shell [fS]:

= $P \cdot R_{xs}(L_R + t_n)$
= $1.034 \cdot 1407.635(53.8 + 15.264)$
= 1025.3 Kgf

Discontinuity Force from External Pressure [fY]:

= $P \cdot R_{xs} \cdot R_{nc}$
= $1.034 \cdot 1407.635 \cdot 26.9$
= 399.3 Kgf

Area Resisting External Pressure [Ap]:

= $(f_S + f_Y + f_N)/P$
= $(1025.3 + 399.3 + 23.2)/1.034$
= 1372.8 cm²

Maximum Allowable Working Pressure Candidate [Pmax1]:

= $S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$
= $247.678 / (2 \cdot 1372.818 / 32.703 - 1407.635 / 44.5)$
= 47.3 bars

Maximum Allowable Working Pressure Candidate [Pmax2]:

= $S[t/R_{xs}]$
= $165.119[44.5/1407.635]$
= 52.2 bars

Maximum Allowable Working Pressure [Pmax]:

= $\min(P_{max1}, P_{max2})$
= $\min(47.332, 52.197)$
= 47.332 bars

Average Primary Membrane Stress [σ_{Avg}]:



$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (23.196 + 1025.271 + 399.337) / 32.703 \\ &= 4.342 \text{ N./mm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 1.034 \cdot 1407.635 / 44.5 \\ &= 3.3 \text{ N./mm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 4.342 - 3.272, 3.272) \\ &= 5.4 \text{ N./mm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	5.41	N./mm ²
Maximum Allowable External Pressure	Pmax	47.33	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	:	-104 °C
User entered Minimum Design Temperature	:	-10 °C
The Curve for this Material is	:	A

As the Stress Ratio < 0.240, the MDMT is	:	-104 °C
The computed stress ratio is	:	0.061

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (0.939 \cdot 1.0) / (16.764 - 1.5 - 0.0) \\ &= 0.061 \end{aligned}$$

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -48 °C

The Drop for this Nozzle is : 0.6424 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 243.4924 mm.

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Input, Nozzle Desc: L3-2"

From: 80

Pressure for Reinforcement Calculations	P	47.332	bars
Temperature for Internal Pressure	Temp	286	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	286	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	165.12	N./mm ²
Parent Allowable Stress At Ambient	Sva	207.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2768.00	mm.
Design Length of Section	L	7468.6665	mm.
Shell Finished (Minimum) Thickness	t	46.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		13700.00	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

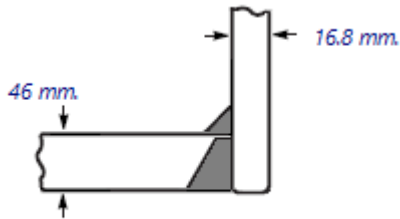
Material		SA-182 F11	
Material UNS Number		K11597	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		120.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Outside Projection	ho	196.8500	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	52.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: L3-2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 53.8 \cdot (\exp(47.332/(138.0 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 0.9387 + 1.5 + 0.0 = 2.4387 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 138.0 \cdot 1.0 \cdot \ln((2 \cdot 15.264 + 53.8) / 53.8) - 0.0$$

$$= 620.192 - 0.0 = 620.1918 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	6.300 mm
Min. required shell thickness, internal p + c :	41.796 mm.
Min. required shell thickness, external p + c :	16.155 mm.
Nozzle neck minimum required thickness :	6.300 mm.

Minimum nozzle neck thickness 16.764 mm., is greater than required thickness 6.300 mm.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[138.0/165.1, 1]$$

$$= 0.836$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.764 - 1.5$$

$$= 15.264 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2771.0/44.5$$

$$= 62.270 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2768.0/2 + 1.5$$

$$= 1385.500 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1385.5 \cdot 44.5}, 2 \cdot 26.9)$$

$$= 53.800 \text{ mm.}$$



Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\ &= \min[1.5 \cdot 44.5, 0.0] + \sqrt{(26.9 \cdot 15.264)} \\ &= 20.263 \text{ mm.} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 196.85 \\ &= 196.850 \text{ mm.} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(44.5 + 0.0) \\ &= 356.000 \text{ mm.} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ &= \min[20.263, 196.85, 356.0] + 44.5 \\ &= 64.763 \text{ mm.} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / LR) \\ &= 44.5 + ((0.0 \cdot 1.0) / 53.8) \\ &= 44.500 \text{ mm.} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}})) \\ &= \min(12, (2 \cdot 26.9 + 15.264) / (\sqrt{(2771.0 + 44.5) \cdot 44.5})) \\ &= 0.195 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 44.5 \cdot 53.8 \cdot \max((0.195/5)^{0.85}, 1) \\ &= 23.941 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 15.264 \cdot 64.763 \\ &= 9.885 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 \\ &= 23.941 + 0.836(9.885 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 32.703 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 15.264 / \ln[1 + 15.264/26.9] \\ &= 33.962 \text{ mm.} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 44.5 / \ln[1 + 44.5/1385.5] \\ &= 1407.635 \text{ mm.} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 165.119 \cdot 1.0 \\ &= 247.7 \text{ N./mm}^2 \end{aligned}$$



Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 47.332 \cdot 33.962 \cdot 64.763$$

$$= 1061.6 \text{ Kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 47.332 \cdot 1407.635 (53.8 + 15.264)$$

$$= 46923.9 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 47.332 \cdot 1407.635 \cdot 26.9$$

$$= 18276.6 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (46923.9 + 18276.6 + 1061.6) / 47.332$$

$$= 1372.8 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 247.678 / (2 \cdot 1372.818 / 32.703 - 1407.635 / 44.5)$$

$$= 47.3 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / R_{xs}$$

$$= 165.119 [44.5 / 1407.635]$$

$$= 52.2 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(47.332, 52.197)$$

$$= 47.332 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (1061.641 + 46923.918 + 18276.574) / 32.703$$

$$= 198.705 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 47.332 \cdot 1407.635 / 44.5$$

$$= 149.7 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 198.705 - 149.731, 149.731)$$

$$= 247.7 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	247.68	N./mm ²
Maximum Allowable Working Pressure	Pmax	47.33	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (26.9 + 15.264) / 26.9$$

$$= 1.567 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (R_n + t_n)$$

$$= \pi / 2 \cdot (26.9 + 15.264)$$

$$= 66.231 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:



= 7.071, 0.000, 0.000, mm.

Required Weld Throat Dimensions :

tc, L41T = 7.071, must be $\geq \min(0.7 * 15.264, 6.000)$ Passed

Weld Load Value [fwelds]:

= $\min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$
= $\min(18277 \cdot 1.57, 1.5 \cdot 138.0(9.885 + 0.0), \pi/4 \cdot 47.3 \cdot 26.9^2 \cdot 1.57^2)$
= 673.947 Kgf

Weld Stress Value [τ]:

= $\text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
= $673.947 / (66.231(0.49 \cdot 7.071 + 0.6 \cdot 44.5 + 0.49 \cdot 0.0))$
= 3.308 < or = to 165.119 Weld Stress Passed

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

= $t_n / \ln[1 + t_n/R_n]$
= $15.264 / \ln[1 + 15.264/26.9]$
= 33.962 mm.

Shell Radius for Force Calculation [Rxs]:

= $t_{eff} / \ln[1 + t_{eff}/R_{eff}]$
= $44.5 / \ln[1 + 44.5/1385.5]$
= 1407.635 mm.

Allowable Local Primary Membrane Stress [Sallow]:

= $1.5 \cdot S$
= $1.5 \cdot 165.119$
= 247.7 N/mm²

Force acting on the Nozzle [fN]:

= $P \cdot R_{xn} \cdot L_H$
= $1.034 \cdot 33.962 \cdot 64.763$
= 23.2 Kgf

Force acting on the Shell [fS]:

= $P \cdot R_{xs} (L_R + t_n)$
= $1.034 \cdot 1407.635 (53.8 + 15.264)$
= 1025.3 Kgf

Discontinuity Force from External Pressure [fY]:

= $P \cdot R_{xs} \cdot R_{nc}$
= $1.034 \cdot 1407.635 \cdot 26.9$
= 399.3 Kgf

Area Resisting External Pressure [Ap]:

= $(f_S + f_Y + f_N) / P$
= $(1025.3 + 399.3 + 23.2) / 1.034$
= 1372.8 cm²

Maximum Allowable Working Pressure Candidate [Pmax1]:

= $S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$
= $247.678 / (2 \cdot 1372.818 / 32.703 - 1407.635 / 44.5)$
= 47.3 bars

Maximum Allowable Working Pressure Candidate [Pmax2]:

= $S[t/R_{xs}]$
= $165.119[44.5/1407.635]$
= 52.2 bars

Maximum Allowable Working Pressure [Pmax]:

= $\min(P_{max1}, P_{max2})$
= $\min(47.332, 52.197)$
= 47.332 bars

Average Primary Membrane Stress [σ_{Avg}]:



$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (23.196 + 1025.271 + 399.337) / 32.703 \\ &= 4.342 \text{ N./mm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 1.034 \cdot 1407.635 / 44.5 \\ &= 3.3 \text{ N./mm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 4.342 - 3.272, 3.272) \\ &= 5.4 \text{ N./mm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	5.41	N./mm ²
Maximum Allowable External Pressure	Pmax	47.33	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	:	-104 °C
User entered Minimum Design Temperature	:	-10 °C
The Curve for this Material is	:	A

As the Stress Ratio < 0.240, the MDMT is	:	-104 °C
The computed stress ratio is	:	0.061

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (0.939 \cdot 1.0) / (16.764 - 1.5 - 0.0) \\ &= 0.061 \end{aligned}$$

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -48 °C

The Drop for this Nozzle is : 0.6424 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 243.4924 mm.

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Input, Nozzle Desc: L5-2"

From: 80

Pressure for Reinforcement Calculations	P	47.332	bars
Temperature for Internal Pressure	Temp	286	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	286	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	165.12	N./mm ²
Parent Allowable Stress At Ambient	Sva	207.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2768.00	mm.
Design Length of Section	L	7468.6665	mm.
Shell Finished (Minimum) Thickness	t	46.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		13700.00	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

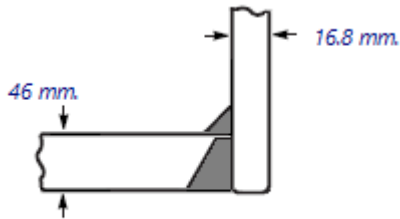
Material		SA-182 F11	
Material UNS Number		K11597	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		150.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Outside Projection	ho	196.8500	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		52.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: L5-2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 53.8 \cdot (\exp(47.332/(138.0 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 0.9387 + 1.5 + 0.0 = 2.4387 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 138.0 \cdot 1.0 \cdot \ln((2 \cdot 15.264 + 53.8) / 53.8) - 0.0$$

$$= 620.192 - 0.0 = 620.1918 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	6.300 mm
Min. required shell thickness, internal p + c :	41.796 mm.
Min. required shell thickness, external p + c :	16.155 mm.

Nozzle neck minimum required thickness :	6.300 mm.
--	-----------

Minimum nozzle neck thickness 16.764 mm., is greater than required thickness 6.300 mm.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[138.0/165.1, 1]$$

$$= 0.836$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.764 - 1.5$$

$$= 15.264 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2771.0/44.5$$

$$= 62.270 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2768.0/2 + 1.5$$

$$= 1385.500 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1385.5 \cdot 44.5}, 2 \cdot 26.9)$$

$$= 53.800 \text{ mm.}$$



Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\ &= \min[1.5 \cdot 44.5, 0.0] + \sqrt{(26.9 \cdot 15.264)} \\ &= 20.263 \text{ mm.} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 196.85 \\ &= 196.850 \text{ mm.} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(44.5 + 0.0) \\ &= 356.000 \text{ mm.} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ &= \min[20.263, 196.85, 356.0] + 44.5 \\ &= 64.763 \text{ mm.} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A_5 \cdot frp) / LR) \\ &= 44.5 + ((0.0 \cdot 1.0) / 53.8) \\ &= 44.500 \text{ mm.} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}})) \\ &= \min(12, (2 \cdot 26.9 + 15.264) / (\sqrt{(2771.0 + 44.5) \cdot 44.5})) \\ &= 0.195 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 44.5 \cdot 53.8 \cdot \max((0.195/5)^{0.85}, 1) \\ &= 23.941 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 15.264 \cdot 64.763 \\ &= 9.885 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 \\ &= 23.941 + 0.836(9.885 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 32.703 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 15.264 / \ln[1 + 15.264/26.9] \\ &= 33.962 \text{ mm.} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 44.5 / \ln[1 + 44.5/1385.5] \\ &= 1407.635 \text{ mm.} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 165.119 \cdot 1.0 \\ &= 247.7 \text{ N./mm}^2 \end{aligned}$$



Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 47.332 \cdot 33.962 \cdot 64.763$$

$$= 1061.6 \text{ Kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 47.332 \cdot 1407.635 (53.8 + 15.264)$$

$$= 46923.9 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 47.332 \cdot 1407.635 \cdot 26.9$$

$$= 18276.6 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (46923.9 + 18276.6 + 1061.6) / 47.332$$

$$= 1372.8 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 247.678 / (2 \cdot 1372.818 / 32.703 - 1407.635 / 44.5)$$

$$= 47.3 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / R_{xs}$$

$$= 165.119 [44.5 / 1407.635]$$

$$= 52.2 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(47.332, 52.197)$$

$$= 47.332 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (1061.641 + 46923.918 + 18276.574) / 32.703$$

$$= 198.705 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 47.332 \cdot 1407.635 / 44.5$$

$$= 149.7 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 198.705 - 149.731, 149.731)$$

$$= 247.7 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	247.68	N./mm ²
Maximum Allowable Working Pressure	Pmax	47.33	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (26.9 + 15.264) / 26.9$$

$$= 1.567 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (R_n + t_n)$$

$$= \pi / 2 \cdot (26.9 + 15.264)$$

$$= 66.231 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:



= 7.071, 0.000, 0.000, mm.

Required Weld Throat Dimensions :

tc, L41T = 7.071, must be $\geq \min(0.7 * 15.264, 6.000)$ Passed

Weld Load Value [fwlds]:

= $\min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$
= $\min(18277 \cdot 1.57, 1.5 \cdot 138.0(9.885 + 0.0), \pi/4 \cdot 47.3 \cdot 26.9^2 \cdot 1.57^2)$
= 673.947 Kgf

Weld Stress Value [τ]:

= $fwlds / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
= $673.947 / (66.231(0.49 \cdot 7.071 + 0.6 \cdot 44.5 + 0.49 \cdot 0.0))$
= 3.308 < or = to 165.119 Weld Stress Passed

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

= $t_n / \ln[1 + t_n/R_n]$
= $15.264 / \ln[1 + 15.264/26.9]$
= 33.962 mm.

Shell Radius for Force Calculation [Rxs]:

= $t_{eff} / \ln[1 + t_{eff}/R_{eff}]$
= $44.5 / \ln[1 + 44.5/1385.5]$
= 1407.635 mm.

Allowable Local Primary Membrane Stress [Sallow]:

= $1.5 \cdot S$
= $1.5 \cdot 165.119$
= 247.7 N/mm²

Force acting on the Nozzle [fN]:

= $P \cdot R_{xn} \cdot L_H$
= $1.034 \cdot 33.962 \cdot 64.763$
= 23.2 Kgf

Force acting on the Shell [fS]:

= $P \cdot R_{xs} (L_R + t_n)$
= $1.034 \cdot 1407.635(53.8 + 15.264)$
= 1025.3 Kgf

Discontinuity Force from External Pressure [fY]:

= $P \cdot R_{xs} \cdot R_{nc}$
= $1.034 \cdot 1407.635 \cdot 26.9$
= 399.3 Kgf

Area Resisting External Pressure [Ap]:

= $(f_S + f_Y + f_N)/P$
= $(1025.3 + 399.3 + 23.2)/1.034$
= 1372.8 cm²

Maximum Allowable Working Pressure Candidate [Pmax1]:

= $Sallow / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$
= $247.678 / (2 \cdot 1372.818 / 32.703 - 1407.635 / 44.5)$
= 47.3 bars

Maximum Allowable Working Pressure Candidate [Pmax2]:

= $S[t/R_{xs}]$
= $165.119[44.5/1407.635]$
= 52.2 bars

Maximum Allowable Working Pressure [Pmax]:

= $\min(P_{max1}, P_{max2})$
= $\min(47.332, 52.197)$
= 47.332 bars

Average Primary Membrane Stress [σ_{Avg}]:



$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (23.196 + 1025.271 + 399.337) / 32.703 \\ &= 4.342 \text{ N./mm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 1.034 \cdot 1407.635 / 44.5 \\ &= 3.3 \text{ N./mm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 4.342 - 3.272, 3.272) \\ &= 5.4 \text{ N./mm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	5.41	N./mm ²
Maximum Allowable External Pressure	Pmax	47.33	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	:	-104 °C
User entered Minimum Design Temperature	:	-10 °C
The Curve for this Material is	:	A

As the Stress Ratio < 0.240, the MDMT is	:	-104 °C
The computed stress ratio is	:	0.061

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (0.939 \cdot 1.0) / (16.764 - 1.5 - 0.0) \\ &= 0.061 \end{aligned}$$

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -48 °C

The Drop for this Nozzle is : 0.6424 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 243.4924 mm.

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Input, Nozzle Desc: L7-2"

From: 80

Pressure for Reinforcement Calculations	P	47.332	bars
Temperature for Internal Pressure	Temp	286	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	286	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	165.12	N./mm ²
Parent Allowable Stress At Ambient	Sva	207.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2768.00	mm.
Design Length of Section	L	7468.6665	mm.
Shell Finished (Minimum) Thickness	t	46.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		13700.00	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

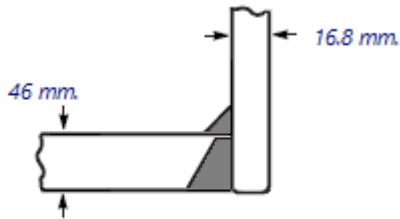
Material		SA-182 F11	
Material UNS Number		K11597	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Outside Projection	ho	196.8500	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		52.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: L7-2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 53.8 \cdot (\exp(47.332/(138.0 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 0.9387 + 1.5 + 0.0 = 2.4387 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 138.0 \cdot 1.0 \cdot \ln((2 \cdot 15.264 + 53.8) / 53.8) - 0.0$$

$$= 620.192 - 0.0 = 620.1918 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	6.300 mm
Min. required shell thickness, internal p + c :	41.796 mm.
Min. required shell thickness, external p + c :	16.155 mm.
Nozzle neck minimum required thickness :	6.300 mm.

Minimum nozzle neck thickness 16.764 mm., is greater than required thickness 6.300 mm.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[138.0/165.1, 1]$$

$$= 0.836$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.764 - 1.5$$

$$= 15.264 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2771.0/44.5$$

$$= 62.270 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2768.0/2 + 1.5$$

$$= 1385.500 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1385.5 \cdot 44.5}, 2 \cdot 26.9)$$

$$= 53.800 \text{ mm.}$$



Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\ &= \min[1.5 \cdot 44.5, 0.0] + \sqrt{(26.9 \cdot 15.264)} \\ &= 20.263 \text{ mm.} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 196.85 \\ &= 196.850 \text{ mm.} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(44.5 + 0.0) \\ &= 356.000 \text{ mm.} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ &= \min[20.263, 196.85, 356.0] + 44.5 \\ &= 64.763 \text{ mm.} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / LR) \\ &= 44.5 + ((0.0 \cdot 1.0) / 53.8) \\ &= 44.500 \text{ mm.} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}})) \\ &= \min(12, (2 \cdot 26.9 + 15.264) / (\sqrt{(2771.0 + 44.5) \cdot 44.5})) \\ &= 0.195 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 44.5 \cdot 53.8 \cdot \max((0.195/5)^{0.85}, 1) \\ &= 23.941 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 15.264 \cdot 64.763 \\ &= 9.885 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 \\ &= 23.941 + 0.836(9.885 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 32.703 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 15.264 / \ln[1 + 15.264/26.9] \\ &= 33.962 \text{ mm.} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 44.5 / \ln[1 + 44.5/1385.5] \\ &= 1407.635 \text{ mm.} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 165.119 \cdot 1.0 \\ &= 247.7 \text{ N./mm}^2 \end{aligned}$$



Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 47.332 \cdot 33.962 \cdot 64.763$$

$$= 1061.6 \text{ Kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 47.332 \cdot 1407.635 (53.8 + 15.264)$$

$$= 46923.9 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 47.332 \cdot 1407.635 \cdot 26.9$$

$$= 18276.6 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (46923.9 + 18276.6 + 1061.6) / 47.332$$

$$= 1372.8 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 247.678 / (2 \cdot 1372.818 / 32.703 - 1407.635 / 44.5)$$

$$= 47.3 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / R_{xs}$$

$$= 165.119 [44.5 / 1407.635]$$

$$= 52.2 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(47.332, 52.197)$$

$$= 47.332 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (1061.641 + 46923.918 + 18276.574) / 32.703$$

$$= 198.705 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 47.332 \cdot 1407.635 / 44.5$$

$$= 149.7 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 198.705 - 149.731, 149.731)$$

$$= 247.7 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	247.68	N./mm ²
Maximum Allowable Working Pressure	Pmax	47.33	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (26.9 + 15.264) / 26.9$$

$$= 1.567 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [L_T]:

$$= \pi / 2 \cdot (R_n + t_n)$$

$$= \pi / 2 \cdot (26.9 + 15.264)$$

$$= 66.231 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:



= 7.071, 0.000, 0.000, mm.

Required Weld Throat Dimensions :

tc, L41T = 7.071, must be $\geq \min(0.7 * 15.264, 6.000)$ Passed

Weld Load Value [fwelds]:

= $\min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$
= $\min(18277 \cdot 1.57, 1.5 \cdot 138.0(9.885 + 0.0), \pi/4 \cdot 47.3 \cdot 26.9^2 \cdot 1.57^2)$
= 673.947 Kgf

Weld Stress Value [τ]:

= $fwelds / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
= $673.947 / (66.231(0.49 \cdot 7.071 + 0.6 \cdot 44.5 + 0.49 \cdot 0.0))$
= 3.308 < or = to 165.119 Weld Stress Passed

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

= $t_n / \ln[1 + t_n/R_n]$
= $15.264 / \ln[1 + 15.264/26.9]$
= 33.962 mm.

Shell Radius for Force Calculation [Rxs]:

= $t_{eff} / \ln[1 + t_{eff}/R_{eff}]$
= $44.5 / \ln[1 + 44.5/1385.5]$
= 1407.635 mm.

Allowable Local Primary Membrane Stress [Sallow]:

= $1.5 \cdot S$
= $1.5 \cdot 165.119$
= 247.7 N/mm²

Force acting on the Nozzle [fN]:

= $P \cdot R_{xn} \cdot L_H$
= $1.034 \cdot 33.962 \cdot 64.763$
= 23.2 Kgf

Force acting on the Shell [fS]:

= $P \cdot R_{xs} (L_R + t_n)$
= $1.034 \cdot 1407.635 (53.8 + 15.264)$
= 1025.3 Kgf

Discontinuity Force from External Pressure [fY]:

= $P \cdot R_{xs} \cdot R_{nc}$
= $1.034 \cdot 1407.635 \cdot 26.9$
= 399.3 Kgf

Area Resisting External Pressure [Ap]:

= $(f_S + f_Y + f_N)/P$
= $(1025.3 + 399.3 + 23.2)/1.034$
= 1372.8 cm²

Maximum Allowable Working Pressure Candidate [Pmax1]:

= $Sallow / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$
= $247.678 / (2 \cdot 1372.818 / 32.703 - 1407.635 / 44.5)$
= 47.3 bars

Maximum Allowable Working Pressure Candidate [Pmax2]:

= $S[t/R_{xs}]$
= $165.119[44.5/1407.635]$
= 52.2 bars

Maximum Allowable Working Pressure [Pmax]:

= $\min(P_{max1}, P_{max2})$
= $\min(47.332, 52.197)$
= 47.332 bars

Average Primary Membrane Stress [σ_{Avg}]:



$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (23.196 + 1025.271 + 399.337) / 32.703 \\ &= 4.342 \text{ N./mm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 1.034 \cdot 1407.635 / 44.5 \\ &= 3.3 \text{ N./mm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 4.342 - 3.272, 3.272) \\ &= 5.4 \text{ N./mm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	5.41	N./mm ²
Maximum Allowable External Pressure	Pmax	47.33	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	:	-104 °C
User entered Minimum Design Temperature	:	-10 °C
The Curve for this Material is	:	A

As the Stress Ratio < 0.240, the MDMT is	:	-104 °C
The computed stress ratio is	:	0.061

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (0.939 \cdot 1.0) / (16.764 - 1.5 - 0.0) \\ &= 0.061 \end{aligned}$$

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -48 °C

The Drop for this Nozzle is : 0.6424 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 243.4924 mm.

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Input, Nozzle Desc: L9-2"

From: 80

Pressure for Reinforcement Calculations	P	47.332	bars
Temperature for Internal Pressure	Temp	286	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	286	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	165.12	N./mm ²
Parent Allowable Stress At Ambient	Sva	207.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2768.00	mm.
Design Length of Section	L	7468.6665	mm.
Shell Finished (Minimum) Thickness	t	46.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		13700.00	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

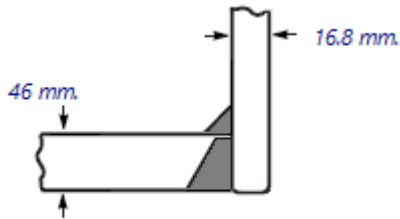
Material		SA-182 F11	
Material UNS Number		K11597	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		210.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Outside Projection	ho	196.8500	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		52.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: L9-2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 53.8 \cdot (\exp(47.332/(138.0 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 0.9387 + 1.5 + 0.0 = 2.4387 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 138.0 \cdot 1.0 \cdot \ln((2 \cdot 15.264 + 53.8) / 53.8) - 0.0$$

$$= 620.192 - 0.0 = 620.1918 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	6.300 mm
Min. required shell thickness, internal p + c :	41.796 mm.
Min. required shell thickness, external p + c :	16.155 mm.

Nozzle neck minimum required thickness :	6.300 mm.
--	-----------

Minimum nozzle neck thickness 16.764 mm., is greater than required thickness 6.300 mm.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[138.0/165.1, 1]$$

$$= 0.836$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.764 - 1.5$$

$$= 15.264 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2771.0/44.5$$

$$= 62.270 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2768.0/2 + 1.5$$

$$= 1385.500 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1385.5 \cdot 44.5}, 2 \cdot 26.9)$$

$$= 53.800 \text{ mm.}$$

Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\ &= \min[1.5 \cdot 44.5, 0.0] + \sqrt{(26.9 \cdot 15.264)} \\ &= 20.263 \text{ mm.} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 196.85 \\ &= 196.850 \text{ mm.} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(44.5 + 0.0) \\ &= 356.000 \text{ mm.} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ &= \min[20.263, 196.85, 356.0] + 44.5 \\ &= 64.763 \text{ mm.} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / LR) \\ &= 44.5 + ((0.0 \cdot 1.0) / 53.8) \\ &= 44.500 \text{ mm.} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + teff) \cdot teff})) \\ &= \min(12, (2 \cdot 26.9 + 15.264) / (\sqrt{(2771.0 + 44.5) \cdot 44.5})) \\ &= 0.195 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 44.5 \cdot 53.8 \cdot \max((0.195/5)^{0.85}, 1) \\ &= 23.941 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 15.264 \cdot 64.763 \\ &= 9.885 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot Leg^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + frn(A2 + A3) + A41 + A42 + A43 \\ &= 23.941 + 0.836(9.885 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 32.703 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 15.264 / \ln[1 + 15.264/26.9] \\ &= 33.962 \text{ mm.} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= teff / \ln[1 + teff/Reff] \\ &= 44.5 / \ln[1 + 44.5/1385.5] \\ &= 1407.635 \text{ mm.} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 165.119 \cdot 1.0 \\ &= 247.7 \text{ N./mm}^2 \end{aligned}$$



Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 47.332 \cdot 33.962 \cdot 64.763$$

$$= 1061.6 \text{ Kgf}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 47.332 \cdot 1407.635 (53.8 + 15.264)$$

$$= 46923.9 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 47.332 \cdot 1407.635 \cdot 26.9$$

$$= 18276.6 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (46923.9 + 18276.6 + 1061.6) / 47.332$$

$$= 1372.8 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 247.678 / (2 \cdot 1372.818 / 32.703 - 1407.635 / 44.5)$$

$$= 47.3 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / R_{xs}$$

$$= 165.119 [44.5 / 1407.635]$$

$$= 52.2 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(47.332, 52.197)$$

$$= 47.332 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (1061.641 + 46923.918 + 18276.574) / 32.703$$

$$= 198.705 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 47.332 \cdot 1407.635 / 44.5$$

$$= 149.7 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 198.705 - 149.731, 149.731)$$

$$= 247.7 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	247.68	N./mm ²
Maximum Allowable Working Pressure	Pmax	47.33	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (26.9 + 15.264) / 26.9$$

$$= 1.567 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (R_n + t_n)$$

$$= \pi / 2 \cdot (26.9 + 15.264)$$

$$= 66.231 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

= 7.071, 0.000, 0.000, mm.

Required Weld Throat Dimensions :

tc, L41T = 7.071, must be $\geq \min(0.7 * 15.264, 6.000)$ Passed

Weld Load Value [fwelds]:

= $\min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$
= $\min(18277 \cdot 1.57, 1.5 \cdot 138.0(9.885 + 0.0), \pi/4 \cdot 47.3 \cdot 26.9^2 \cdot 1.57^2)$
= 673.947 Kgf

Weld Stress Value [τ]:

= $fwelds / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$
= $673.947 / (66.231(0.49 \cdot 7.071 + 0.6 \cdot 44.5 + 0.49 \cdot 0.0))$
= 3.308 < or = to 165.119 Weld Stress Passed

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

= $t_n / \ln[1 + t_n/R_n]$
= $15.264 / \ln[1 + 15.264/26.9]$
= 33.962 mm.

Shell Radius for Force Calculation [Rxs]:

= $t_{eff} / \ln[1 + t_{eff}/R_{eff}]$
= $44.5 / \ln[1 + 44.5/1385.5]$
= 1407.635 mm.

Allowable Local Primary Membrane Stress [Sallow]:

= $1.5 \cdot S$
= $1.5 \cdot 165.119$
= 247.7 N/mm²

Force acting on the Nozzle [fN]:

= $P \cdot R_{xn} \cdot L_H$
= $1.034 \cdot 33.962 \cdot 64.763$
= 23.2 Kgf

Force acting on the Shell [fS]:

= $P \cdot R_{xs} (L_R + t_n)$
= $1.034 \cdot 1407.635 (53.8 + 15.264)$
= 1025.3 Kgf

Discontinuity Force from External Pressure [fY]:

= $P \cdot R_{xs} \cdot R_{nc}$
= $1.034 \cdot 1407.635 \cdot 26.9$
= 399.3 Kgf

Area Resisting External Pressure [Ap]:

= $(f_S + f_Y + f_N)/P$
= $(1025.3 + 399.3 + 23.2)/1.034$
= 1372.8 cm²

Maximum Allowable Working Pressure Candidate [Pmax1]:

= $S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$
= $247.678 / (2 \cdot 1372.818 / 32.703 - 1407.635 / 44.5)$
= 47.3 bars

Maximum Allowable Working Pressure Candidate [Pmax2]:

= $S[t/R_{xs}]$
= $165.119[44.5/1407.635]$
= 52.2 bars

Maximum Allowable Working Pressure [Pmax]:

= $\min(P_{max1}, P_{max2})$
= $\min(47.332, 52.197)$
= 47.332 bars

Average Primary Membrane Stress [σ_{Avg}]:



$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (23.196 + 1025.271 + 399.337) / 32.703 \\ &= 4.342 \text{ N./mm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 1.034 \cdot 1407.635 / 44.5 \\ &= 3.3 \text{ N./mm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 4.342 - 3.272, 3.272) \\ &= 5.4 \text{ N./mm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	5.41	N./mm ²
Maximum Allowable External Pressure	Pmax	47.33	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	:	-104 °C
User entered Minimum Design Temperature	:	-10 °C
The Curve for this Material is	:	A

As the Stress Ratio < 0.240, the MDMT is	:	-104 °C
The computed stress ratio is	:	0.061

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (0.939 \cdot 1.0) / (16.764 - 1.5 - 0.0) \\ &= 0.061 \end{aligned}$$

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -48 °C

The Drop for this Nozzle is : 0.6424 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 243.4924 mm.

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Input, Nozzle Desc: M2-24"

From: 80

Pressure for Reinforcement Calculations	P	47.332	bars
Temperature for Internal Pressure	Temp	286	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	286	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	165.12	N./mm ²
Parent Allowable Stress At Ambient	Sva	207.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2768.00	mm.
Design Length of Section	L	7468.6665	mm.
Shell Finished (Minimum) Thickness	t	46.0000	mm.
Shell Internal Corrosion Allowance	cas	1.5000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		13700.00	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

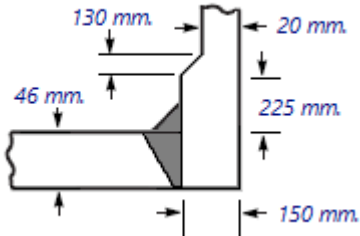
Material		SA-336 F12	
Material UNS Number		K11564	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	132.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		22.4252	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm.
Flange Material		SA-182 F11	
Hub Height of Integral Nozzle	h	225.0000	mm.
Height of Beveled Transition	L'	130.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		150.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Outside Projection	ho	196.8500	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	52.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
This is a Manway, Coupling or Access Opening.			
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: M2-24"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	22.425	in.
Actual Thickness Used in Calculation	0.787	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 572.6 \cdot (\exp(47.332/(131.998 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 10.4532 + 1.5 + 0.0 = 11.9532 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 131.998 \cdot 1.0 \cdot \ln((2 \cdot 18.5 + 572.6) / 572.6) - 0.0$$

$$= 82.646 - 0.0 = 82.6463 \text{ bars}$$

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[132.0/165.1, 1]$$

$$= 0.799$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 150.0 - 1.5$$

$$= 148.500 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 20.0 - 1.5$$

$$= 18.500 \text{ mm.}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2771.0/44.5$$

$$= 62.270 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2768.0/2 + 1.5$$

$$= 1385.500 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1385.5 \cdot 44.5}, 2 \cdot 286.3)$$

$$= 248.304 \text{ mm.}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 44.5, 0.0] + \sqrt{286.3 \cdot 148.5}$$



= 206.193 mm.

Thickness Limit Candidate [LH2]:

= Lpr1
= 196.85
= 196.850 mm.

Thickness Limit Candidate [LH3]:

= 8(t + te)
= 8(44.5 + 0.0)
= 356.000 mm.

Effective Nozzle Wall Length Outside the Vessel [LH]:

= min[LH1, LH2, LH3] + t for set-in nozzles
= min[206.193, 196.85, 356.0] + 44.5
= 241.350 mm.

Effective Vessel Thickness [teff]:

= t + ((A5 • frp) / LR)
= 44.5 + ((0.0 • 1.0) / 248.304)
= 44.500 mm.

Parameter [λ]:

= min(12, (2 • Rn + tn) / (√((Di + teff) • teff)))
= min(12, (2 • 286.3 + 148.5) / (√((2771.0 + 44.5) • 44.5)))
= 2.037

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

= t • LR • max((λ / 5)^{0.85}, 1)
= 44.5 • 248.304 • max((2.037 / 5)^{0.85}, 1)
= 110.495 cm²

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

= tn • LH
= 148.5 • 241.35
= 358.405 cm²

Area Contributed by the Outside Fillet Weld [A41]:

= 0.5 • Leg⁴¹²
= 0.5 • 10.0²
= 0.500 cm²

The total area contributed by A1 through A43 [AT]:

= A1 + frn(A2 + A3) + A41 + A42 + A43
= 110.495 + 0.799(358.405 + 0.0) + 0.5 + 0.0 + 0.0
= 397.507 cm²

Nozzle Radius for Force Calculation [Rxn]:

= tn / ln[1 + tn / Rn]
= 148.5 / ln[1 + 148.5 / 286.3]
= 355.394 mm.

Shell Radius for Force Calculation [Rxs]:

= teff / ln[1 + teff / Reff]
= 44.5 / ln[1 + 44.5 / 1385.5]
= 1407.635 mm.

Allowable Local Primary Membrane Stress [Sallow]:

= 1.5 • S • E
= 1.5 • 165.119 • 1.0
= 247.7 N./mm²

Force acting on the Nozzle [fN]:

= P • Rxn • LH
= 47.332 • 355.394 • 241.35
= 41400.9 Kgf

Force acting on the Shell [fS]:

$$= P \cdot Rxs(LR + tn)$$

$$= 47.332 \cdot 1407.635(248.304 + 148.5)$$

$$= 269599.0 \text{ Kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$

$$= 47.332 \cdot 1407.635 \cdot 286.3$$

$$= 194519.8 \text{ Kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN)/P$$

$$= (269599.0 + 194519.8 + 41400.9)/47.332$$

$$= 10473.4 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap/AT - Rxs/teff)$$

$$= 247.678 / (2 \cdot 10473.351/397.507 - 1407.635/44.5)$$

$$= 117.6 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/Rxs]$$

$$= 165.119[44.5/1407.635]$$

$$= 52.2 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(117.583, 52.197)$$

$$= 52.197 \text{ bars}$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (41400.938 + 269599.0 + 194519.828)/397.507$$

$$= 124.716 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / teff$$

$$= 47.332 \cdot 1407.635/44.5$$

$$= 149.7 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 124.716 - 149.731, 149.731)$$

$$= 149.7 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	149.73	N./mm ²
Maximum Allowable Working Pressure	Pmax	52.20	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (286.3 + 148.5)/286.3$$

$$= 1.519 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi/2 \cdot (Rn + tn)$$

$$= \pi/2 \cdot (286.3 + 148.5)$$

$$= 682.982 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 148.500, 6.000) \text{ Passed}$$



Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(194520 \cdot 1.52, 1.5 \cdot 132.0(358.405 + 0.0), \pi/4 \cdot 47.3 \cdot 286.3^2 \cdot 1.52^2)$$

$$= 71667.375 \text{ Kgf}$$

Weld Stress Value [τ]:

$$= fwelds / (L \cdot \tau(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$

$$= 71667.375 / (682.982(0.49 \cdot 7.071 + 0.6 \cdot 44.5 + 0.49 \cdot 0.0))$$

$$= 34.115 < \text{or} = \text{to } 165.119 \text{ Weld Stress Passed}$$

Nozzle Calculations per Section 4.5: External Pressure Case:Nozzle Radius for Force Calculation [R_{xn}]:

$$= t_n / \ln[1 + t_n / R_n]$$

$$= 148.5 / \ln[1 + 148.5 / 286.3]$$

$$= 355.394 \text{ mm.}$$

Shell Radius for Force Calculation [R_{xs}]:

$$= t_{eff} / \ln[1 + t_{eff} / R_{eff}]$$

$$= 44.5 / \ln[1 + 44.5 / 1385.5]$$

$$= 1407.635 \text{ mm.}$$

Allowable Local Primary Membrane Stress [S_{allow}]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 165.119$$

$$= 247.7 \text{ N./mm}^2$$

Force acting on the Nozzle [f_N]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 1.034 \cdot 355.394 \cdot 241.35$$

$$= 904.6 \text{ Kgf}$$

Force acting on the Shell [f_S]:

$$= P \cdot R_{xs}(L_R + t_n)$$

$$= 1.034 \cdot 1407.635(248.304 + 148.5)$$

$$= 5890.6 \text{ Kgf}$$

Discontinuity Force from External Pressure [f_Y]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 1.034 \cdot 1407.635 \cdot 286.3$$

$$= 4250.2 \text{ Kgf}$$

Area Resisting External Pressure [A_p]:

$$= (f_S + f_Y + f_N) / P$$

$$= (5890.6 + 4250.2 + 904.6) / 1.034$$

$$= 10473.3 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [P_{max1}]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff})$$

$$= 247.678 / (2 \cdot 10473.35 / 397.507 - 1407.635 / 44.5)$$

$$= 117.6 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [P_{max2}]:

$$= S / [t / R_{xs}]$$

$$= 165.119 [44.5 / 1407.635]$$

$$= 52.2 \text{ bars}$$

Maximum Allowable Working Pressure [P_{max}]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(117.583, 52.197)$$

$$= 52.197 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / AT$$

$$= (904.596 + 5890.642 + 4250.189) / 397.507$$

$$= 2.725 \text{ N./mm}^2$$



General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_{xs} / t_{eff} \\ &= 1.034 \cdot 1407.635 / 44.5 \\ &= 3.3 \text{ N./mm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 2.725 - 3.272, 3.272) \\ &= 3.3 \text{ N./mm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	3.27	N./mm ²
Maximum Allowable External Pressure	Pmax	52.20	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	:	36 °C
User entered Minimum Design Temperature	:	-10 °C
The Curve for this Material is	:	A
From Figure 3.7M- Non PWHT, the MDMT is	:	48 °C
Temperature reduction per Graph 3.12	:	11 °C
The computed stress ratio is	:	0.720

The computed thickness ratio [sr]:

$$\begin{aligned} &= (t_r \cdot E^*) / (t_{nom} - c_i - c_{ext}) \\ &= (32.048 \cdot 1.0) / (46.0 - 1.5 - 0.0) \\ &= 0.720 \end{aligned}$$

Warning: Impact Testing Is Mandatory!

Gov. MDMT of the nozzle to shell joint welded assembly : 36 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -48 °C

The Drop for this Nozzle is : 70.0727 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 312.9227 mm.

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Input, Nozzle Desc: B2-14"

From: 90

Pressure for Reinforcement Calculations	P	47.332	bars
Temperature for Internal Pressure	Temp	286	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	286	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	165.12	N./mm ²
Parent Allowable Stress At Ambient	Sva	207.00	N./mm ²
Inside Diameter of Hemispherical Head	D	2812.00	mm.
Head Finished (Minimum) Thickness	t	24.0000	mm.
Head Internal Corrosion Allowance	cas	1.5000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

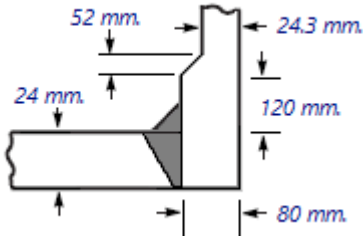
Material		SA-182 F11	
Material UNS Number		K11597	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		14.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		120	
Hub Height of Integral Nozzle	h	120.0000	mm.
Height of Beveled Transition	L'	52.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		80.0000	mm.
Corrosion Allowance	can	1.5000	mm.
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		27.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: B2-14"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	12.085	in.
Actual Thickness Used in Calculation	0.957	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 309.972 \cdot (\exp(47.332/(138.0 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 5.4083 + 1.5 + 0.0 = 6.9083 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 138.0 \cdot 1.0 \cdot \ln((2 \cdot 22.814 + 309.972) / 309.972) - 0.0$$

$$= 189.498 - 0.0 = 189.4982 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	9.840 mm
Min. required shell thickness, internal p + c :	21.820 mm.
Min. required shell thickness, external p + c :	6.881 mm.

Nozzle neck minimum required thickness :	9.840 mm.
--	-----------

Minimum nozzle neck thickness 24.308 mm., is greater than required thickness 9.840 mm.

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	29.5,	Allowable :	138.0 N./mm ²	Passed
Expansion :	0.0,	Allowable :	315.5 N./mm ²	Passed
Occasional :	15.0,	Allowable :	183.5 N./mm ²	Passed
Shear :	9.3,	Allowable :	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[138.0/165.1, 1]$$

$$= 0.836$$

Thickness of Nozzle at Shell [tn]:

$$= \text{hub thickness} - \text{corrosion allowance}$$

$$= 80.0 - 1.5$$

$$= 78.500 \text{ mm.}$$

Thickness of Nozzle at Top [tn2]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 24.314 - 1.5$$

$$= 22.814 \text{ mm.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2812.0/2 + 1.5$$

$$= 1407.500 \text{ mm.}$$



Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot Rn)$$

$$= \min(\sqrt{1407.5 \cdot 22.5}, 2 \cdot 154.986)$$

$$= 177.957 \text{ mm.}$$

Intermediate value [Cn]:

$$= \min((t + te)/tn)^{0.35}, 1)$$

$$= \min((22.5 + 0.0)/78.5)^{0.35}, 1)$$

$$= \min(0.646, 1)$$

$$= 0.646$$

Intermediate value [Cp]:

$$= 9999999.000$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min(t + te + Fp \cdot \sqrt{Rn \cdot tn}, Lpr1 + t)$$

$$= \min(22.5 + 0.0 + 0.646 \cdot \sqrt{154.986 \cdot 78.5}, 300.0 + 22.5)$$

$$= 93.726 \text{ mm.}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 22.5 + ((0.0 \cdot 1.0) / 177.957)$$

$$= 22.500 \text{ mm.}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR$$

$$= 22.5 \cdot 177.957$$

$$= 40.040 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= tn \cdot LH$$

$$= 78.5 \cdot 93.726$$

$$= 73.575 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + frn(A2 + A3) + A41 + A42 + A43$$

$$= 40.04 + 0.836(73.575 + 0.0) + 0.5 + 0.0 + 0.0$$

$$= 102.032 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= tn / \ln[1 + tn/Rn]$$

$$= 78.5 / \ln[1 + 78.5/154.986]$$

$$= 191.563 \text{ mm.}$$

Shell Radius for Force Calculation [Rxs]:

$$= teff / \ln[1 + teff/Reff]$$

$$= 22.5 / \ln[1 + 22.5/1407.5]$$

$$= 1418.715 \text{ mm.}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 165.119 \cdot 1.0$$

$$= 247.7 \text{ N./mm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot Rxn \cdot LH$$

$$= 47.332 \cdot 191.563 \cdot 93.726$$

$$= 8666.1 \text{ Kg}$$

Force acting on the Shell [fS]:

$$= P \cdot Rxs \cdot (LR + tn) / 2$$

$$= 47.332 \cdot 1418.715 \cdot (177.957 + 78.5) / 2$$

$$= 87807.7 \text{ Kg}$$

Discontinuity Force from Internal Pressure [fY]:

$$= (P \cdot Rxs \cdot Rnc) / 2$$

$$= (47.332 \cdot 1418.715 \cdot 154.986) / 2$$

$$= 53065.2 \text{ Kg}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (87807.7 + 53065.2 + 8666.1) / 47.332$$

$$= 3098.1 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap / AT - Rxs / (2 \cdot teff))$$

$$= 247.678 / (2 \cdot 3098.148 / 102.032 - 1418.715 / (2 \cdot 22.5))$$

$$= 84.8 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S[t / Rxs]$$

$$= 2 \cdot 165.119 [22.5 / 1418.715]$$

$$= 52.4 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$

$$= \min(84.81, 52.371)$$

$$= 52.371 \text{ bars}$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (8666.122 + 87807.719 + 53065.215) / 102.032$$

$$= 143.730 \text{ N./mm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot Rxs / (2 \cdot teff)$$

$$= 47.332 \cdot 1418.715 / (2 \cdot 22.5)$$

$$= 149.2 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ)$$

$$= \max(2 \cdot 143.73 - 149.233, 149.233)$$

$$= 149.2 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	149.23	N./mm ²
Maximum Allowable Working Pressure	Pmax	52.37	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (Rnc + tn) / Rnc$$

$$= (154.986 + 78.5) / 154.986$$

$$= 1.506 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$= \pi / 2 \cdot (Rn + tn)$$

$$= \pi / 2 \cdot (154.986 + 78.5)$$

$$= 366.759 \text{ mm.}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 78.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:



$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(53065 \cdot 1.51, 1.5 \cdot 138.0(104.249+0.0), \pi/4 \cdot 47.3 \cdot 154.99^2 \cdot 1.51^2) \\ &= 20666.336 \text{ Kgf} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= f_{welds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T})) \\ &= 20666.336 / (366.759(0.49 \cdot 7.071 + 0.6 \cdot 22.5 + 0.49 \cdot 0.0)) \\ &= 32.573 < \text{or} = \text{to } 165.119 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 78.5 / \ln[1 + 78.5/154.986] \\ &= 191.563 \text{ mm.} \end{aligned}$$

Shell Radius for Force Calculation [R_{xs}]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 22.5 / \ln[1 + 22.5/1407.5] \\ &= 1418.715 \text{ mm.} \end{aligned}$$

Allowable Local Primary Membrane Stress [S_{allow}]:

$$\begin{aligned} &= 1.5 \cdot S \\ &= 1.5 \cdot 165.119 \\ &= 247.7 \text{ N./mm}^2 \end{aligned}$$

Force acting on the Nozzle [f_N]:

$$\begin{aligned} &= P \cdot R_{xn} \cdot L_H \\ &= 1.034 \cdot 191.563 \cdot 93.726 \\ &= 189.4 \text{ Kgf} \end{aligned}$$

Force acting on the Shell [f_S]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot (L_R + t_n) / 2 \\ &= 1.034 \cdot 1418.715 \cdot (177.957 + 78.5) / 2 \\ &= 1918.6 \text{ Kgf} \end{aligned}$$

Discontinuity Force from External Pressure [f_Y]:

$$\begin{aligned} &= (P \cdot R_{xs} \cdot R_{nc}) / 2 \\ &= (1.034 \cdot 1418.715 \cdot 154.986) / 2 \\ &= 1159.5 \text{ Kgf} \end{aligned}$$

Area Resisting External Pressure [A_p]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (1918.6 + 1159.5 + 189.4) / 1.034 \\ &= 3098.1 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [P_{max1}]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / (2 \cdot t_{eff})) \\ &= 247.678 / (2 \cdot 3098.148 / 102.032 - 1418.715 / (2 \cdot 22.5)) \\ &= 84.8 \text{ bars} \end{aligned}$$

Maximum Allowable Working Pressure Candidate [P_{max2}]:

$$\begin{aligned} &= 2 \cdot S / [t / R_{xs}] \\ &= 2 \cdot 165.119 [22.5 / 1418.715] \\ &= 52.4 \text{ bars} \end{aligned}$$

Maximum Allowable Working Pressure [P_{max}]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(84.81, 52.371) \\ &= 52.371 \text{ bars} \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / A_T \\ &= (189.352 + 1918.567 + 1159.456) / 102.032 \\ &= 3.140 \text{ N./mm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / (2 \cdot t_{eff})$$

$$= 1.034 \cdot 1418.715 / (2 \cdot 22.5)$$

$$= 3.3 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$$

$$= \max(2 \cdot 3.14 - 3.261, 3.261)$$

$$= 3.3 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	3.26	N./mm ²
Maximum Allowable External Pressure	Pmax	52.37	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	:	22 °C
User entered Minimum Design Temperature	:	-10 °C
The Curve for this Material is	:	A
From Figure 3.7M- Non PWHT, the MDMT is	:	34 °C
Temperature reduction per Graph 3.12	:	12 °C
The computed stress ratio is	:	0.719

The computed thickness ratio [sr]:

$$= (tr \cdot E^*) / (tnom - ci - cext)$$

$$= (16.185 \cdot 1.0) / (24.0 - 1.5 - 0.0)$$

$$= 0.719$$

Warning: Impact Testing Is Mandatory!

Gov. MDMT of the nozzle to shell joint welded assembly : 22 °C

The Drop for this Nozzle is : 19.5224 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 343.5223 mm.

Input Echo, WRC107/537 Item 1, Description: B2-14" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.5000	mm.
Vessel Diameter	Dv	2812.000	mm.
Vessel Thickness	Tv	24.000	mm.
Design Temperature	T1	286.0	C
Vessel Material		SA-387 11	
Vessel UNS Number		K11789	
Vessel Cold S.I. Allowable	Smc	207.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	165.12	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	1.5000	mm.
Nozzle Diameter	Dn	306.972	mm.
Nozzle Thickness	Tn	80.000	mm.
Nozzle Material		SA-182 F11	
Nozzle UNS Number		K11597	
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm ²
Design Internal Pressure	Dp	47.332	bars
Include Pressure Thrust		No	

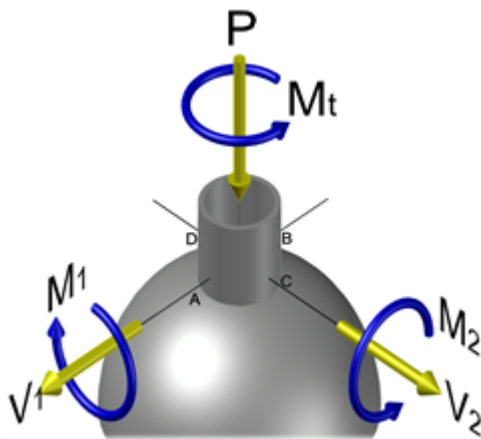
External Forces and Moments in WRC 107/537 Convention:

Radial Load	(SUS)	P	-2284.3	Kgf
Longitudinal Shear	(SUS)	(V1) V1	1713.2	Kgf
Circumferential Shear	(SUS)	(Vc) V2	2284.3	Kgf
Circumferential Moment	(SUS)	(Mc) M1	2077.9	Kg-m.
Longitudinal Moment	(SUS)	(Ml) M2	2077.9	Kg-m.
Torsional Moment	(SUS)	Mt	2397.6	Kg-m.

Use Interactive Control No
WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
Compute Pressure Stress per WRC-368 No
Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(\text{t} - \text{ca}))}$$

$$= 466.972 + 2 \cdot 1.65 \cdot \sqrt{(1418.75 (24.0 - 1.5))}$$

$$= 1056.573 \text{ mm.}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load	P	-2284.3	Kgf
Circumferential Shear	(VC) V2	2284.3	Kgf
Longitudinal Shear	(VL) V1	1713.2	Kgf
Circumferential Moment	(MC) M1	2077.9	Kg-m.
Longitudinal Moment	(ML) M2	2077.9	Kg-m.
Torsional Moment	MT	2397.6	Kg-m.

Dimensionless Param: U = 1.31 TAU = 5.00 (2.47) RHO = 0.29

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 1	0.03036	(A,B,C,D)
$M(x) / P$	SP 1	0.03099	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 1	0.03105	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 1	0.04760	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 1	0.03105	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 1	0.04760	(A,B,C,D)



N(y) * T / P	SP 1	0.04469	(A,B,C,D)
M(y) / P	SP 1	0.00848	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 1	0.04211	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 1	0.01396	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 1	0.04211	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 1	0.01396	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		1.3	1.3	1.3	1.3	1.3	1.3	1.3	1.3
Rad. Bend. P		8.2	-8.2	8.2	-8.2	8.2	-8.2	8.2	-8.2
Rad. Memb. MC		0.0	0.0	0.0	0.0	-7.0	-7.0	7.0	7.0
Rad. Bend. MC		0.0	0.0	0.0	0.0	-64.3	64.3	64.3	-64.3
Rad. Memb. ML		-7.0	-7.0	7.0	7.0	0.0	0.0	0.0	0.0
Rad. Bend. ML		-64.3	64.3	64.3	-64.3	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-61.8	50.5	80.9	-64.2	-61.8	50.5	80.9	-64.2
Tang. Memb. P		2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
Tang. Bend. P		2.3	-2.3	2.3	-2.3	2.3	-2.3	2.3	-2.3
Tang. Memb. MC		0.0	0.0	0.0	0.0	-9.5	-9.5	9.5	9.5
Tang. Bend. MC		0.0	0.0	0.0	0.0	-18.9	18.9	18.9	-18.9
Tang. Memb. ML		-9.5	-9.5	9.5	9.5	0.0	0.0	0.0	0.0
Tang. Bend. ML		-18.9	18.9	18.9	-18.9	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-24.1	9.1	32.6	-9.7	-24.1	9.1	32.6	-9.7
Shear VC		1.4	1.4	-1.4	-1.4	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-1.0	-1.0	1.0	1.0
Shear MT		3.1	3.1	3.1	3.1	3.1	3.1	3.1	3.1
Tot. Shear		4.4	4.4	1.7	1.7	2.0	2.0	4.1	4.1
Str. Int.		62.3	50.9	81.0	64.3	61.9	50.6	81.3	64.5

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		149.2	149.2	149.2	149.2	149.2	149.2	149.2	149.2
Rad. Pl (SUS)		-5.7	-5.7	8.3	8.3	-5.7	-5.7	8.3	8.3
Rad. Q (SUS)		-56.1	56.1	72.6	-72.6	-56.1	56.1	72.6	-72.6
Long. Pm (SUS)		149.2	149.2	149.2	149.2	149.2	149.2	149.2	149.2
Long. Pl (SUS)		-7.5	-7.5	11.5	11.5	-7.5	-7.5	11.5	11.5
Long. Q (SUS)		-16.6	16.6	21.1	-21.1	-16.6	16.6	21.1	-21.1
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		1.4	1.4	-1.4	-1.4	-1.0	-1.0	1.0	1.0
Shear Q (SUS)		3.1	3.1	3.1	3.1	3.1	3.1	3.1	3.1
Pm (SUS)		149.2	149.2	149.2	149.2	149.2	149.2	149.2	149.2
Pm+Pl (SUS)		144.3	144.3	161.2	161.2	144.0	144.0	161.0	161.0
Pm+Pl+Q (Total)		125.6	200.2	230.2	139.6	125.2	199.8	230.5	139.9

STEAM DRUM (WHB)
(A150NRJ10EQ001)



HEXAGON

PV Elite 28 U2 Licensee: SL User

FileName : Steam drum (WHB) - division 2

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Nozzle Calcs.: B2-14"

Nozl: 47 6:47pm Sep 16,2026

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	149.23	165.12	Passed
Pm+Pl (SUS)	161.21	247.68	Passed
Pm+Pl+Q (TOTAL)	230.49	495.38	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

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Input, Nozzle Desc: P1-2"

From: 90

Pressure for Reinforcement Calculations	P	47.332	bars
Temperature for Internal Pressure	Temp	286	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	286	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	165.12	N./mm ²
Parent Allowable Stress At Ambient	Sva	207.00	N./mm ²
Inside Diameter of Hemispherical Head	D	2812.00	mm.
Head Finished (Minimum) Thickness	t	24.0000	mm.
Head Internal Corrosion Allowance	cas	1.5000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	600.0000	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

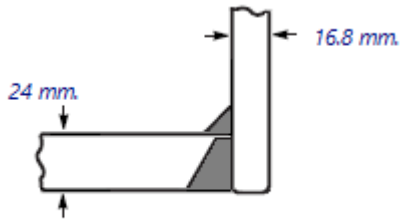
Material		SA-182 F11	
Material UNS Number		K11597	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Outside Projection	ho	400.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	27.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: P1-2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 53.8 \cdot (\exp(47.332/(138.0 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 0.9387 + 1.5 + 0.0 = 2.4387 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 138.0 \cdot 1.0 \cdot \ln((2 \cdot 15.264 + 53.8) / 53.8) - 0.0$$

$$= 620.192 - 0.0 = 620.1918 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	6.300 mm
Min. required shell thickness, internal p + c :	21.820 mm.
Min. required shell thickness, external p + c :	6.881 mm.

Nozzle neck minimum required thickness :	6.300 mm.
--	-----------

Minimum nozzle neck thickness 16.764 mm., is greater than required thickness 6.300 mm.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[138.0/165.1, 1]$$

$$= 0.836$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.764 - 1.5$$

$$= 15.264 \text{ mm.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2812.0/2 + 1.5$$

$$= 1407.500 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1407.5 \cdot 22.5}, 2 \cdot 26.9)$$

$$= 53.800 \text{ mm.}$$

Intermediate value [Cn]:

$$= \min(((t + t_e)/t_n)^{0.35}, 1)$$

$$= \min(((22.5 + 0.0)/15.264)^{0.35}, 1)$$

$$= \min(1.145, 1)$$

$$= 1.000$$

Intermediate value [Cp]:

$$= 999999.000$$

Effective Nozzle Wall Length Outside the Vessel [LH]:



$$\begin{aligned} &= \min(t + t_e + F_p \cdot \sqrt{(R_n \cdot t_n)}, L_{pr1} + t) \\ &= \min(22.5 + 0.0 + 1.0 \cdot \sqrt{(26.9 \cdot 15.264)}, 400.0 + 22.5) \\ &= 42.763 \text{ mm.} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A_5 \cdot frp) / LR) \\ &= 22.5 + ((0.0 \cdot 1.0) / 53.8) \\ &= 22.500 \text{ mm.} \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \\ &= 22.5 \cdot 53.8 \\ &= 12.105 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 15.264 \cdot 42.763 \\ &= 6.527 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot Leg_1^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + frn(A2 + A3) + A41 + A42 + A43 \\ &= 12.105 + 0.836(6.527 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 18.060 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 15.264 / \ln[1 + 15.264/26.9] \\ &= 33.962 \text{ mm.} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 22.5 / \ln[1 + 22.5/1407.5] \\ &= 1418.715 \text{ mm.} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 165.119 \cdot 1.0 \\ &= 247.7 \text{ N./mm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot Rxn \cdot LH \\ &= 47.332 \cdot 33.962 \cdot 42.763 \\ &= 701.0 \text{ Kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot Rxs \cdot (LR + t_n)/2 \\ &= 47.332 \cdot 1418.715 \cdot (53.8 + 15.264)/2 \\ &= 23646.6 \text{ Kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= (P \cdot Rxs \cdot R_{nc})/2 \\ &= (47.332 \cdot 1418.715 \cdot 29.737)/2 \\ &= 10181.7 \text{ Kgf} \end{aligned}$$

Note: Rnc in the above formula is divided by sin(theta).

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN)/P \\ &= (23646.6 + 10181.7 + 701.0)/47.332 \\ &= 715.4 \text{ cm}^2 \end{aligned}$$



Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= \text{Sallow} / (2 \cdot \text{Ap}/\text{AT} - \text{Rxs}/(2 \cdot \text{teff}))$$

$$= 247.678 / (2 \cdot 715.378/18.06 - 1418.715/(2 \cdot 22.5))$$

$$= 51.9 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot \text{S}[t/\text{Rxs}]$$

$$= 2 \cdot 165.119[22.5/1418.715]$$

$$= 52.4 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(\text{Pmax1}, \text{Pmax2})$$

$$= \min(51.928, 52.371)$$

$$= 51.928 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (\text{fN} + \text{fS} + \text{fY}) / \text{AT}$$

$$= (701.003 + 23646.648 + 10181.68) / 18.06$$

$$= 187.496 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= \text{P} \cdot \text{Rxs} / (2 \cdot \text{teff})$$

$$= 47.332 \cdot 1418.715 / (2 \cdot 22.5)$$

$$= 149.2 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$$

$$= \max(2 \cdot 187.496 - 149.233, 149.233)$$

$$= 225.8 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	225.76	N./mm ²
Maximum Allowable Working Pressure	Pmax	51.93	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (\text{Rnc} + \text{tn}) / \text{Rnc}$$

$$= (29.737 + 15.264) / 29.737$$

$$= 1.513 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld for non-radial Nozzles [LT]:

$$= \pi/2 \cdot \sqrt{((\text{Rnc} + \text{Tn})^2 + (\text{Rn} + \text{Tn})^2)/2}$$

$$= \pi/2 \cdot \sqrt{((29.737 + 15.264)^2 + (26.9 + 15.264)^2)/2}$$

$$= 68.496 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$\text{tc, L41T} = 7.071, \text{ must be } \geq \min(0.7 \cdot 15.264, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(\text{fy} \cdot \text{ky}, 1.5 \cdot \text{Sn}(\text{A2} + \text{A3}), \pi/4 \cdot \text{P} \cdot \text{Rn}^2 \cdot \text{ky}^2)$$

$$= \min(10182 \cdot 1.51, 1.5 \cdot 138.0(6.527 + 0.0), \pi/4 \cdot 47.3 \cdot 26.9^2 \cdot 1.51^2)$$

$$= 628.194 \text{ Kg}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (\text{LT}(0.49 \cdot \text{L41T} + 0.6 \cdot \text{tw1} + 0.49 \cdot \text{L43T}))$$

$$= 628.194 / (68.496(0.49 \cdot 7.071 + 0.6 \cdot 22.5 + 0.49 \cdot 0.0))$$

$$= 5.302 < \text{or} = \text{to } 165.119 \text{ Weld Stress Passed}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$= \text{tn} / \ln[1 + \text{tn}/\text{Rn}]$$

$$= 15.264 / \ln[1 + 15.264/26.9]$$

= 33.962 mm.

Shell Radius for Force Calculation [Rxs]:

= $t_{eff} / \ln[1 + t_{eff}/R_{eff}]$
= $22.5 / \ln[1 + 22.5/1407.5]$
= 1418.715 mm.

Allowable Local Primary Membrane Stress [Sallow]:

= $1.5 \cdot S$
= $1.5 \cdot 165.119$
= 247.7 N./mm²

Force acting on the Nozzle [fN]:

= $P \cdot R_{xn} \cdot L_H$
= $1.034 \cdot 33.962 \cdot 42.763$
= 15.3 Kgf

Force acting on the Shell [fS]:

= $P \cdot R_{xs} \cdot (LR + t_n)/2$
= $1.034 \cdot 1418.715 \cdot (53.8 + 15.264)/2$
= 516.7 Kgf

Discontinuity Force from External Pressure [fY]:

= $(P \cdot R_{xs} \cdot R_{nc})/2$
= $(1.034 \cdot 1418.715 \cdot 29.737)/2$
= 222.5 Kgf

Note: Rnc in the above formula is divided by sin(theta).

Area Resisting External Pressure [Ap]:

= $(f_S + f_Y + f_N)/P$
= $(516.7 + 222.5 + 15.3)/1.034$
= 715.4 cm²

Maximum Allowable Working Pressure Candidate [Pmax1]:

= $Sallow / (2 \cdot A_p / AT - R_{xs} / (2 \cdot t_{eff}))$
= $247.678 / (2 \cdot 715.378 / 18.06 - 1418.715 / (2 \cdot 22.5))$
= 51.9 bars

Maximum Allowable Working Pressure Candidate [Pmax2]:

= $2 \cdot S[t/R_{xs}]$
= $2 \cdot 165.119[22.5/1418.715]$
= 52.4 bars

Maximum Allowable Working Pressure [Pmax]:

= min(Pmax1, Pmax2)
= min(51.928, 52.371)
= 51.928 bars

Average Primary Membrane Stress [σAvg]:

= $(f_N + f_S + f_Y) / AT$
= $(15.317 + 516.671 + 222.466) / 18.06$
= 4.097 N./mm²

General Primary Membrane Stress [σCirc]:

= $P \cdot R_{xs} / (2 \cdot t_{eff})$
= $1.034 \cdot 1418.715 / (2 \cdot 22.5)$
= 3.3 N./mm²

Maximum Local Primary Membrane Stress [PL]:

= max($2 \cdot \sigma_{Avg} - \sigma_{Circ}$, σ_{Circ})
= max($2 \cdot 4.097 - 3.261$, 3.261)
= 4.9 N./mm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	4.93	N./mm ²
Maximum Allowable External Pressure	Pmax	51.93	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	:	-104 °C
User entered Minimum Design Temperature	:	-10 °C
The Curve for this Material is	:	A

As the Stress Ratio < 0.240, the MDMT is	:	-104 °C
The computed stress ratio is	:	0.061

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (0.939 \cdot 1.0) / (16.764 - 1.5 - 0.0) \\ &= 0.061 \end{aligned}$$

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

The Drop for this Nozzle is : 20.7644 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 447.2527 mm.

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Input, Nozzle Desc: P2-2"

From: 90

Pressure for Reinforcement Calculations	P	47.332	bars
Temperature for Internal Pressure	Temp	286	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	286	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	165.12	N./mm ²
Parent Allowable Stress At Ambient	Sva	207.00	N./mm ²
Inside Diameter of Hemispherical Head	D	2812.00	mm.
Head Finished (Minimum) Thickness	t	24.0000	mm.
Head Internal Corrosion Allowance	cas	1.5000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	600.0000	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

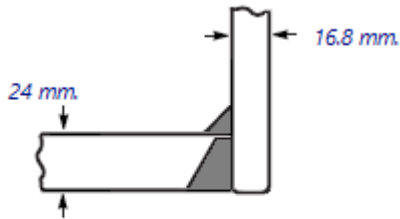
Material		SA-182 F11	
Material UNS Number		K11597	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Outside Projection	ho	400.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	27.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: P2-2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 53.8 \cdot (\exp(47.332/(138.0 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 0.9387 + 1.5 + 0.0 = 2.4387 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 138.0 \cdot 1.0 \cdot \ln((2 \cdot 15.264 + 53.8) / 53.8) - 0.0$$

$$= 620.192 - 0.0 = 620.1918 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	6.300 mm
Min. required shell thickness, internal p + c :	21.820 mm.
Min. required shell thickness, external p + c :	6.881 mm.

Nozzle neck minimum required thickness :	6.300 mm.
--	-----------

Minimum nozzle neck thickness 16.764 mm., is greater than required thickness 6.300 mm.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[138.0/165.1, 1]$$

$$= 0.836$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.764 - 1.5$$

$$= 15.264 \text{ mm.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2812.0/2 + 1.5$$

$$= 1407.500 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1407.5 \cdot 22.5}, 2 \cdot 26.9)$$

$$= 53.800 \text{ mm.}$$

Intermediate value [Cn]:

$$= \min(((t + t_e)/t_n)^{0.35}, 1)$$

$$= \min(((22.5 + 0.0)/15.264)^{0.35}, 1)$$

$$= \min(1.145, 1)$$

$$= 1.000$$

Intermediate value [Cp]:

$$= 999999.000$$

Effective Nozzle Wall Length Outside the Vessel [LH]:



$$\begin{aligned}
 &= \min(t + t_e + F_p \cdot \sqrt{(R_n \cdot t_n)}, L_{pr1} + t) \\
 &= \min(22.5 + 0.0 + 1.0 \cdot \sqrt{(26.9 \cdot 15.264)}, 400.0 + 22.5) \\
 &= 42.763 \text{ mm.}
 \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned}
 &= t + ((A_5 \cdot f_{rp}) / LR) \\
 &= 22.5 + ((0.0 \cdot 1.0) / 53.8) \\
 &= 22.500 \text{ mm.}
 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned}
 &= t \cdot LR \\
 &= 22.5 \cdot 53.8 \\
 &= 12.105 \text{ cm}^2
 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned}
 &= t_n \cdot LH \\
 &= 15.264 \cdot 42.763 \\
 &= 6.527 \text{ cm}^2
 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned}
 &= 0.5 \cdot \text{Leg}^2 \\
 &= 0.5 \cdot 10.0^2 \\
 &= 0.500 \text{ cm}^2
 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned}
 &= A1 + f_{rn}(A2 + A3) + A41 + A42 + A43 \\
 &= 12.105 + 0.836(6.527 + 0.0) + 0.5 + 0.0 + 0.0 \\
 &= 18.060 \text{ cm}^2
 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned}
 &= t_n / \ln[1 + t_n/R_n] \\
 &= 15.264 / \ln[1 + 15.264/26.9] \\
 &= 33.962 \text{ mm.}
 \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned}
 &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\
 &= 22.5 / \ln[1 + 22.5/1407.5] \\
 &= 1418.715 \text{ mm.}
 \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned}
 &= 1.5 \cdot S \cdot E \\
 &= 1.5 \cdot 165.119 \cdot 1.0 \\
 &= 247.7 \text{ N./mm}^2
 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned}
 &= P \cdot R_{xn} \cdot LH \\
 &= 47.332 \cdot 33.962 \cdot 42.763 \\
 &= 701.0 \text{ Kg}
 \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned}
 &= P \cdot R_{xs} \cdot (LR + t_n)/2 \\
 &= 47.332 \cdot 1418.715 \cdot (53.8 + 15.264)/2 \\
 &= 23646.6 \text{ Kg}
 \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned}
 &= (P \cdot R_{xs} \cdot R_{nc})/2 \\
 &= (47.332 \cdot 1418.715 \cdot 29.737)/2 \\
 &= 10181.7 \text{ Kg}
 \end{aligned}$$

Note: Rnc in the above formula is divided by sin(theta).

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned}
 &= (f_S + f_Y + f_N)/P \\
 &= (23646.6 + 10181.7 + 701.0)/47.332 \\
 &= 715.4 \text{ cm}^2
 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:
= Sallow / (2 • Ap/AT - Rxs/(2 • teff))
= 247.678/(2 • 715.378/18.06 - 1418.715/(2•22.5))
= 51.9 bars

Maximum Allowable Working Pressure Candidate [Pmax2]:
= 2 • S[t/Rxs]
= 2 • 165.119[22.5/1418.715]
= 52.4 bars

Maximum Allowable Working Pressure [Pmax]:
= min(Pmax1, Pmax2)
= min(51.928, 52.371)
= 51.928 bars

Average Primary Membrane Stress [σAvg]:
= (fN + fS + fY) / AT
= (701.003 + 23646.648 + 10181.68)/18.06
= 187.496 N./mm²

General Primary Membrane Stress [σCirc]:
= P • Rxs / (2 • teff)
= 47.332 • 1418.715/(2 • 22.5)
= 149.2 N./mm²

Maximum Local Primary Membrane Stress [PL]:
= max(2 • σAvg - σCirc, σCirc)
= max(2 • 187.496 - 149.233, 149.233)
= 225.8 N./mm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	225.76	N./mm ²
Maximum Allowable Working Pressure	Pmax	51.93	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:
= (Rnc + tn) / Rnc
= (29.737 + 15.264)/29.737
= 1.513 For set-in Nozzles

Weld Length of Nozzle to Shell Weld for non-radial Nozzles [Lt]:
= $\pi/2 \cdot \sqrt{((Rnc+Tn)^2 + (Rn+Tn)^2)/2}$
= $\pi/2 \cdot \sqrt{((29.737+15.264)^2 + (26.9+15.264)^2)/2}$
= 68.496 mm.

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:
= 7.071, 0.000, 0.000, mm.

Required Weld Throat Dimensions :
tc, L41T = 7.071, must be >= min(0.7 • 15.264, 6.000) Passed

Weld Load Value [fwelds]:
= min(fy • ky, 1.5 • Sn(A2 + A3), $\pi/4 \cdot P \cdot Rn^2 \cdot ky^2$)
= min(10182•1.51, 1.5•138.0(6.527+0.0), $\pi/4 \cdot 47.3 \cdot 26.9^2 \cdot 1.51^2$)
= 628.194 Kgf

Weld Stress Value [τ]:
= fwelds/(Lt(0.49•L41T + 0.6•tw1 + 0.49•L43T))
= 628.194/(68.496(0.49•7.071 + 0.6•22.5 + 0.49•0.0))
= 5.302 < or = to 165.119 Weld Stress Passed

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:
= tn / ln[1 + tn/Rn]
= 15.264/ln[1 + 15.264/26.9]

= 33.962 mm.

Shell Radius for Force Calculation [Rxs]:

= $t_{eff} / \ln[1 + t_{eff}/R_{eff}]$
= $22.5 / \ln[1 + 22.5/1407.5]$
= 1418.715 mm.

Allowable Local Primary Membrane Stress [Sallow]:

= $1.5 \cdot S$
= $1.5 \cdot 165.119$
= 247.7 N./mm²

Force acting on the Nozzle [fN]:

= $P \cdot R_{xn} \cdot L_H$
= $1.034 \cdot 33.962 \cdot 42.763$
= 15.3 Kgf

Force acting on the Shell [fS]:

= $P \cdot R_{xs} \cdot (LR + t_n)/2$
= $1.034 \cdot 1418.715 \cdot (53.8 + 15.264)/2$
= 516.7 Kgf

Discontinuity Force from External Pressure [fY]:

= $(P \cdot R_{xs} \cdot R_{nc})/2$
= $(1.034 \cdot 1418.715 \cdot 29.737)/2$
= 222.5 Kgf

Note: Rnc in the above formula is divided by sin(theta).

Area Resisting External Pressure [Ap]:

= $(f_S + f_Y + f_N)/P$
= $(516.7 + 222.5 + 15.3)/1.034$
= 715.4 cm²

Maximum Allowable Working Pressure Candidate [Pmax1]:

= $Sallow / (2 \cdot A_p / AT - R_{xs} / (2 \cdot t_{eff}))$
= $247.678 / (2 \cdot 715.378 / 18.06 - 1418.715 / (2 \cdot 22.5))$
= 51.9 bars

Maximum Allowable Working Pressure Candidate [Pmax2]:

= $2 \cdot S[t/R_{xs}]$
= $2 \cdot 165.119[22.5/1418.715]$
= 52.4 bars

Maximum Allowable Working Pressure [Pmax]:

= min(Pmax1, Pmax2)
= min(51.928, 52.371)
= 51.928 bars

Average Primary Membrane Stress [σAvg]:

= $(f_N + f_S + f_Y) / AT$
= $(15.317 + 516.671 + 222.466) / 18.06$
= 4.097 N./mm²

General Primary Membrane Stress [σCirc]:

= $P \cdot R_{xs} / (2 \cdot t_{eff})$
= $1.034 \cdot 1418.715 / (2 \cdot 22.5)$
= 3.3 N./mm²

Maximum Local Primary Membrane Stress [PL]:

= max($2 \cdot \sigma_{Avg} - \sigma_{Circ}$, σ_{Circ})
= max($2 \cdot 4.097 - 3.261$, 3.261)
= 4.9 N./mm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	4.93	N./mm ²
Maximum Allowable External Pressure	Pmax	51.93	bars

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	: -104 °C
User entered Minimum Design Temperature	: -10 °C
The Curve for this Material is	: A

As the Stress Ratio < 0.240, the MDMT is	: -104 °C
The computed stress ratio is	: 0.061

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (0.939 \cdot 1.0) / (16.764 - 1.5 - 0.0) \\ &= 0.061 \end{aligned}$$

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

The Drop for this Nozzle is : 20.7644 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 447.2527 mm.

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Input, Nozzle Desc: V1-2"

From: 90

Pressure for Reinforcement Calculations	P	47.332	bars
Temperature for Internal Pressure	Temp	286	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	286	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	165.12	N./mm ²
Parent Allowable Stress At Ambient	Sva	207.00	N./mm ²
Inside Diameter of Hemispherical Head	D	2812.00	mm.
Head Finished (Minimum) Thickness	t	24.0000	mm.
Head Internal Corrosion Allowance	cas	1.5000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	600.0000	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

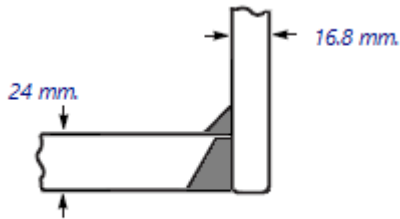
Material		SA-182 F11	
Material UNS Number		K11597	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Outside Projection	ho	400.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	27.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: V1-2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 53.8 \cdot (\exp(47.332/(138.0 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 0.9387 + 1.5 + 0.0 = 2.4387 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 138.0 \cdot 1.0 \cdot \ln((2 \cdot 15.264 + 53.8) / 53.8) - 0.0$$

$$= 620.192 - 0.0 = 620.1918 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	6.300 mm
Min. required shell thickness, internal p + c :	21.820 mm.
Min. required shell thickness, external p + c :	6.881 mm.

Nozzle neck minimum required thickness :	6.300 mm.
--	-----------

Minimum nozzle neck thickness 16.764 mm., is greater than required thickness 6.300 mm.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[138.0/165.1, 1]$$

$$= 0.836$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.764 - 1.5$$

$$= 15.264 \text{ mm.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2812.0/2 + 1.5$$

$$= 1407.500 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1407.5 \cdot 22.5}, 2 \cdot 26.9)$$

$$= 53.800 \text{ mm.}$$

Intermediate value [Cn]:

$$= \min(((t + t_e)/t_n)^{0.35}, 1)$$

$$= \min(((22.5 + 0.0)/15.264)^{0.35}, 1)$$

$$= \min(1.145, 1)$$

$$= 1.000$$

Intermediate value [Cp]:

$$= 999999.000$$

Effective Nozzle Wall Length Outside the Vessel [LH]:



$$\begin{aligned} &= \min(t + t_e + F_p \cdot \sqrt{(R_n \cdot t_n)}, L_{pr1} + t) \\ &= \min(22.5 + 0.0 + 1.0 \cdot \sqrt{(26.9 \cdot 15.264)}, 400.0 + 22.5) \\ &= 42.763 \text{ mm.} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A_5 \cdot frp) / LR) \\ &= 22.5 + ((0.0 \cdot 1.0) / 53.8) \\ &= 22.500 \text{ mm.} \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \\ &= 22.5 \cdot 53.8 \\ &= 12.105 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 15.264 \cdot 42.763 \\ &= 6.527 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot Leg_1^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + frn(A2 + A3) + A41 + A42 + A43 \\ &= 12.105 + 0.836(6.527 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 18.060 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 15.264 / \ln[1 + 15.264/26.9] \\ &= 33.962 \text{ mm.} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= teff / \ln[1 + teff/Reff] \\ &= 22.5 / \ln[1 + 22.5/1407.5] \\ &= 1418.715 \text{ mm.} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 165.119 \cdot 1.0 \\ &= 247.7 \text{ N./mm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot Rxn \cdot LH \\ &= 47.332 \cdot 33.962 \cdot 42.763 \\ &= 701.0 \text{ Kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot Rxs \cdot (LR + t_n)/2 \\ &= 47.332 \cdot 1418.715 \cdot (53.8 + 15.264)/2 \\ &= 23646.6 \text{ Kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= (P \cdot Rxs \cdot R_{nc})/2 \\ &= (47.332 \cdot 1418.715 \cdot 29.737)/2 \\ &= 10181.7 \text{ Kgf} \end{aligned}$$

Note: Rnc in the above formula is divided by sin(theta).

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN)/P \\ &= (23646.6 + 10181.7 + 701.0)/47.332 \\ &= 715.4 \text{ cm}^2 \end{aligned}$$



Maximum Allowable Working Pressure Candidate [Pmax1]:
 = Sallow / (2 • Ap/AT - Rxs/(2 • teff))
 = 247.678/(2 • 715.378/18.06 - 1418.715/(2•22.5))
 = 51.9 bars

Maximum Allowable Working Pressure Candidate [Pmax2]:
 = 2 • S[t/Rxs]
 = 2 • 165.119[22.5/1418.715]
 = 52.4 bars

Maximum Allowable Working Pressure [Pmax]:
 = min(Pmax1, Pmax2)
 = min(51.928, 52.371)
 = 51.928 bars

Average Primary Membrane Stress [σAvg]:
 = (fN + fS + fY) / AT
 = (701.003 + 23646.648 + 10181.68)/18.06
 = 187.496 N./mm²

General Primary Membrane Stress [σCirc]:
 = P • Rxs / (2 • teff)
 = 47.332 • 1418.715/(2 • 22.5)
 = 149.2 N./mm²

Maximum Local Primary Membrane Stress [PL]:
 = max(2 • σAvg - σCirc, σCirc)
 = max(2 • 187.496 - 149.233, 149.233)
 = 225.8 N./mm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	225.76	N./mm ²
Maximum Allowable Working Pressure	Pmax	51.93	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:
 = (Rnc + tn) / Rnc
 = (29.737 + 15.264)/29.737
 = 1.513 For set-in Nozzles

Weld Length of Nozzle to Shell Weld for non-radial Nozzles [Lt]:
 = $\pi/2 \cdot \sqrt{((Rnc+Tn)^2 + (Rn+Tn)^2)/2}$
 = $\pi/2 \cdot \sqrt{((29.737+15.264)^2 + (26.9+15.264)^2)/2}$
 = 68.496 mm.

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:
 = 7.071, 0.000, 0.000, mm.

Required Weld Throat Dimensions :
 tc, L41T = 7.071, must be >= min(0.7 • 15.264, 6.000) Passed

Weld Load Value [fwelds]:
 = min(fy • ky, 1.5 • Sn(A2 + A3), $\pi/4 \cdot P \cdot Rn^2 \cdot ky^2$)
 = min(10182•1.51, 1.5•138.0(6.527+0.0), $\pi/4 \cdot 47.3 \cdot 26.9^2 \cdot 1.51^2$)
 = 628.194 Kgf

Weld Stress Value [τ]:
 = fwelds/(Lt(0.49•L41T + 0.6•tw1 + 0.49•L43T))
 = 628.194/(68.496(0.49•7.071 + 0.6•22.5 + 0.49•0.0))
 = 5.302 < or = to 165.119 Weld Stress Passed

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:
 = tn / ln[1 + tn/Rn]
 = 15.264/ln[1 + 15.264/26.9]

= 33.962 mm.

Shell Radius for Force Calculation [Rxs]:

= $t_{eff} / \ln[1 + t_{eff}/R_{eff}]$
= $22.5 / \ln[1 + 22.5/1407.5]$
= 1418.715 mm.

Allowable Local Primary Membrane Stress [Sallow]:

= $1.5 \cdot S$
= $1.5 \cdot 165.119$
= 247.7 N./mm²

Force acting on the Nozzle [fN]:

= $P \cdot R_{xn} \cdot L_H$
= $1.034 \cdot 33.962 \cdot 42.763$
= 15.3 Kgf

Force acting on the Shell [fS]:

= $P \cdot R_{xs} \cdot (LR + t_n)/2$
= $1.034 \cdot 1418.715 \cdot (53.8 + 15.264)/2$
= 516.7 Kgf

Discontinuity Force from External Pressure [fY]:

= $(P \cdot R_{xs} \cdot R_{nc})/2$
= $(1.034 \cdot 1418.715 \cdot 29.737)/2$
= 222.5 Kgf

Note: Rnc in the above formula is divided by sin(theta).

Area Resisting External Pressure [Ap]:

= $(f_S + f_Y + f_N)/P$
= $(516.7 + 222.5 + 15.3)/1.034$
= 715.4 cm²

Maximum Allowable Working Pressure Candidate [Pmax1]:

= $Sallow / (2 \cdot A_p / AT - R_{xs} / (2 \cdot t_{eff}))$
= $247.678 / (2 \cdot 715.378 / 18.06 - 1418.715 / (2 \cdot 22.5))$
= 51.9 bars

Maximum Allowable Working Pressure Candidate [Pmax2]:

= $2 \cdot S[t/R_{xs}]$
= $2 \cdot 165.119[22.5/1418.715]$
= 52.4 bars

Maximum Allowable Working Pressure [Pmax]:

= min(Pmax1, Pmax2)
= min(51.928, 52.371)
= 51.928 bars

Average Primary Membrane Stress [σAvg]:

= $(f_N + f_S + f_Y) / AT$
= $(15.317 + 516.671 + 222.466) / 18.06$
= 4.097 N./mm²

General Primary Membrane Stress [σCirc]:

= $P \cdot R_{xs} / (2 \cdot t_{eff})$
= $1.034 \cdot 1418.715 / (2 \cdot 22.5)$
= 3.3 N./mm²

Maximum Local Primary Membrane Stress [PL]:

= max($2 \cdot \sigma_{Avg} - \sigma_{Circ}$, σ_{Circ})
= max($2 \cdot 4.097 - 3.261$, 3.261)
= 4.9 N./mm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	4.93	N./mm ²
Maximum Allowable External Pressure	Pmax	51.93	bars



Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	: -104 °C
User entered Minimum Design Temperature	: -10 °C
The Curve for this Material is	: A

As the Stress Ratio < 0.240, the MDMT is	: -104 °C
The computed stress ratio is	: 0.061

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (0.939 \cdot 1.0) / (16.764 - 1.5 - 0.0) \\ &= 0.061 \end{aligned}$$

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

The Drop for this Nozzle is : 20.7644 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 447.2527 mm.

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Input, Nozzle Desc: S1-2"

From: 90

Pressure for Reinforcement Calculations	P	47.332	bars
Temperature for Internal Pressure	Temp	286	°C
Design External Pressure	Pext	1.03	bars
Temperature for External Pressure	Tempex	286	°C
Parent Material		SA-387 11	
Parent Allowable Stress at Temperature	Sv	165.12	N./mm ²
Parent Allowable Stress At Ambient	Sva	207.00	N./mm ²
Inside Diameter of Hemispherical Head	D	2812.00	mm.
Head Finished (Minimum) Thickness	t	24.0000	mm.
Head Internal Corrosion Allowance	cas	1.5000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	600.0000	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

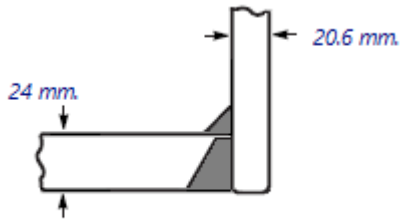
Material		SA-182 F11	
Material UNS Number		K11597	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.5740	mm.
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.5000	mm.
Outside Projection	ho	190.5000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	27.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.

The Pressure Design option was MAWP + static head (to the nozzle).

Reinforcement CALCULATION, Description: S1-2"

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	3.000	in.
Actual Thickness Used in Calculation	0.810	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 79.2 \cdot (\exp(47.332/(138.0 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 1.3819 + 1.5 + 0.0 = 2.8819 \text{ mm. Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 138.0 \cdot 1.0 \cdot \ln((2 \cdot 19.074 + 79.2) / 79.2) - 0.0$$

$$= 542.54 - 0.0 = 542.5397 \text{ bars}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	7.230 mm
Min. required shell thickness, internal p + c :	21.820 mm.
Min. required shell thickness, external p + c :	6.881 mm.

Nozzle neck minimum required thickness :	7.230 mm.
--	-----------

Minimum nozzle neck thickness 20.574 mm., is greater than required thickness 7.230 mm.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[138.0/165.1, 1]$$

$$= 0.836$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 20.574 - 1.5$$

$$= 19.074 \text{ mm.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2812.0/2 + 1.5$$

$$= 1407.500 \text{ mm.}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1407.5 \cdot 22.5}, 2 \cdot 39.6)$$

$$= 79.200 \text{ mm.}$$

Intermediate value [Cn]:

$$= \min(((t + t_e)/t_n)^{0.35}, 1)$$

$$= \min(((22.5 + 0.0)/19.074)^{0.35}, 1)$$

$$= \min(1.06, 1)$$

$$= 1.000$$

Intermediate value [Cp]:

$$= 999999.000$$

Effective Nozzle Wall Length Outside the Vessel [LH]:



$$\begin{aligned}
 &= \min(t + t_e + F_p \cdot \sqrt{(R_n \cdot t_n)}, L_{pr1} + t) \\
 &= \min(22.5 + 0.0 + 1.0 \cdot \sqrt{(39.6 \cdot 19.074)}, 190.5 + 22.5) \\
 &= 49.983 \text{ mm.}
 \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned}
 &= t + ((A_5 \cdot f_{rp}) / LR) \\
 &= 22.5 + ((0.0 \cdot 1.0) / 79.2) \\
 &= 22.500 \text{ mm.}
 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned}
 &= t \cdot LR \\
 &= 22.5 \cdot 79.2 \\
 &= 17.820 \text{ cm}^2
 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned}
 &= t_n \cdot LH \\
 &= 19.074 \cdot 49.983 \\
 &= 9.534 \text{ cm}^2
 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned}
 &= 0.5 \cdot \text{Leg}^2 \\
 &= 0.5 \cdot 10.0^2 \\
 &= 0.500 \text{ cm}^2
 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned}
 &= A1 + f_{rn}(A2 + A3) + A41 + A42 + A43 \\
 &= 17.82 + 0.836(9.534 + 0.0) + 0.5 + 0.0 + 0.0 \\
 &= 26.288 \text{ cm}^2
 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned}
 &= t_n / \ln[1 + t_n/R_n] \\
 &= 19.074 / \ln[1 + 19.074/39.6] \\
 &= 48.514 \text{ mm.}
 \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned}
 &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\
 &= 22.5 / \ln[1 + 22.5/1407.5] \\
 &= 1418.715 \text{ mm.}
 \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned}
 &= 1.5 \cdot S \cdot E \\
 &= 1.5 \cdot 165.119 \cdot 1.0 \\
 &= 247.7 \text{ N./mm}^2
 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned}
 &= P \cdot R_{xn} \cdot LH \\
 &= 47.332 \cdot 48.514 \cdot 49.983 \\
 &= 1170.4 \text{ Kg}
 \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned}
 &= P \cdot R_{xs} \cdot (LR + t_n)/2 \\
 &= 47.332 \cdot 1418.715 \cdot (79.2 + 19.074)/2 \\
 &= 33647.8 \text{ Kg}
 \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned}
 &= (P \cdot R_{xs} \cdot R_{nc})/2 \\
 &= (47.332 \cdot 1418.715 \cdot 43.777)/2 \\
 &= 14988.6 \text{ Kg}
 \end{aligned}$$

Note: Rnc in the above formula is divided by sin(theta).

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned}
 &= (f_S + f_Y + f_N)/P \\
 &= (33647.8 + 14988.6 + 1170.4)/47.332 \\
 &= 1031.9 \text{ cm}^2
 \end{aligned}$$



Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= \text{Sallow} / (2 \cdot \text{Ap}/\text{AT} - \text{Rxs}/(2 \cdot \text{teff}))$$

$$= 247.678 / (2 \cdot 1031.897/26.288 - 1418.715/(2 \cdot 22.5))$$

$$= 52.7 \text{ bars}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot \text{S}[t/\text{Rxs}]$$

$$= 2 \cdot 165.119[22.5/1418.715]$$

$$= 52.4 \text{ bars}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(\text{Pmax1}, \text{Pmax2})$$

$$= \min(52.717, 52.371)$$

$$= 52.371 \text{ bars}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / \text{AT}$$

$$= (1170.419 + 33647.785 + 14988.645) / 26.288$$

$$= 185.806 \text{ N./mm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot \text{Rxs} / (2 \cdot \text{teff})$$

$$= 47.332 \cdot 1418.715 / (2 \cdot 22.5)$$

$$= 149.2 \text{ N./mm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$$

$$= \max(2 \cdot 185.806 - 149.233, 149.233)$$

$$= 222.4 \text{ N./mm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	222.38	N./mm ²
Maximum Allowable Working Pressure	Pmax	52.37	bars

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= (\text{Rnc} + \text{tn}) / \text{Rnc}$$

$$= (43.777 + 19.074) / 43.777$$

$$= 1.436 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld for non-radial Nozzles [LT]:

$$= \pi/2 \cdot \sqrt{((\text{Rnc} + \text{Tn})^2 + (\text{Rn} + \text{Tn})^2)/2}$$

$$= \pi/2 \cdot \sqrt{((43.777 + 19.074)^2 + (39.6 + 19.074)^2)/2}$$

$$= 95.502 \text{ mm.}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm.}$$

Required Weld Throat Dimensions :

$$\text{tc, L41T} = 7.071, \text{ must be } \geq \min(0.7 \cdot 19.074, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot \text{Sn}(\text{A2} + \text{A3}), \pi/4 \cdot P \cdot \text{Rn}^2 \cdot k_y^2)$$

$$= \min(14989 \cdot 1.44, 1.5 \cdot 138.0(9.534 + 0.0), \pi/4 \cdot 47.3 \cdot 39.6^2 \cdot 1.44^2)$$

$$= 1225.367 \text{ Kg}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (\text{LT}(0.49 \cdot \text{L41T} + 0.6 \cdot \text{tw1} + 0.49 \cdot \text{L43T}))$$

$$= 1225.367 / (95.502(0.49 \cdot 7.071 + 0.6 \cdot 22.5 + 0.49 \cdot 0.0))$$

$$= 7.417 < \text{or} = \text{to } 165.119 \text{ Weld Stress Passed}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$= \text{tn} / \ln[1 + \text{tn}/\text{Rn}]$$

$$= 19.074 / \ln[1 + 19.074/39.6]$$



= 48.514 mm.

Shell Radius for Force Calculation [Rxs]:

= $t_{eff} / \ln[1 + t_{eff}/R_{eff}]$
= $22.5 / \ln[1 + 22.5/1407.5]$
= 1418.715 mm.

Allowable Local Primary Membrane Stress [Sallow]:

= $1.5 \cdot S$
= $1.5 \cdot 165.119$
= 247.7 N./mm²

Force acting on the Nozzle [fN]:

= $P \cdot R_{xn} \cdot L_H$
= $1.034 \cdot 48.514 \cdot 49.983$
= 25.6 Kgf

Force acting on the Shell [fS]:

= $P \cdot R_{xs} \cdot (L_R + t_n)/2$
= $1.034 \cdot 1418.715 \cdot (79.2 + 19.074)/2$
= 735.2 Kgf

Discontinuity Force from External Pressure [fY]:

= $(P \cdot R_{xs} \cdot R_{nc})/2$
= $(1.034 \cdot 1418.715 \cdot 43.777)/2$
= 327.5 Kgf

Note: Rnc in the above formula is divided by sin(theta).

Area Resisting External Pressure [Ap]:

= $(f_S + f_Y + f_N)/P$
= $(735.2 + 327.5 + 25.6)/1.034$
= 1031.9 cm²

Maximum Allowable Working Pressure Candidate [Pmax1]:

= $Sallow / (2 \cdot A_p / AT - R_{xs} / (2 \cdot t_{eff}))$
= $247.678 / (2 \cdot 1031.897 / 26.288 - 1418.715 / (2 \cdot 22.5))$
= 52.7 bars

Maximum Allowable Working Pressure Candidate [Pmax2]:

= $2 \cdot S[t/R_{xs}]$
= $2 \cdot 165.119[22.5/1418.715]$
= 52.4 bars

Maximum Allowable Working Pressure [Pmax]:

= min(Pmax1, Pmax2)
= min(52.717, 52.371)
= 52.371 bars

Average Primary Membrane Stress [σAvg]:

= $(f_N + f_S + f_Y) / AT$
= $(25.573 + 735.192 + 327.497) / 26.288$
= 4.060 N./mm²

General Primary Membrane Stress [σCirc]:

= $P \cdot R_{xs} / (2 \cdot t_{eff})$
= $1.034 \cdot 1418.715 / (2 \cdot 22.5)$
= 3.3 N./mm²

Maximum Local Primary Membrane Stress [PL]:

= max($2 \cdot \sigma_{Avg} - \sigma_{Circ}$, σ_{Circ})
= max($2 \cdot 4.06 - 3.261$, 3.261)
= 4.9 N./mm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	247.68	N./mm ²
Local Primary Membrane Stress	PL	4.86	N./mm ²
Maximum Allowable External Pressure	Pmax	52.37	bars



Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: A

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	: -104 °C
User entered Minimum Design Temperature	: -10 °C
The Curve for this Material is	: A

As the Stress Ratio < 0.240, the MDMT is	: -104 °C
The computed stress ratio is	: 0.072

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (1.382 \cdot 1.0) / (20.574 - 1.5 - 0.0) \\ &= 0.072 \end{aligned}$$

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

The Drop for this Nozzle is : 29.3795 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 246.3678 mm.

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STEAM DRUM (WHB)
(A150NRJ10EQ001)

PV Elite 28 U2 Licensee: SL User

FileName : Steam drum (WHB) - division 2

Problems/Failures Summary:

Step: 51

6:47pm

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HEXAGON

Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: The computed vessel MAWP seems to be less than the design pressure!**

Please review all reports carefully!

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